CONFIDENTIAL

STi5512

SET TOP BOX BACKEND DECODER

REGISTER MANUAL

7145562 B 14/4/00

1 Introduction

This manual describes the bit formats and functionality of the STi5512 registers. The registers are grouped into chapters according to their associated module. A full list of all registers with their absolute addresses is given in the Register map on page 4; an alphabetical index is in the Index of registers on page 203.

The STi5512 can be used as an advanced set top box backend decoder, or it can be used to replace the STi5510. The Revision history on page 201 describes the changes made to the STi5510 Register Manual version 7112462A to create this STi5512 Register Manual. The changes listed in the Revision history are hypertext-linked into the document where they are also identified by change bars.

The registers can be examined and set by the *devlw* (device load word) and *devsw* (device store word) instructions; they cannot be accessed using memory instructions. The registers are all in the peripheral address space in region 2 of the address space. The registers of each module are also grouped in the address space, usually in a 4 Kbyte block.

In the register descriptions, the addresses are given as offsets from the base address. The register block base addresses are given in Table 1 .

Any unused register map locations, and unused register bits are reserved. Only the value 0 must be written to these locations or bits. The values which are read from these locations or bits are undefined. The reset state is defined as the state existing after a hard reset.

Variable	Value	Block
ASC0BaseAddress	0x20003000	Asynchronous serial controller (ASC) 0.
ASC1BaseAddress	0x20004000	Asynchronous serial controller (ASC) 1.
ASC2BaseAddress	0x20005000	Asynchronous serial controller (ASC) 2.
ASC3BaseAddress	0x20006000	Asynchronous serial controller (ASC) 3.
AudioBaseAddress	0x00000800	MPEG audio decoder.
BCK Base Address	0x00000098	Background color registers.
BMBaseAddress	0x20026000	Block move DMA controller.
CacheBaseAddress	0x00004000	Cache configuration.
CFG Base Address	0x00000000	Configuration and control registers.
CKG Base Address	0x00000030	Clock generator registers.
DENCBaseAddress	0x00001800	PAL/NTSC/SECAM encoder.
EMIBaseAddress	0x00002000	External memory interface (EMI).
1284BaseAddress	0x20025000	IEEE 1284 parallel port.
IntControllerBase	0x20000000	Interrupt controller.
InterruptLevelBase	0x20011000	Interrupt level controller.
LLIBaseAddress	0x20027000	IEEE 1394 link layer interface (LLI).
LPM Base Address	0x20000400	Low power module registers.
MPEG Base Address	0x00005000	MPEG control register.
OSD Base Address	0x0000002A	On screen display registers.
PES Base Address	0x00000040	PES parser registers
PIOBaseAddress	0x2000C000	PIO port controller.
PTIBaseAddress	0x20030000	Programmable transport interface (PTI).
PWMBaseAddress	0x2000B000	PWM and counter module.
SmartCard0BaseAddress	0x20007000	SmartCard interface 0.
SmartCard1BaseAddress	0x20008000	SmartCard interface 1.
SSC0BaseAddress	0x20009000	Synchronous serial controller (SSC) 0.
SSC1BaseAddress	0x2000A000	Synchronous serial controller (SSC) 1.
SubPictureBaseAddress	0x00001000	Sub-picture decoder.
TDL Base Address	0x00000080	Still picture plane registers.
TtxtBaseAddress	0x20024000	Teletext interface.
USD Base Registers	0x00000088	Block move registers
VideoBaseAddress	0x00000000	MPEG video decoder.

Table 1 Register block base variables

1.1 The MPEG control register

The MPEG control register **MPEGcontrol** enables the user to control the speed of CPU accesses to the audio, video, sub-picture and DENC registers. It can also be used to control the nature and speed of accesses to the CD FIFOs. For normal applications this register should be left in its default state.

Bits 14:12 are used to slow down register accesses by adding extra cycles on the internal register select signals. Bits 10:8 are used to slow down CD FIFO accesses by adding extra cycles on the internal FIFO strobe signals.

The user has two mechanisms to aid in the control of traffic to CD FIFOs:

- a 5-bit hold-off value;
- enable FIFO fullness polling.

The 5-bit hold-off value enables the user to define a minimum wait time between two CD FIFO accesses. (The delay is in units of system clock cycles.)

If the FIFO fullness polling enable bit is set, the CD FIFO fullness is examined on every byte transfer. If the FIFO is full then the transfer will be stalled. If the enable is not set, then data will be transferred regardless of the FIFO fullness. Hence there is a risk of FIFO overflow if the incoming bandwidth is too high.

The register can be accessed by 8-bit word reads or word writes at the address 0x00005000 [7:0] and 0x00005001 14:8].

Bit Field	Bits	Function	Reset
Register select hold time	14	Extra cycle with register select high.	0
Register select setup time	13:12	Number of extra cycles with registers select low.	0
CD strobes hold time	10	Extra cycle with FIFO strobe high.	0
CD strobes setup time	9:8	Number of extra cycles with FIFO strobe low.	0
Hold-off value	7:3	Minimum wait time before sending next CD FIFO access.	0
Enable sub-picture FIFO fullness polling	2	Enable FIFO fullness polling. Use of the FIFO fullness polling mechanism will cause the chip to hang up in cases where it is necessary to access the AV registers to empty the cd FIFOs. Hence we advise NOT to use this feature	0
Enable audio FIFO fullness polling	1		0
Enable video FIFO fullness polling	0		0

Table 2 MPEGcontrol AV buffer control register

1.2 Synchronization of video decoder registers

There are two types of video decoder register: *synchronized* and *unsynchronized*.

- *Synchronized* registers only change value in response to an internal event, either DSYNC or VSYNC, depending on the register. These registers are double-banked; during the write cycle the new value is loaded into a master register, and on the occurrence of the synchronizing event this value is loaded into a slave register, at which time the new value is available to the circuit. If a synchronized register is read, the value returned is that held in the master register.
- *Unsynchronized* registers change their value immediately they are written to.

Some registers are non synchronized registers with edge triggered write (on end of write cycle). This is to avoid glitches on internal signals during write cycles.

2 Register map

This chapter lists all the memory-mapped registers, where each table is given in address order. Details of the contents of each register are given in the remainder of this manual.

The registers are grouped into modules for ease of reference. The addresses are absolute addresses. The column headed **Bits** gives the number of significant bits; in the case of serial registers, only 8 bits should be read or written at each access.

Register	Address	Bits	Access	Description
ASC_0BaudRate	0x20003000	16	R/W	ASC 0 Baud rate generator/reload.
ASC_0TxBuffer	0x20003004	9	W	ASC 0 Output buffer.
ASC_0RxBuffer	0x20003008	9	R	ASC 0 Input buffer.
ASC_0Control	0x2000300C	13	R/W	ASC 0 Control register.
ASC_0IntEnable	0x20003010	9	R/W	ASC 0 Enable interrupts.
ASC_0Status	0x20003014	10	R	ASC 0 Interrupt status.
ASC_0Guardtime	0x20003018	8	R/W	ASC 0 Delay before transmitter empty flag.
ASC_0Timeout	0x2000301C	8	R/W	ASC 0 Time out register.
ASC_0TxReset	0x20003020	0	W	ASC 0 Reset output FIFO.
ASC_0RxReset	0x20003024	0	W	ASC 0 Reset input FIFO.
ASC_0_Retries	0x20003028	8	R/W	ASC 0 number of retries on transmission
ASC_1BaudRate	0x20004000	16	R/W	ASC 1 Baud rate generator/reload.
ASC_1TxBuffer	0x20004004	9	W	ASC 1 Output buffer.
ASC_1RxBuffer	0x20004008	9	R	ASC 1 Input buffer.
ASC_1Control	0x2000400C	13	R/W	ASC 1 Control register.
ASC_1IntEnable	0x20004010	9	R/W	ASC 1 Enable interrupts.
ASC_1Status	0x20004014	10	R	ASC 1 Interrupt status.
ASC_1Guardtime	0x20004018	8	R/W	ASC 1 Delay before transmitter empty flag.
ASC_1Timeout	0x2000401C	8	R/W	ASC 1 Time out register.
ASC_1TxReset	0x20004020	0	W	ASC 1 Reset output FIFO.
ASC_1RxReset	0x20004024	0	W	ASC 1 Reset input FIFO.
ASC_1_Retries	0x20004028	8	R/W	ASC 1 number of retries on transmission
ASC_2BaudRate	0x20005000	16	R/W	ASC 2 Baud rate generator/reload.
ASC_2TxBuffer	0x20005004	9	W	ASC 2 Output buffer.
ASC_2RxBuffer	0x20005008	9	R	ASC 2 Input buffer.
ASC_2Control	0x2000500C	13	R/W	ASC 2 Control register.
ASC_2IntEnable	0x20005010	9	R/W	ASC 2 Enable interrupts.
ASC_2Status	0x20005014	10	R	ASC 2 Interrupt status.
ASC_2Guardtime	0x20005018	8	R/W	ASC 2 Delay before transmitter empty flag.
ASC_2Timeout	0x2000501C	8	R/W	ASC 2 Time out register.
ASC_2TxReset	0x20005020	0	W	ASC 2 Reset output FIFO.
ASC_2RxReset	0x20005024	0	W	ASC 2 Reset input FIFO.
ASC_2_Retries	0x20005028	8	R/W	ASC 2 number of retries on transmission
ASC_3BaudRate	0x20006000	16	R/W	ASC 3 Baud rate generator/reload.
ASC_3TxBuffer	0x20006004	9	W	ASC 3 Output buffer.
ASC_3RxBuffer	0x20006008	9	R	ASC 3 Input buffer.

Table 3 Asynchronous serial control registers (ASC)

Register	Address	Bits	Access	Description
ASC_3Control	0x2000600C	13	R/W	ASC 3 Control register.
ASC_3IntEnable	0x20006010	9	R/W	ASC 3 Enable interrupts.
ASC_3Status	0x20006014	10	R	ASC 3 Interrupt status.
ASC_Guardtime	0x20006018	8	R/W	ASC 3 Delay before transmitter empty flag.
ASC_3Timeout	0x2000601C	8	R/W	ASC 3 Time out register.
ASC_3TxReset	0x20006020	0	W	ASC 3 Reset output FIFO.
ASC_3RxReset	0x20006024	0	W	ASC 3 Reset input FIFO.
ASC_3_Retries	0x20006028	8	R/W	ASC 3 number of retries on transmission

Table 3 Asynchronous serial control registers (ASC)

Register	Address	Bits	Access	Description
AUD_ANC[7:0]	0x00000806	8	R	Ancillary Data Buffer.
AUD_ANC[15:8]	0x00000807	8	R	Ancillary Data Buffer.
AUD_ANC[23:16]	0x00000808	8	R	Ancillary Data Buffer.
AUD_ANC[31:24]	0x00000809	8	R	Ancillary Data Buffer.
AUD_ESC[7:0]	0x0000080A	8	R	Elementary Stream Clock Reference.
AUD_ESC[15:8]	0x0000080B	8	R	Elementary Stream Clock Reference.
AUD_ESC[23:16]	0x0000080C	8	R	Elementary Stream Clock Reference.
AUD_ESC[31:24]	0x0000080D	8	R	Elementary Stream Clock Reference.
AUD_ESC[33:32]	0x0000080E	2	R	Elementary Stream Clock Reference.
AUD_ESCX[7:0]	0x0000080F	8	R	Elementary Stream Clock Reference.
AUD_LRP	0x00000811	1	R/W	LRCK Polarity.
AUD_FFL[7:0]	0x00000814	8	R/W	Free-Format Frame Length.
AUD_FFL[15:8]	0x00000815	8	R/W	Free-Format Frame Length.
AUD_P18	0x00000816	1	R/W	PCM Output Precision.
AUD_CDI	0x00000818	8	W	Compressed Data Input.
AUD_FOR	0x00000819	1	R/W	PCM Output Format.
AUD_ITR[7:0]	0x0000081A	8	R	Interrupt Status Request Register.
AUD_ITR[14:8]	0x0000081B	7	R	Interrupt Status Request Register.
AUD_ITM[7:0]	0x0000081C	8	R/W	Interrupt Mask Register.
AUD_ITM[14:8]	0x0000081D	7	R/W	Interrupt Mask Register.
AUD_LCA	0x0000081E	6	R/W	Left Channel Attenuation.
AUD_EXT	0x0000081F	2	R/W	Decoding Mode Extension.
AUD_RCA	0x00000820	6	R/W	Right Channel Attenuation.
AUD_SID	0x00000822	5	R/W	Audio Stream ID.
AUD_SYN	0x00000823	2	R/W	Packet Sync Mode.
AUD_IDE	0x00000824	1	R/W	Audio Stream ID Enable.
AUD_SCM	0x00000825	1	R/W	Sync Confirmation Mode.
AUD_SYS	0x00000826	2	R	Synchronization Status.
AUD_SYE	0x00000827	8	R/W	Sync Word Extension.
AUD_LCK	0x00000828	2	R/W	Sync Words Until Lock.
AUD_CRC	0x0000082A	2	R/W	CRC Error Concealment Mode.

Table 4 Audio controller registers (AUD)

Register	Address	Bits	Access	Description
AUD_SEM	0x0000082C	2	R/W	Sync Error Concealment Mode.
AUD_PLY	0x0000082E	1	R/W	Play.
AUD_MUT	0x00000830	1	R/W	Mute.
AUD_SKP	0x00000832	1	R/W	Skip Next Frame.
AUD_ISS	0x00000836	3	R/W	Input Stream Selection.
AUD_ORD	0x00000838	1	R/W	PCM Output Bit Order.
AUD_LAT	0x0000083C	1	R/W	Latency Selection.
AUD_RES	0x00000840	1	R/W	Audio Decoder Software Reset.
AUD_RST	0x00000842	1	R/W	Restart.
AUD_SFR	0x00000844	2	R	Sampling Frequency.
AUD_DEM	0x00000846	2	R	De-Emphasis Mode.
AUD_IFT	0x00000852	8	R/W	Input FIFO Threshold.
AUD_SCP	0x00000853	1	R/W	SCLK Polarity.
AUD_ITS	0x0000085B	2	R/W	Audio Interrupt Extension.
AUD_IMS	0x0000085C	2	R/W	Audio Interrupt Extension Mask.
AUD_HDR[7:0]	0x0000085E	8	R	Frame Header.
AUD_HDR[15:8]	0x0000085F	8	R	Frame Header.
AUD_HDR[23:16]	0x00000860	8	R	Frame Header.
AUD_HDR[31:24]	0x00000861	8	R	Frame Header.
AUD_PTS[7:0]	0x00000862	8	R	Presentation Time Stamp.
AUD_PTS[15:8]	0x00000863	8	R	Presentation Time Stamp.
AUD_PTS[23:16]	0x00000864	8	R	Presentation Time Stamp.
AUD_PTS[31:24]	0x00000865	8	R	Presentation Time Stamp.
AUD_PTS[32]	0x00000866	1	R	Presentation Time Stamp.
AUD_ADA	0x0000086C	6	R	Ancillary Data Buffer Size.
AUD_REV	0x0000086D	8	R	Audio Decoder Revision.
AUD_DIV	0x0000086E	6	R/W	PCM Clock Divider.
AUD_DIF	0x0000086F	1	R/W	PCM Output Justification.
AUD_BBE	0x00000870	1	R/W	Bit Buffer Enable.

Table 4 Audio controller registers (AUD)

Register	Address	Bits	Access	Description
BCK_Y	0x00000098	8	R/W	Background Color Y.
BCK_U	0x00000099	8	R/W	Background Color U.
BCK_V	0x0000009A	8	R/W	Background Color V.

Table 5 Background colour registers (BCK)

Register	Address	Bits	Access	Description
BMDMA_SrcAddress	0x20026000	32	W	Block move DMA source address.
BMDMA_DestAddress	0x20026004	32	W	Block move DMA destination address.
BMDMA_Count	0x20026008	16	W	Block move DMA counts.

Table 6 Block move DMA registers (BMDMA)

Register	Address	Bits	Access	Description
BMDMA_IntEn	0x2002600C	1	R/W	Block move DMA interrupt enable.
BMDMA_Status	0x20026010	2	R	Block move DMA status.
BMDMA_IntAck	0x20026014	32	W	Block move DMA interrupt acknowledge.
BMDMA_Abort	0x20026018	32	W	Block move DMA abort.

Table 6 Block move DMA registers (BMDMA)

Register	Address	Bits	Access	Description
CAC_CacheControl0	0x00004000	9	R/W	Region 1 Cache control 0
CAC_CacheControl1	0x00004100	8	R/W	Region 1 Cache control 1
CAC_EMIBlock0	0x00004200	8	R/W	EMI Block 0-7 cacheability
CAC_EMIBank0-3	0x00004300	4	R/W	EMI Bank 0-3 cacheability
CAC_DCacheNotSRAM	0x00004400	1	W	DCache as a cache or SRAM
CAC_InvalidateDCACHE	0x00004500	1	W	Invalidate DCache
CAC_FlushingDCACHE	0x00004600	1	W	Flushing DCache
CAC_EnableICACHE	0x00004700	1	W	Enable ICache
CAC_InvalidateICACHE	0x00004800	1	W	Invalidate ICache
CAC_STATUS	0x00004900	7	R	Internal register status
CAC_CacheControlLock	0x00004A00	1	R/W	Cache control lock

Table 7 Cache Control Registers (CAC)

Register	Address	Bits	Access	Description
CFG_MCF	0x00000000	7	R/W	Memory Refresh Interval.
CFG_CCF	0x00000001	8	R/W	Chip Configuration.
CFG_DRC	0x00000038	6	R/W	DRAM Configuration.
CFG_GCF	0x0000003A	8	R/W	General Configuration.
CFG_CDR	0x00000044	8	W	Compressed Data Input.

Table 8 Configuration and control registers (CFG)

Register	Address	Bits	Access	Description
CKG_PLL	0x00000030	8	R/W	Clock Generator PLL Parameters.
CKG_CFG	0x00000031	7	R/W	Clock Generator Configuration.
CKG_PCM1	0x000000D0	6	R/W	Clock Generator PCMCLK Divider.
CKG_PCM2	0x000000D1	7	R/W	Clock Generator PCMCLK Divider.
CKG_PCM3	0x000000D2	7	R/W	Clock Generator PCMCLK Divider.
CKG_PCM4	0x000000D3	7	R/W	Clock Generator PCMCLK Divider.
CKG_MCK1	0x000000D8	7	R/W	Clock Generator MEMCLK Divider.
CKG_MCK2	0x000000D9	7	R/W	Clock Generator MEMCLK Divider.
CKG_MCK3	0x000000DA	7	R/W	Clock Generator MEMCLK Divider.
CKG_MCK4	0x000000DB	7	R/W	Clock Generator MEMCLK Divider.
CKG_AUX1	0x000000DC	6	R/W	Clock Generator for the Auxiliary Clock Divider.

Table 9 Clock generator registers (CKG)

Register	Address	Bits	Access	Description
CKG_AUX2	0x000000DD	7	R/W	Clock Generator for the Auxiliary Clock Divider.
CKG_AUX3	0x000000DE	7	R/W	Clock Generator for the Auxiliary Clock Divider.
CKG_AUX4	0x000000DF	7	R/W	Clock Generator for the Auxiliary Clock Divider.

Table 9 Clock generator registers (CKG)

Register	Address	Bits	Access	Description
DEN_CFG0	0x00001800	8	R/W	DENC configuration 0.
DEN_CFG1	0x00001801	8	R/W	DENC configuration 1.
DEN_CFG2	0x00001802	8	R/W	DENC configuration 2.
DEN_CFG3	0x00001803	8	R/W	DENC configuration 3.
DEN_CFG4	0x00001804	8	R/W	DENC configuration 4.
DEN_CFG5	0x00001805	7	R/W	DENC configuration 5.
DEN_CFG6	0x00001806	8	R/W	DENC configuration 6.
DEN_CFG7	0x00001807	8	R/W	DENC configuration 7.
DEN_CFG8	0x00001808	8	R/W	DENC configuration 8.
DEN_STA	0x00001809	8	R	DENC Status.
DEN_IDFS1	0x0000180A	8	R/W	DENC Increment for Digital Frequency Synthesizer.
DEN_IDFS2	0x0000180B	8	R/W	DENC Increment for Digital Frequency Synthesizer.
DEN_IDFS3	0x0000180C	8	R/W	DENC Increment for Digital Frequency Synthesizer.
DEN_PDFS1	0x0000180D	8	R/W	Static phase offset for digital frequency synthesizer.
DEN_PDFS2	0x0000180E	8	R/W	Static phase offset for digital frequency synthesizer.
DEN_WSS1	0x0000180F	8	R/W	WSS data registers.
DEN_WSS2	0x00001810	8	R/W	WSS data registers.
DEN_DAC13	0x00001811	8	R/W	DAC1 and DAC3 multiplying factors.
DEN_DAC45	0x00001812	8	R/W	DAC4 and DAC5 multiplying factors.
DEN_DAC6C	0x00001813	8	R/W	DAC6 and C multiplying factors.
DEN_LJMP1	0x00001815	8	R/W	Line jump.
DEN_LJMP2	0x00001816	8	R/W	Line jump.
DEN_LJMP3	0x00001817	8	R/W	Line jump.
DEN_CID	0x00001818	8	R	DENC identification number.
DEN_VPS1	0x00001819	8	R/W	VPS data registers.
DEN_VPS2	0x0000181A	8	R/W	VPS data registers.
DEN_VPS3	0x0000181B	8	R/W	VPS data registers.
DEN_VPS4	0x0000181C	8	R/W	VPS data registers.
DEN_VPS5	0x0000181D	8	R/W	VPS data registers.
DEN_VPS6	0x0000181E	8	R/W	VPS data registers.
DEN_CGMS1	0x0000181F	4	R/W	CGMS data registers.
DEN_CGMS2	0x00001820	8	R/W	CGMS data registers.
DEN_CGMS3	0x00001821	8	R/W	CGMS data registers.
DEN_TTX1	0x00001822	8	R/W	Teletext block definition.
DEN_TTX2	0x00001823	8	R/W	Teletext block definition.
DEN_TTX3	0x00001824	8	R/W	Teletext block definition.

Table 10 Digital encoder registers (DEN)

Register	Address	Bits	Access	Description
DEN_TTX4	0x00001825	8	R/W	Teletext block definition.
DEN_TTXM	0x00001826	8	R/W	Teletext Block Mapping.
DEN_CCF1	0x00001827	8	R/W	Closed caption characters/extended data for field 1.
	0x00001828	8	R/W	Closed caption characters/extended data for field 1.
DEN_CCF2	0x00001829	8	R/W	Closed caption characters/extended data for field 2.
	0x0000182A	8	R/W	Closed caption characters/extended data for field 2.
DEN_CLF1	0x0000182B	5	R/W	Closed caption/extended data line insert for field 1.
DEN_CLF2	0x0000182C	5	R/W	Closed caption/extended data line insert for field 2.
DEN_REG_45...63				Reserved
DEN_REG_64	0x00001840	3	R/W	TTX conference
DEN_REG_65	0x00001841	7	R/W	DAC2 Mult and TTXS
DEN_REG_69	0x00001845	8	R/W	Brightness
DEN_REG_70	0x00001846	8	R/W	Contrast
DEN_REG_71	0x00001847	8	R/W	Saturation

Table 10 Digital encoder registers (DEN)

Register	Address	Bits	Access	Description
EMI_ConfigData0	0x00002000	14	R/W	EMI bank 0 configuration data register 0
EMI_ConfigData1	0x00002004	16	R/W	EMI bank 0 configuration data register 1
EMI_ConfigData2	0x00002008	16	R/W	EMI bank 0 configuration data register 2
EMI_ConfigData3	0x0000200C	16	R/W	EMI bank 0 configuration data register 3
EMI_ConfigData0	0x00002010	16	R/W	EMI bank 1 configuration data register 0
EMI_ConfigData1	0x00002014	16	R/W	EMI bank 1 configuration data register 1
EMI_ConfigData2	0x00002018	16	R/W	EMI bank 1 configuration data register 2
EMI_ConfigData3	0x0000201C	16	R/W	EMI bank 1 configuration data register 3
EMI_ConfigData0	0x00002020	16	R/W	EMI bank 2 configuration data register 0
EMI_ConfigData1	0x00002024	16	R/W	EMI bank 2 configuration data register 1
EMI_ConfigData2	0x00002028	16	R/W	EMI bank 2 configuration data register 2
EMI_ConfigData3	0x0000202C	16	R/W	EMI bank 2 configuration data register 3
EMI_ConfigData0	0x00002030	16	R/W	EMI bank 3 configuration data register 0
EMI_ConfigData1	0x00002034	16	R/W	EMI bank 3 configuration data register 1
EMI_ConfigData2	0x00002038	16	R/W	EMI bank 3 configuration data register 2
EMI_ConfigData3	0x0000203C	16	R/W	EMI bank 3 configuration data register 3
EMI_ConfigLockBank3-0	0x00002040	1	W	Write protection bit for bank 0
EMI_ConfigLockBank3-0	0x00002044	1	W	Write protection bit for bank 1
EMI_ConfigLockBank3-0	0x00002048	1	W	Write protection bit for bank 2
EMI_ConfigLockBank3-0	0x0000204C	1	W	Write protection bit for bank 3
EMI_ConfigStatus	0x00002050	8	R	EMI configuration status information
EMI_SDRAMModeReg0	0x00002058	7	W	SDRAM Mode register
EMI_SDRAMModeReg1	0x0000205C	7	W	SDRAM Mode register
EMI_DRAMInitialize	0x00002060	1	W	Initialize any DRAM/SDRAM in the system
EMI_ConfigPadlogic	0x00002070	12	R/W	Padlogic configuration data register

Table 11 EMI configuration registers (EMI)

Register	Address	Bits	Access	Description
1284_ModeEnable	0x20025000	6	W	1284 mode enable.
1284_PulseWidth	0x20025004	8	W	1284 pulse width.
1284_Control	0x20025008	9	W	1284 control.
1284_Status	0x2002500C	9	R	1284 status.
1284_PinIn	0x20025010	5	R	1284 pin input.
1284_PinInEnable	0x20025014	5	R/W	1284 pin input enable.
1284_PinInValue	0x20025018	5	R/W	1284 pin input value.
1284_PinOut	0x2002501C	6	R/W	1284 pin output.
1284_DataIn	0x20025020	9	R	1284 data input.
1284_DataOut	0x20025024	9	W	1284 data output.
1284_Checksum	0x20025028	8	R	1284 check sum.
1284_PacketSize	0x2002502C	12	W	1284 packet size.
1284_DmaToken	0x20025030	2	R/W	1284 DMA token.
1284_DmaAddress	0x20025040	32	W	1284 DMA address.
1284_DmaCount	0x20025044	16	R/W	1284 DMA count.
1284_DmaControl	0x20025048	3	R/W	1284 DMA control.
1284_IntEnable	0x2002504C	10	R/W	1284 interrupt enable.
1284_IntStatus	0x20025050	10	R	1284 interrupt status.
1284_IntAck	0x20025054	10	W	1284 interrupt acknowledge.

Table 12 IEEE 1284 (PC parallel port) Registers (1284)

Register	Address	Bits	Access	Description
INC_HandlerWptr0	0x20000000	32	R/W	Interrupt handler 0 work space pointer.
INC_HandlerWptr1	0x20000004	32	R/W	Interrupt handler 1 work space pointer.
INC_HandlerWptr2	0x20000008	32	R/W	Interrupt handler 2 work space pointer.
INC_HandlerWptr3	0x2000000C	32	R/W	Interrupt handler 3 work space pointer.
INC_HandlerWptr4	0x20000010	32	R/W	Interrupt handler 4 work space pointer.
INC_HandlerWptr5	0x20000014	32	R/W	Interrupt handler 5 work space pointer.
INC_HandlerWptr6	0x20000018	32	R/W	Interrupt handler 6 work space pointer.
INC_HandlerWptr7	0x2000001C	32	R/W	Interrupt handler 7 work space pointer.
INC_TriggerMode0	0x20000040	3	R/W	Interrupt 0 trigger mode.
INC_TriggerMode1	0x20000044	3	R/W	Interrupt 1 trigger mode.
INC_TriggerMode2	0x20000048	3	R/W	Interrupt 2 trigger mode.
INC_TriggerMode3	0x2000004C	3	R/W	Interrupt trigger mode.
INC_TriggerMode4	0x20000050	3	R/W	Interrupt 3 trigger mode.
INC_TriggerMode5	0x20000054	3	R/W	Interrupt 4 trigger mode.
INC_TriggerMode6	0x20000058	3	R/W	Interrupt 5 trigger mode.
INC_TriggerMode7	0x2000005C	3	R/W	Interrupt 6 trigger mode.
INC_Mask	0x200000C0	17	R/W	Interrupt enable mask.
INC_Set_Mask	0x200000C4	17	W	Set a bit of the interrupt enable mask.
INC_Clear_Mask	0x200000C8	17	W	Clear a bit of the interrupt enable mask.

Table 13 Interrupt controller registers (INC)

Register	Address	Bits	Access	Description
INC_Pending	0x20000080	8	R/W	Interrupt pending.
INC_Set_Pending	0x20000084	8	W	Set a bit of the **Pending** register.
INC_Clear_Pending	0x20000088	8	W	Clear a bit of the **Pending** register.
INC_Exec	0x20000100	8	R/W	Interrupts executing.
INC_Set_Exec	0x20000104	8	W	Set a bit of the **Exec** register.
INC_Clear_Exec	0x20000108	8	W	Clear a bit of the **Exec** register.

Table 13 Interrupt controller registers (INC)

Register	Address	Bits	Access	Description
INL_Int0Priority	0x20011000	3	R/W	Interrupt 0 priority.
INL_Int1Priority	0x20011004	3	R/W	Interrupt 1 priority.
INL_Int2Priority	0x20011008	3	R/W	Interrupt 2 priority.
INL_Int3Priority	0x2001100C	3	R/W	Interrupt 3 priority.
INL_Int4Priority	0x20011010	3	R/W	Interrupt 4 priority.
INL_Int5Priority	0x20011014	3	R/W	Interrupt 5 priority.
INL_Int6Priority	0x20011018	3	R/W	Interrupt 6 priority.
INL_Int7Priority	0x2001101C	3	R/W	Interrupt 7 priority.
INL_Int8Priority	0x20011020	3	R/W	Interrupt 8 priority.
INL_Int9Priority	0x20011024	3	R/W	Interrupt 9 priority.
INL_Int10Priority	0x20011028	3	R/W	Interrupt 10 priority.
INL_Int11Priority	0x2001102C	3	R/W	Interrupt 11 priority.
INL_Int12Priority	0x20011030	3	R/W	Interrupt 12 priority.
INL_Int13Priority	0x20011034	3	R/W	Interrupt 13 priority.
INL_Int14Priority	0x20011038	3	R/W	Interrupt 14 priority.
INL_Int15Priority	0x2001103C	3	R/W	Interrupt 15 priority.
INL_Int16Priority	0x20011040	3	R/W	Interrupt 16 priority.
INL_Int17Priority	0x20011044	3	R/W	Interrupt 17 priority.
INL_Int18Priority	0x20011048	3	R/W	Interrupt 18 priority.
INL_Int19Priority	0x2001104C	3	R/W	Interrupt 19 priority.
INL_Int20Priority	0x20011050	3	R/W	Interrupt 20 priority.
INL_Int21Priority	0x20011054	3	R/W	Interrupt 21 priority.
INL_Int22Priority	0x20011058	3	R/W	Interrupt 22 priority.
INL_SelectnotInv	0x20011060	4	R/W	External interrupt sense for low power controller
INL_ExtIntEnable	0x20011064	4	R/W	Enable external interrupt for low power controller
INL_InputInterrupts	0x2001107C	23	R	Input interrupt status.

Table 14 Interrupt level control registers (INL)

Register	Address	Bits	Access	Description
LLI_Control	0x20027000	6	R/W	LLI control register.
LLI_ByteClockRatio	0x20027004	4	R/W	Local byte clock speed ratio.

Table 15 IEEE1394 link layer interface (LLI)

Register	Address	Bits	Access	Description
LLI_ClockDiv	0x20027010	9	R/W	Fast clock divider configuration.
LLI_SyncLock	0x20027014	4	W	Sync lock value.
LLI_SyncDrop	0x20027016	4	W	Sync drop value.
LLI_SyncPeriod	0x20027018	8	W	Sync period.

Table 15 IEEE1394 link layer interface (LLI)

Register	Address	Bits	Access	Description
LPM_LPTimer[31:0]	0x20000400	32	R/W	Low power timer least significant word.
LPM_LPTimer[32]	0x20000404	1	R/W	Low power timer most significant word.
LPM_LPTimerStart	0x20000408	1	W	Low power timer start.
LPM_LPAlarm[31:0]	0x20000410	32	R/W	Low power alarm least significant word.
LPM_LPAlarm[32]	0x20000414	1	R/W	Low power alarm most significant word.
LPM_LPAlarmStart	0x20000418	1	W	Low power alarm start.
LPM_LPSysPll	0x20000420	2	R/W	System clock PLL.
LPM_SysRatio	0x20000500	6	R	System clock ratio.
LPM_WdEnable	0x20000510	1	R/W	Watchdog enable.
LPM_WdFlag	0x20000514	1	R	Watchdog flag.

Table 16 Low power module registers (LPM)

Register	Address	Bits	Access	Description
MPEG_control	0x00005000	15	R/W	MPEG AV buffer control.

Table 17 MPEG control register (MPEG)

Register	Address	Bits	Access	Description
OSD_TOP	0x0000002A	15	Ser.R/W	OSD Top Field Pointer.
OSD_BOT	0x0000002B	15	Ser. R/W	OSD Bottom Field Pointer.
OSD_ACT	0x0000003E	7	R/W	Active Signal.
OSD_CFG	0x00000091	4	R/W	OSD Configuration.
OSD_BDW	0x00000092	6	R/W	OSD Boundary Weight.

Table 18 On Screen Display Registers (OSD)

Register	Address	Bits	Access	Description
PES_CF1	0x00000040	8	R/W	PES audio decoding control
PES_CF2	0x00000041	8	R/W	PES video parser control
PES_TM1	0x00000042	8	R	DSM trick mode
PES_TM2	0x00000043	2	R	PES Parser Status
PES_TS[7:0]	0x00000049	8	R	PES time stamps
PES_TS[15:8]	0x0000004A	8	R	PES time stamps
PES_TS[23:16]	0x0000004B	8	R	PES time stamps
PES_TS[31:24]	0x0000004C	8	R	PES time stamps
PES_TS[32]	0x0000004D	2	R	PES time stamps

Table 19 PES parser registers (PES)

Register	Address	Bits	Access	Description
PIO_P0Out	0x2000C000	8	R/W	PIO 0 output.
PIO_Set_P0Out	0x2000C004	8	W	Set bits of **P0Out**.
PIO_Clear_P0Out	0x2000C008	8	W	Clear bits of **P0Out**.
PIO_P0In	0x2000C010	8	R	PIO 0 input.
PIO_P0C0	0x2000C020	8	R/W	PIO 0 configuration 0.
PIO_Set_P0C0	0x2000C024	8	W	Set bits of **P0C0**.
PIO_Clear_P0C0	0x2000C028	8	W	Clear bits of **P0C0**.
PIO_P0C1	0x2000C030	8	R/W	PIO 0 configuration 1.
PIO_Set_P0C1	0x2000C034	8	W	Set bits of **P0C1**.
PIO_Clear_P0C1	0x2000C038	8	W	Clear bits of **P0C1**.
PIO_P0C2	0x2000C040	8	R/W	PIO 0 configuration 2.
PIO_Set_P0C2	0x2000C044	8	W	Set bits of **P0C2**.
PIO_Clear_P0C2	0x2000C048	8	W	Clear bits of **P0C2**.
PIO_P0Comp	0x2000C050	8	R/W	PIO 0 input comparison.
PIO_Set_P0Comp	0x2000C054	8	W	Set bits of **P0Comp**.
PIO_Clear_P0Comp	0x2000C058	8	W	Clear bits of **P0Comp**.
PIO_P0Mask	0x2000C060	8	R/W	PIO 0 input comparison mask.
PIO_Set_P0Mask	0x2000C064	8	W	Set bits of **P0Mask**.
PIO_Clear_P0Mask	0x2000C068	8	W	Clear bits of **P0Mask**.
PIO_P1Out	0x2000D000	8	R/W	PIO 1 output.
PIO_Set_P1Out	0x2000D004	8	W	Set bits of **P1Out**.
PIO_Clear_P1Out	0x2000D008	8	W	Clear bits of **P1Out**.
PIO_P1In	0x2000D010	8	R	PIO 1 input.
PIO_P1C0	0x2000D020	8	R/W	PIO 1 configuration 0.
PIO_Set_P1C0	0x2000D024	8	W	Set bits of **P1C0**.
PIO_Clear_P1C0	0x2000D028	8	W	Clear bits of **P1C0**.
PIO_P1C1	0x2000D030	8	R/W	PIO 1 configuration 1.
PIO_Set_P1C1	0x2000D034	8	W	Set bits of **P1C1**.

Table 20 Parallel input/output registers (PIO)

Register	Address	Bits	Access	Description
PIO_Clear_P1C1	0x2000D038	8	W	Clear bits of **P1C1**.
PIO_P1C2	0x2000D040	8	R/W	PIO 1 configuration 2.
PIO_Set_P1C2	0x2000D044	8	W	Set bits of **P1C2**.
PIO_Clear_P1C2	0x2000D048	8	W	Clear bits of **P1C2**.
PIO_P1Comp	0x2000D050	8	R/W	PIO 1 input comparison.
PIO_Set_P1Comp	0x2000D054	8	W	Set bits of **P1Comp**.
PIO_Clear_P1Comp	0x2000D058	8	W	Clear bits of **P1Comp**.
PIO_P1Mask	0x2000D060	8	R/W	PIO 1 input comparison mask.
PIO_Set_P1Mask	0x2000D064	8	W	Set bits of **P1Mask**.
PIO_Clear_P1Mask	0x2000D068	8	W	Clear bits of **P1Mask**.
PIO_P2Out	0x2000E000	8	R/W	PIO 2 output.
PIO_Set_P2Out	0x2000E004	8	W	Set bits of **P2Out**.
PIO_Clear_P2Out	0x2000E008	8	W	Clear bits of **P2Out**.
PIO_P2In	0x2000E010	8	R	PIO 2 input.
PIO_P2C0	0x2000E020	8	R/W	PIO 2 configuration 0.
PIO_Set_P2C0	0x2000E024	8	W	Set bits of **P2C0**.
PIO_Clear_P2C0	0x2000E028	8	W	Clear bits of **P2C0**.
PIO_P2C1	0x2000E030	8	R/W	PIO 2 configuration 1.
PIO_Set_P2C1	0x2000E034	8	W	Set bits of **P2C1**.
PIO_Clear_P2C1	0x2000E038	8	W	Clear bits of **P2C1**.
PIO_P2C2	0x2000E040	8	R/W	PIO 2 configuration 2.
PIO_Set_P2C2	0x2000E044	8	W	Set bits of **P2C2**.
PIO_Clear_P2C2	0x2000E048	8	W	Clear bits of **P2C2**.
PIO_2Comp	0x2000E050	8	R/W	PIO 2 input comparison.
PIO_Set_P2Comp	0x2000E054	8	W	Set bits of **P2Comp**.
PIO_Clear_P2Comp	0x2000E058	8	W	Clear bits of **P2Comp**.
PIO_P2Mask	0x2000E060	8	R/W	PIO 2 input comparison mask.
PIO_Set_P2Mask	0x2000E064	8	W	Set bits of **P2Mask**.
PIO_Clear_P2Mask	0x2000E068	8	W	Clear bits of **P2Mask**.
PIO_P3Out	0x2000F000	8	R/W	PIO 3 output.
PIO_Set_P3Out	0x2000F004	8	W	Set bits of **P3Out**.
PIO_Clear_P3Out	0x2000F008	8	W	Clear bits of **P3Out**.
PIO_P3In	0x2000F010	8	R	PIO 3 input.
PIO_P3C0	0x2000F020	8	R/W	PIO 3 configuration 0.
PIO_Set_P3C0	0x2000F024	8	W	Set bits of **P3C0**.
PIO_Clear_P3C0	0x2000F028	8	W	Clear bits of **P3C0**.
PIO_P3C1	0x2000F030	8	R/W	PIO 3 configuration 1.
PIO_Set_P3C1	0x2000F034	8	W	Set bits of **P3C1**.
PIO_Clear_P3C1	0x2000F038	8	W	Clear bits of **P3C1**.
PIO_P3C2	0x2000F040	8	R/W	PIO 3 configuration 2.
PIO_Set_P3C2	0x2000F044	8	W	Set bits of **P3C2**.
PIO_Clear_P3C2	0x2000F048	8	W	Clear bits of **P3C2**.
PIO_P3Comp	0x2000F050	8	R/W	PIO 3 input comparison.

Table 20 Parallel input/output registers (PIO)

Register	Address	Bits	Access	Description
PIO_Set_P3Comp	0x2000F054	8	W	Set bits of **P3Comp**.
PIO_Clear_P3Comp	0x2000F058	8	W	Clear bits of **P3Comp**.
PIO_P3Mask	0x2000F060	8	R/W	PIO 3 input comparison mask.
PIO_Set_P3Mask	0x2000F064	8	W	Set bits of **P3Mask**.
PIO_Clear_P3Mask	0x2000F068	8	W	Clear bits of **P3Mask**.
PIO_P4Out	0x20010000	8	R/W	PIO 4 output.
PIO_Set_P4Out	0x20010004	8	W	Set bits of **P4Out**.
PIO_Clear_P4Out	0x20010008	8	W	Clear bits of **P4Out**.
PIO_4In	0x20010010	8	R	PIO 4 input.
PIO_P4C0	0x20010020	8	R/W	PIO 4 configuration 0.
PIO_Set_P4C0	0x20010024	8	W	Set bits of **P4C0**.
PIO_Clear_P4C0	0x20010028	8	W	Clear bits of **P4C0**.
PIO_P4C1	0x20010030	8	R/W	PIO 4 configuration 1.
PIO_Set_P4C1	0x20010034	8	W	Set bits of **P4C1**.
PIO_Clear_P4C1	0x20010038	8	W	Clear bits of **P4C1**.
PIO_P4C2	0x20010040	8	R/W	PIO 4 configuration 2.
PIO_Set_P4C2	0x20010044	8	W	Set bits of **P4C2**.
PIO_Clear_P4C2	0x20010048	8	W	Clear bits of **P4C2**.
PIO_P4Comp	0x20010050	8	R/W	PIO 4 input comparison.
PIO_Set_P4Comp	0x20010054	8	W	Set bits of **P4Comp**.
PIO_Clear_P4Comp	0x20010058	8	W	Clear bits of **P4Comp**.
PIO_P4Mask	0x20010060	8	R/W	PIO 4 input comparison mask.
PIO_Set_P4Mask	0x20010064	8	W	Set bits of **P4Mask**.
PIO_Clear_P4Mask	0x20010068	8	W	Clear bits of **P4Mask**.

Table 20 Parallel input/output registers (PIO)

Register	Address	Bits	Access	Description
PTI_IntStatus0	0x20030000	16	R	Interrupt events.
PTI_IntStatus1	0x20030004	16	R	Interrupt events.
PTI_IntStatus2	0x20030008	16	R	Interrupt events.
PTI_ntStatus3	0x2003000C	16	R	Interrupt events.
PTI_IntEnable0	0x20030010	16	R/W	Enable interrupt events.
PTI_IntEnable1	0x20030014	16	R/W	Enable interrupt events.
PTI_IntEnable2	0x20030018	16	R/W	Enable interrupt events.
PTI_IntEnable3	0x2003001C	16	R/W	Enable interrupt events.
PTI_IntAck0	0x20030020	16	W	Clear interrupt events.
PTI_IntAck1	0x20030024	16	W	Clear interrupt events.
PTI_IntAck2	0x20030028	16	W	Clear interrupt events.
PTI_IntAck3	0x2003002C	16	W	Clear interrupt events.
PTI_TCMode	0x20030030	3	R/W	Debugging and reset.
PTI_AudPTS[31:0]	0x20030040	32	R	Audio PTS least significant 32 bits.
PTI_AudPTS[32]	0x20030044	1	R	Audio PTS most significant bit.

Table 21 PTI Registers (PTI)

Register	Address	Bits	Access	Description
PTI_VidPTS[31:0]	0x20030048	32	R	Video PTS least significant 32 bits.
PTI_VidPTS[32]	0x2003004C	1	R	Video PTS most significant bit.
PTI_STCTimer0	0x20030050	32	W	Load bits 0:31 of the timer and start.
PTI_STCTimer1	0x20030054	1	W	Load bit 32 of the 33-bit timer.
PTI_DMA0Base	0x20031000	32	W	Base address of the channel 0 circular buffer.
PTI_DMA1Base	0x20031004	32	W	Base address of the channel 1 circular buffer.
PTI_DMA2Base	0x20031008	32	W	Base address of the channel 2 circular buffer.
PTI_DMA3Base	0x2003100C	32	W	Reserved.
PTI_DMA0Top	0x20031010	32	W	Address of the top byte of the channel 0 buffer.
PTI_DMA1Top	0x20031014	32	W	Address of the top byte of the channel 1 buffer.
PTI_DMA2Top	0x20031018	32	W	Address of the top byte of the channel 2 buffer.
PTI_DMA3Top	0x2003101C	32	W	Reserved.
PTI_DMA0Write	0x20031020	32	R/W	Absolute write pointer for channel 0.
PTI_DMA1Write	0x20031024	32	W	Absolute write pointer for channel 1.
PTI_DMA2Write	0x20031028	32	W	Absolute write pointer for channel 2.
PTI_DMA3Write	0x2003102C	32	W	Reserved
PTI_DMA0Read	0x20031030	32	W	Absolute read pointer for channel 0.
PTI_DMA1Read	0x20031034	32	R/W	Absolute read pointer for channel 1.
PTI_DMA2Read	0x20031038	32	R/W	Absolute read pointer for channel 2.
PTI_DMA3Read	0x2003103C	32	R/W	Reserved
PTI_DMA0Status	0x20031040	2	R/W	The current status of channel 0.
PTI_DMA1Burst	0x20031044	1	W	Select byte or word transfers to channel 1.
PTI_DMA2Burst	0x20031048	1	W	Select byte or word transfers to channel 2.
PTI_DMA3Burst	0x2003104C	1	W	Reserved
PTI_DMAEnable	0x20031050	4	W	Enable or disable the four DMA channels.
PTI_DMA1Holdoff	0x20031054	8	W	Clock cycles to wait before polling **notCDReq1**.
PTI_DMA2Holdoff	0x20031058	8	W	Clock cycles to wait before polling **notCDReq2**.
PTI_DMA3Holdoff	0x2003105C	8	W	Clock cycles to wait before polling **notCDReq3**.
PTI_DMACDAddr	0x20031060	32	W	Configuration of CD FIFO addresses.
PTI_IIFFIFOCount	0x20032000	8	R	The number of bytes in the input FIFO.
PTI_IIFAltFIFOCount	0x20032004	8	R	The number of bytes in the alternative output FIFO.
PTI_IIFFIFOEnable	0x20032008	8	R/W	Allow data to be written into the input FIFO.
PTI_IIFAltLatency	0x20032010	8	W	The delay from a packet arriving to alternative output.
PTI_IIFSyncLock	0x20032014	4	W	The number of correct SYNC bytes to lock.
PTI_IIFSyncDrop	0x20032018	4	W	The number of wrong SYNC bytes to lose lock.
Reserved	0x200320E0	8	W	When not in ICAM mode, set this register to 0x00000001
PTI_SFFilterDataLS0	0x20034000	32	R/W	Section filter 0 data least significant word.
PTI_SFFilterMaskLS0	0x20034004	32	R/W	Section filter 0 mask least significant word.
PTI_SFFilterDataMS0	0x20034008	32	R/W	Section filter 0 data most significant word.
PTI_SFFilterMaskMS0	0x2003400C	32	R/W	Section filter 0 mask most significant word.
PTI_SFFilterDataLS1	0x20034010	32	R/W	Section filter 1 data least significant word.
PTI_SFFilterMaskLS1	0x20034014	32	R/W	Section filter 1 mask least significant word.
PTI_SFFilterDataMS1	0x20034018	32	R/W	Section filter 1 data most significant word.

Table 21 PTI Registers (PTI)

Register	Address	Bits	Access	Description
PTI_SFFilterMaskMS1	0x2003401C	32	R/W	Section filter 1 mask most significant word.
PTI_SFFilterDataLS2	0x20034020	32	R/W	Section filter 2 data least significant word.
PTI_SFFilterMaskLS2	0x20034024	32	R/W	Section filter 2 mask least significant word.
PTI_SFFilterDataMS2	0x20034028	32	R/W	Section filter 2 data most significant word.
PTI_SFFilterMaskMS2	0x2003402C	32	R/W	Section filter 2 mask most significant word.
PTI_SFFilterDataLS3	0x20034030	32	R/W	Section filter 3 data least significant word.
PTI_SFFilterMaskLS3	0x20034034	32	R/W	Section filter 3 mask least significant word.
PTI_SFFilterDataMS3	0x20034038	32	R/W	Section filter 3 data most significant word.
PTI_SFFilterMaskMS3	0x2003403C	32	R/W	Section filter 3 mask most significant word.
PTI_SFFilterDataLS4	0x20034040	32	R/W	Section filter 4 data least significant word.
PTI_SFFilterMaskLS4	0x20034044	32	R/W	Section filter 4 mask least significant word.
PTI_SFFilterDataMS4	0x20034048	32	R/W	Section filter 4 data most significant word.
PTI_SFFilterMaskMS4	0x2003404C	32	R/W	Section filter 4 mask most significant word.
PTI_SFFilterDataLS5	0x20034050	32	R/W	Section filter 5 data least significant word.
PTI_SFFilterMaskLS5	0x20034054	32	R/W	Section filter 5 mask least significant word.
PTI_SFFilterDataMS5	0x20034058	32	R/W	Section filter 5 data most significant word.
PTI_SFFilterMaskMS5	0x2003405C	32	R/W	Section filter 5 mask most significant word.
PTI_SFFilterDataLS6	0x20034060	32	R/W	Section filter 6 data least significant word.
PTI_SFFilterMaskLS6	0x20034064	32	R/W	Section filter 6 mask least significant word.
PTI_SFFilterDataMS6	0x20034068	32	R/W	Section filter 6 data most significant word.
PTI_SFFilterMaskMS6	0x2003406C	32	R/W	Section filter 6 mask most significant word.
PTI_SFFilterDataLS7	0x20034070	32	R/W	Section filter 7 data least significant word.
PTI_SFFilterMaskLS7	0x20034074	32	R/W	Section filter 7 mask least significant word.
PTI_SFFilterDataMS7	0x20034078	32	R/W	Section filter 7 data most significant word.
PTI_SFFilterMaskMS7	0x2003407C	32	R/W	Section filter 7 mask most significant word.
PTI_SFFilterDataLS8	0x20034080	32	R/W	Section filter 8 data least significant word.
PTI_SFFilterMaskLS8	0x20034084	32	R/W	Section filter 8 mask least significant word.
PTI_SFFilterDataMS8	0x20034088	32	R/W	Section filter 8 data most significant word.
PTI_SFFilterMaskMS8	0x2003408C	32	R/W	Section filter 8 mask most significant word.
PTI_SFFilterDataLS9	0x20034090	32	R/W	Section filter 9 data least significant word.
PTI_SFFilterMaskLS9	0x20034094	32	R/W	Section filter 9 mask least significant word.
PTI_SFFilterDataMS9	0x20034098	32	R/W	Section filter 9 data most significant word.
PTI_SFFilterMaskMS9	0x2003409C	32	R/W	Section filter 9 mask most significant word.
PTI_SFFilterDataLS10	0x200340A0	32	R/W	Section filter 10 data least significant word.
PTI_SFFilterMaskLS10	0x200340A4	32	R/W	Section filter 10 mask least significant word.
PTI_SFFilterDataMS10	0x200340A8	32	R/W	Section filter 10 data most significant word.
PTI_SFFilterMaskMS10	0x200340AC	32	R/W	Section filter 10 mask most significant word.
PTI_SFFilterDataLS11	0x200340B0	32	R/W	Section filter 11 data least significant word.
PTI_SFFilterMaskLS11	0x200340B4	32	R/W	Section filter 11 mask least significant word.
PTI_SFFilterDataMS11	0x200340B8	32	R/W	Section filter 11 data most significant word.
PTI_SFFilterMaskMS11	0x200340BC	32	R/W	Section filter 11 mask most significant word.
PTI_SFFilterDataLS12	0x200340C0	32	R/W	Section filter 12 data least significant word.
PTI_SFFilterMaskLS12	0x200340C4	32	R/W	Section filter 12 mask least significant word.

Table 21 PTI Registers (PTI)

Register	Address	Bits	Access	Description
PTI_SFFilterDataMS12	0x200340C8	32	R/W	Section filter 12 data most significant word.
PTI_SFFilterMaskMS12	0x200340CC	32	R/W	Section filter 12 mask most significant word.
PTI_SFFilterDataLS13	0x200340D0	32	R/W	Section filter 13 data least significant word.
PTI_SFFilterMaskLS13	0x200340D4	32	R/W	Section filter 13 mask least significant word.
PTI_SFFilterDataMS13	0x200340D8	32	R/W	Section filter 13 data most significant word.
PTI_SFFilterMaskMS13	0x200340DC	32	R/W	Section filter 13 mask most significant word.
PTI_SFFilterDataLS14	0x200340E0	32	R/W	Section filter 14 data least significant word.
PTI_SFFilterMaskLS14	0x200340E4	32	R/W	Section filter 14 mask least significant word.
PTI_SFFilterDataMS14	0x200340E8	32	R/W	Section filter 14 data most significant word.
PTI_SFFilterMaskMS14	0x200340EC	32	R/W	Section filter 14 mask most significant word.
PTI_SFFilterDataLS15	0x200340F0	32	R/W	Section filter 15 data least significant word.
PTI_SFFilterMaskLS15	0x200340F4	32	R/W	Section filter 15 mask least significant word.
PTI_SFFilterDataMS15	0x200340F8	32	R/W	Section filter 15 data most significant word.
PTI_SFFilterMaskMS15	0x200340FC	32	R/W	Section filter 15 mask most significant word.
PTI_SFFilterDataLS16	0x20034100	32	R/W	Section filter 16 data least significant word.
PTI_SFFilterMaskLS16	0x20034104	32	R/W	Section filter 16 mask least significant word.
PTI_SFFilterDataMS16	0x20034108	32	R/W	Section filter 16 data most significant word.
PTI_SFFilterMaskMS16	0x2003410C	32	R/W	Section filter 16 mask most significant word.
PTI_SFFilterDataLS17	0x20034110	32	R/W	Section filter 17 data least significant word.
PTI_SFFilterMaskLS17	0x20034114	32	R/W	Section filter 17 mask least significant word.
PTI_SFFilterDataMS17	0x20034118	32	R/W	Section filter 17 data most significant word.
PTI_SFFilterMaskMS17	0x2003411C	32	R/W	Section filter 17 mask most significant word.
PTI_SFFilterDataLS18	0x20034120	32	R/W	Section filter 18 data least significant word.
PTI_SFFilterMaskLS18	0x20034124	32	R/W	Section filter 18 mask least significant word.
PTI_SFFilterDataMS18	0x20034128	32	R/W	Section filter 18 data most significant word.
PTI_SFFilterMaskMS18	0x2003412C	32	R/W	Section filter 18 mask most significant word.
PTI_SFFilterDataLS19	0x20034130	32	R/W	Section filter 19 data least significant word.
PTI_SFFilterMaskLS19	0x20034134	32	R/W	Section filter 19 mask least significant word.
PTI_SFFilterDataMS19	0x20034138	32	R/W	Section filter 19 data most significant word.
PTI_SFFilterMaskMS19	0x2003413C	32	R/W	Section filter 19 mask most significant word.
PTI_SFFilterDataLS20	0x20034140	32	R/W	Section filter 20 data least significant word.
PTI_SFFilterMaskLS20	0x20034144	32	R/W	Section filter 20 mask least significant word.
PTI_SFFilterDataMS20	0x20034148	32	R/W	Section filter 20 data most significant word.
PTI_SFFilterMaskMS20	0x2003414C	32	R/W	Section filter 20 mask most significant word.
PTI_SFFilterDataLS21	0x20034150	32	R/W	Section filter 21 data least significant word.
PTI_SFFilterMaskLS21	0x20034154	32	R/W	Section filter 21 mask least significant word.
PTI_SFFilterDataMS21	0x20034158	32	R/W	Section filter 21 data most significant word.
PTI_SFFilterMaskMS21	0x2003415C	32	R/W	Section filter 21 mask most significant word.
PTI_SFFilterDataLS22	0x20034160	32	R/W	Section filter 22 data least significant word.
PTI_SFFilterMaskLS22	0x20034164	32	R/W	Section filter 22 mask least significant word.
PTI_SFFilterDataMS22	0x20034168	32	R/W	Section filter 22 data most significant word.
PTI_SFFilterMaskMS22	0x2003416C	32	R/W	Section filter 22 mask most significant word.
PTI_SFFilterDataLS23	0x20034170	32	R/W	Section filter 23 data least significant word.

Table 21 PTI Registers (PTI)

Register	Address	Bits	Access	Description
PTI_SFFilterMaskLS23	0x20034174	32	R/W	Section filter 23 mask least significant word.
PTI_SFFilterDataMS23	0x20034178	32	R/W	Section filter 23 data most significant word.
PTI_SFFilterMaskMS23	0x2003417C	32	R/W	Section filter 23 mask most significant word.
PTI_SFFilterDataLS24	0x20034180	32	R/W	Section filter 24 data least significant word.
PTI_SFFilterMaskLS24	0x20034184	32	R/W	Section filter 24 mask least significant word.
PTI_SFFilterDataMS24	0x20034188	32	R/W	Section filter 24 data most significant word.
PTI_SFFilterMaskMS24	0x2003418C	32	R/W	Section filter 24 mask most significant word.
PTI_SFFilterDataLS25	0x20034190	32	R/W	Section filter 25 data least significant word.
PTI_SFFilterMaskLS25	0x20034194	32	R/W	Section filter 25 mask least significant word.
PTI_SFFilterDataMS25	0x20034198	32	R/W	Section filter 25 data most significant word.
PTI_SFFilterMaskMS25	0x2003419C	32	R/W	Section filter 25 mask most significant word.
PTI_SFFilterDataLS26	0x200341A0	32	R/W	Section filter 26 data least significant word.
PTI_SFFilterMaskLS26	0x200341A4	32	R/W	Section filter 26 mask least significant word.
PTI_SFFilterDataMS26	0x200341A8	32	R/W	Section filter 26 data most significant word.
PTI_SFFilterMaskMS26	0x200341AC	32	R/W	Section filter 26 mask most significant word.
PTI_SFFilterDataLS27	0x200341B0	32	R/W	Section filter 27 data least significant word.
PTI_SFFilterMaskLS27	0x200341B4	32	R/W	Section filter 27 mask least significant word.
PTI_SFFilterDataMS27	0x200341B8	32	R/W	Section filter 27 data most significant word.
PTI_SFFilterMaskMS27	0x200341BC	32	R/W	Section filter 27 mask most significant word.
PTI_SFFilterDataLS28	0x200341C0	32	R/W	Section filter 28 data least significant word.
PTI_SFFilterMaskLS28	0x200341C4	32	R/W	Section filter 28 mask least significant word.
PTI_SFFilterDataMS28	0x200341C8	32	R/W	Section filter 28 data most significant word.
PTI_SFFilterMaskMS28	0x200341CC	32	R/W	Section filter 28 mask most significant word.
PTI_SFFilterDataLS29	0x200341D0	32	R/W	Section filter 29 data least significant word.
PTI_SFFilterMaskLS29	0x200341D4	32	R/W	Section filter 29 mask least significant word.
PTI_SFFilterDataMS29	0x200341D8	32	R/W	Section filter 29 data most significant word.
PTI_SFFilterMaskMS29	0x200341DC	32	R/W	Section filter 29 mask most significant word.
PTI_SFFilterDataLS30	0x200341E0	32	R/W	Section filter 30 data least significant word.
PTI_SFFilterMaskLS30	0x200341E4	32	R/W	Section filter 30 mask least significant word.
PTI_SFFilterDataMS30	0x200341E8	32	R/W	Section filter 30 data most significant word.
PTI_SFFilterMaskMS30	0x200341EC	32	R/W	Section filter 30 mask most significant word.
PTI_SFFilterDataLS31	0x200341F0	32	R/W	Section filter 31 data least significant word.
PTI_SFFilterMaskLS31	0x200341F4	32	R/W	Section filter 31 mask least significant word.
PTI_SFFilterDataMS31	0x200341F8	32	R/W	Section filter 31 data most significant word.
PTI_SFFilterMaskMS31	0x200341FC	32	R/W	Section filter 31 mask most significant word.
PTI_Transport controller	0x20036000 - 7FFF	9×32	R	Internal TC registers, read for debug.
PTI_Data shared memory	0x20038000 - 8BFF	768×32	R/W	Data memory.
PTI_Instruction shared memory	0x2003C000 - CFFF	1024×32	R/W	Instruction memory.

Table 21 PTI Registers (PTI)

Register	Address	Bits	Access	Description
PWM_0Val	0x2000B000	9	R/W	PWM 0 pulse width.
PWM_1Val	0x2000B004	9	R/W	PWM 1 pulse width.
PWM_2Val	0x2000B008	9	R/W	PWM 2 pulse width.
PWM_3Val	0x2000B00C	9	R/W	PWM 3 pulse width.
PWM_IntEnable	0x2000B054	9	R/W	PWM interrupt enable.
PWM_IntStatus	0x2000B058	9	R	PWM interrupt status.
PWM_IntAck	0x2000B05C	9	W	PWM interrupt acknowledge.
PWM_Control	0x2000B050	11	R/W	PWM control register.
PWM_Count	0x2000B060	8	R/W	PWM output counter.
PWM_0CaptureEdge	0x2000B030	2	R/W	PWM 0 capture event definition.
PWM_1CaptureEdge	0x2000B034	2	R/W	PWM 1 capture event definition.
PWM_2CaptureEdge	0x2000B038	2	R/W	PWM 2 capture event definition.
PWM_3CaptureEdge	0x2000B03C	2	R/W	PWM 3 capture event definition.
PWM_0CaptureVal	0x2000B010	32	R	PWM 0 capture value.
PWM_1CaptureVal	0x2000B014	32	R	PWM 1 capture value.
PWM_2CaptureVal	0x2000B018	32	R	PWM 2 capture value.
PWM_3CaptureVal	0x2000B01C	32	R	PWM 3 capture value.
PWM_0CompareVal	0x2000B020	32	R/W	PWM 0 compare value.
PWM_1CompareVal	0x2000B024	32	R/W	PWM 1 compare value.
PWM_2CompareVal	0x2000B028	32	R/W	PWM 2 compare value.
PWM_3CompareVal	0x2000B02C	32	R/W	PWM 3 compare value.
PWM_0CompareOutVal	0x2000B040	1	R/W	PWM 0 compare output value.
PWM_1CompareOutVal	0x2000B044	1	R/W	PWM 1 compare output value.
PWM_2CompareOutVal	0x2000B048	1	R/W	PWM 2 compare output value.
PWM_3CompareOutVal	0x2000B04C	1	R/W	PWM 3 compare output value.
PWM_CaptureCount	0x2000B064	32	R/W	PWM capture/compare counter.

Table 22 PWM and counter module registers (PWM)

Register	Address	Bits	Access	Description
SC_0ClkVal	0x20007000	5	W	SmartCard 0 clock.
SC_0ClkCon	0x20007004	2	W	SmartCard 0 clock control.
SC_1ClkVal	0x20008000	5	W	SmartCard 1 clock.
SC_1ClkCon	0x20008004	2	W	SmartCard 1 clock control.

Table 23 Smart Card Interface Registers (SC)

Register	Address	Bits	Access	Description
SPD_CTL1	0x00001000	8	R/W	Control Register 1.
SPD_SPR	0x00001001	1	R/W	Soft Reset.
SPD_CTL2	0x00001002	2	R/W	Control Register 2.

Table 24 Sub picture decoder registers (SPD)

Register	Address	Bits	Access	Description
SPD_LUT	0x00001003	8	W	Main Lookup Table.
SPD_XD0[9:8]	0x00001004	2	R/W	Sub-picture X Offset.
SPD_XD0[7:0]	0x00001005	8	R/W	Sub-picture X Offset.
SPD_YD0[9:8]	0x00001006	2	R/W	Sub-picture Y Offset.
SPD_YD0[7:0]	0x00001007	8	R/W	Sub-picture Y Offset.
SPD_HLSX[9:8]	0x0000100C	2	R/W	Highlight Region Start X.
SPD_HLSX[7:0]	0x0000100D	8	R/W	Highlight Region Start X.
SPD_HLSY[9:8]	0x0000100E	2	R/W	Highlight Region Start Y.
SPD_HLSY[7:0]	0x0000100F	8	R/W	Highlight Region Start Y.
SPD_HLEX[9:8]	0x00001010	2	R/W	Highlight Region End X.
SPD_HLEX[7:0]	0x00001011	8	R/W	Highlight Region End X.
SPD_HLEY[9:8]	0x00001012	2	R/W	Highlight Region End Y.
SPD_HLEY[7:0]	0x00001013	8	R/W	Highlight Region End Y.
SPD_HCOL1	0x00001014	8	R/W	Highlight Region Color.
SPD_HCOL2	0x00001015	8	R/W	Highlight Region Color.
SPD_HCN1	0x00001016	8	R/W	Highlight Region Contrast.
SPD_HCN2	0x00001017	8	R/W	Highlight Region Contrast.
SPD_SXD0[9:8]	0x00001024	2	R/W	Sub-picture Display Area.
SPD_SXD0[7:0]	0x00001025	8	R/W	Sub-picture Display Area.
SPD_SYD0[9:8]	0x00001026	2	R/W	Sub-picture Display Area.
SPD_SYD0[7:0]	0x00001027	8	R/W	Sub-picture Display Area.
SPD_SXD1[9:8]	0x00001028	2	R/W	Sub-picture Display Area.
SPD_SXD1[7:0]	0x00001029	8	R/W	Sub-picture Display Area.
SPD_SYD1[9:8]	0x0000102A	2	R/W	Sub-picture Display Area.
SPD_SYD1[7:0]	0x0000102B	8	R/W	Sub-picture Display Area.

Table 24 Sub picture decoder registers (SPD)

Register	**Address**	**Bits**	**Access**	**Description**
SSC_0BRG	0x20009000	10	R/W	SSC 0 baud rate generation.
SSC_0TBuf	0x20009004	16	W	SSC 0 transmit buffer.
SSC_0RBuf	0x20009008	16	R	SSC 0 receive buffer.
SSC_0Con	0x2000900C	11	R/W	SSC 0 control.
SSC_0IEn	0x20009010	9	R/W	SSC 0 interrupt enable.
SSC_0Stat	0x20009014	10	R	SSC 0 status.
SSC_0I2C	0x20009018	5	R/W	SSC 0 I^2C control.
SSC_0SlAd	0x2000901C	16	W	SSC 0 slave address.
SSC_1Con	0x2000A00C	11	R/W	SSC 1 control.
SSC_1I2C	0x2000A018	5	R/W	SSC 1 I^2C control.
SSC_1TBuf	0x2000A004	16	W	SSC 1 transmit buffer.
SSC_1RBuf	0x2000A008	16	R	SSC 1 receive buffer.
SSC_1SlAd	0x2000A01C	16	W	SSC 1 slave address.
SSC_1BRG	0x2000A000	10	R/W	SSC 1 baud rate generation.

Table 25 Synchronous serial controller registers (SSC)

Register	Address	Bits	Access	Description
SSC_1Stat	0x2000A014	10	R	SSC 1 status.
SSC_1IEn	0x2000A010	9	R/W	SSC 1 interrupt enable.

Table 25 Synchronous serial controller registers (SSC)

Register	Address	Bits	Access	Description
TDL_TOP[16]	0x00000080	1	R/W	TDL Odd Pointer.
TDL_TOP[15:8]	0x00000081	8	R/W	TDL Odd Pointer.
TDL_TOP[7:0]	0x00000082	8	R/W	TDL Odd Pointer.
TDL_TEP[16]	0x00000083	1	R/W	TDL Even Pointer.
TDL_TEP[15:8]	0x00000084	8	R/W	TDL Even Pointer.
TDL_TEP[7:0]	0x00000085	8	R/W	TDL Even Pointer.
TDL_TDW	0x00000086	7	R/W	Third Display Width.
TDL_SCN[15:8]	0x000000B0	8	R/W	TDL scan vector
TDL_SCN[7:0]	0x000000B1	8	R/W	TDL scan vector
TDL_TOP2[16]	0x000000C0	1	R/W	TDL Odd Pointer 2.
TDL_TOP2[15:8]	0x000000C1	8	R/W	TDL Odd Pointer 2.
TDL_TOP2[7:0]	0x000000C2	8	R/W	TDL Odd Pointer 2.
TDL_TEP2[16]	0x000000C3	1	R/W	TDL Even Pointer 2.
TDL_TEP2[15:8]	0x000000C4	8	R/W	TDL Even Pointer 2.
TDL_TEP2[7:0]	0x000000C5	8	R/W	TDL Even Pointer 2.
TDL_YDS[8]	0x000000C6	1	R/W	TDL Y End.
TDL_YDS[7:0]	0x000000C7	8	R/W	TDL Y End.
TDL_SWT[8]	0x000000C8	1	R/W	Switch Line Number.
TDL_SWT [7:0]	0x000000C9	8	R/W	Switch Line Number.
TDL_LSR[7:0]	0x000000EB	8	R/W	SRC Luma Resolution.
TDL_LSR[8]	0x000000ED	1	R/W	SRC Luma Resolution.
TDL_YDO[8]	0x000000EE	1	R/W	TDL Y Offset.
TDL_YDO[7:0]	0x000000EF	8	R/W	TDL Y Offset.
TDL_XDO[9:8]	0x000000F0	2	R/W	TDL X Offset.
TDL_XDO[7:0]	0x000000F1	8	R/W	TDL X Offset.
TDL_XDS[9:8]	0x000000F2	2	R/W	TDL X End.
TDL_XDS[7:0]	0x000000F3	8	R/W	TDL X End.
TDL_DCF	0x000000F4	5	R/W	TDL Configuration.

Table 26 Still picture plane registers (TDL)

Register	Address	Bits	Access	Description
Ttxt_DmaAddress	0x20024000	32	R/W	Teletext DMA address.
Ttxt_DmaCount	0x20024004	11	R/W	Teletext DMA count.
Ttxt_OutDelay	0x20024008	9	R/W	Teletext output delay.
Ttxt_InStartLine	0x2002400C	9	R/W	Teletext input start line.
Ttxt_InCbDelay	0x20024010	9	R/W	Teletext input color burst delay.
Ttxt_Mode	0x20024014	2	R/W	Teletext mode.

Table 27 Teletext interface registers (Ttxt)

Register	Address	Bits	Access	Description
Ttxt_IntStatus	0x20024018	3	R	Teletext interrupt status.
Ttxt_IntEnable	0x2002401C	3	R/W	Teletext interrupt enable.
Ttxt_AckOddEven	0x20024020	1	W	Teletext acknowledge odd or even.
Ttxt_Abort	0x20024024	1	W	Teletext abort.

Table 27 Teletext interface registers (Ttxt)

Register	Address	Bits	Access	Description
USD_BRP	0x00000088	19	R/W	Memory read pointer
USD_BWP	0x0000008C	19	R/W	Memory write pointer
USD_BMC1	0x0000009E	7	R/W	Block move control
USD_BMC2	0x0000009F	7	R/W	Block move control
USD_BMW	0x000000A0	16	R/W	Block move width
USD_BMH	0x000000A2	11	R/W	Block move height
USD_BSK	0x000000A4	19	R/W	Block skip
USD_PAT1	0x000000A7	32	R/W	Block move pattern
USD_PAT2	0x000000AB	32	R/W	Block move pattern

Table 28 Block move registers (USD)

Register	Address	Bits	Access	Description
VID_CTL	0x00000002	8	R/W	Decoding Control.
VID_TIS	0x00000003	6	W	Task Instruction.
VID_PFH	0x00000004	8	R/W	Picture F-parameters Horizontal.
VID_PFV	0x00000005	8	R/W	Picture F-parameters Vertical.
VID_PPR1	0x00000006	7	R/W	Picture Parameters 1.
VID_PPR2	0x00000007	7	R/W	Picture Parameters 2.
VID_DFP[13:8]	0x0000000C	6	R/W	Displayed Luma Frame Pointer.
VID_DFP[7:0]	0x0000000D	8	R/W	Displayed Luma Frame Pointer.
VID_RFP[13:8]	0x0000000E	6	R/W	Reconstructed Frame Pointer.
VID_RFP[7:0]	0x0000000F	8	R/W	Reconstructed Frame Pointer.
VID_FFP[13:8]	0x00000010	6	R/W	Forward Luma Frame Pointer.
VID_FFP[7:0]	0x00000011	8	R/W	Forward Luma Frame Pointer.
VID_BFP[13:8]	0x00000012	6	R/W	Backward Frame Pointer.
VID_BFP[7:0]	0x00000013	8	R/W	Backward Frame Pointer.
VID_VBG[13:8]	0x00000014	6	R/W	Start of Video Bit Buffer.
VID_VBG[7:0]	0x00000015	8	R/W	Start of Video Bit Buffer.
VID_VBL[13:8]	0x00000016	6	R	Video Bit Buffer Level.
VID_VBL[7:0]	0x00000017	8	R	Video Bit Buffer Level.
VID_VBS[13:8]	0x00000018	6	R/W	Video Bit Buffer Stop.
VID_VBS[7:0]	0x00000019	8	R/W	Video Bit Buffer Stop.
VID_VBT[13:8]	0x0000001A	6	R/W	Video Bit Buffer Threshold.

Table 29 Video decoder registers (VID)

Register	Address	Bits	Access	Description
VID_VBT[7:0]	0x0000001B	8	R/W	Video Bit Buffer Threshold.
VID_ABG[13:8]	0x0000001C	6	R/W	Start of audio bit buffer.
VID_ABG[7:0]	0x0000001D	8	R/W	Start of audio bit buffer.
VID_ABL[13:8]	0x0000001E	6	R	Audio Bit Buffer Level.
VID_ABL[7:0]	0x0000001F	8	R	Audio Bit Buffer Level.
VID_ABS[13:8]	0x00000020	6	R/W	Audio Bit Buffer Stop.
VID_ABS[7:0]	0x00000021	8	R/W	Audio Bit Buffer Stop.
VID_ABT[13:8]	0x00000022	6	R/W	Audio Bit Buffer Threshold.
VID_ABT[7:0]	0x00000023	8	R/W	Audio Bit Buffer Threshold.
VID_DFS	0x00000024	15	Serial R/W	Decoded Frame Size.
VID_DFW	0x00000025	8	R/W	Decoded Frame Width.
VID_XFW	0x00000028	8	R/W	Displayed Frame Width.
VID_PAN[10:8]	0x0000002C	3	R/W	Pan/Scan Horizontal Vector Integer Part.
VID_PAN[7:0]	0x0000002D	8	R/W	Pan/Scan Horizontal Vector Integer Part.
VID_PTH[13:8]	0x0000002E	6	R/W	Panic threshold.
VID_PTH[7:0]	0x0000002F	8	R/W	Panic threshold.
VID_SRA	0x00000035	1	R/W	Audio Soft Reset.
VID_SRV	0x00000039	1	R/W	Video Soft Reset.
VID_STA1	0x0000003B	8	R	Status.
VID_ITM1	0x0000003C	8	R/W	Interrupt Mask.
VID_ITS1	0x0000003D	8	R	Interrupt Status.
VID_LDP	0x0000003F	1	R/W	Load Pointer.
VID_YDS[8]	0x00000046	1	R/W	Display Y End.
VID_YDS[7:0]	0x00000047	8	R/W	Display Y End.
VID_SPRead	0x0000004E	19	Serial R/W	Sub-picture Read Pointer.
VID_SPWrite	0x0000004F	19	Serial R/W	Sub-picture Write Pointer.
VID_SPB[10:8]	0x00000050	3	R/W	Sub-picture Buffer Begin.
VID_SPB[7:0]	0x00000051	8	R/W	Sub-picture Buffer Begin.
VID_SPE[10:8]	0x00000052	3	R/W	Sub-picture Buffer End.
VID_SPE[7:0]	0x00000053	8	R/W	Sub-picture Buffer End.
VID_VFL	0x00000054	5	R/W	Luma Vertical Filter.
VID_VFC	0x00000055	5	R/W	Chroma Vertical Filter.
VID_TRF[12:8]	0x00000056	5	R/W	Temporal Reference.
VID_TRF[7:0]	0x00000057	8	R/W	Temporal Reference.
VID_DFC[13:8]	0x00000058	6	R/W	Displayed Chroma Frame Pointer.
VID_DFC[7:0]	0x00000059	8	R/W	Displayed Chroma Frame Pointer.
VID_RFC[13:8]	0x0000005A	6	R/W	Reconstructed Chroma Frame Pointer.
VID_RFC[7:0]	0x0000005B	8	R/W	Reconstructed Chroma Frame Pointer.
VID_FFC[13:8]	0x0000005C	6	R/W	Forward Chroma Frame Pointer.
VID_FFC[7:0]	0x0000005D	8	R/W	Forward Chroma Frame Pointer.
VID_BFC[13:8]	0x0000005E	6	R/W	Backward Chroma Pointer.
VID_BFC[7:0]	0x0000005F	8	R/W	Backward Chroma Pointer.
VID_ITM2	0x00000060	8	R/W	Interrupt Mask.

Table 29 Video decoder registers (VID)

Register	Address	Bits	Access	Description
VID_ITM3	0x00000061	8	R/W	Interrupt Mask.
VID_ITS2	0x00000062	8	R	Interrupt Status.
VID_ITS3	0x00000063	8	R	Interrupt Status.
VID_STA2	0x00000064	8	R	Status.
VID_STA3	0x00000065	8	R	Status.
VID_HDF	0x00000066	16	Serial R	Header Data FIFO.
VID_CDcount	0x00000067	24	Serial R	Bit Buffer Input Counter.
VID_SCDcount	0x00000068	24	Serial R	Bit Buffer Output Counter.
VID_HDS	0x00000069	4	R/W	Header Search.
VID_LSO	0x0000006A	8	R/W	SRC Luminance Offset.
VID_LSR[7:0]	0x0000006B	8	R/W	SRC Luma Resolution.
VID_CSO	0x0000006C	8	R/W	SRC Chrominance Offset.
VID_LSR[8]	0x0000006D	1	R/W	SRC Luma Resolution.
VID_YDO[13:8]	0x0000006E	4	R/W	Display Y Offset. & Pan Scan Vertical filter line precision
VID_YDO[7:0]	0x0000006F	8	R/W	Display Y Offset.
VID_XDO[9:8]	0x00000070	2	R/W	Display X Offset.
VID_XDO[7:0]	0x00000071	8	R/W	Display X Offset.
VID_XDS[9:8]	0x00000072	2	R/W	Display X End.
VID_XDS[7:0]	0x00000073	8	R/W	Display X End.
VID_DCF	0x00000074	6	R/W	Display Configuration.
VID_DCF	0x00000075	5	R/W	Display Configuration.
VID_QMW	0x00000076	8	W	Quantization Matrix Data.
VID_REV	0x00000078	8	R	Revision
VID_LCK	0x0000007B	1	W	VID_MCF, VID_DRC & VID_CCF lock bit
VID_SCN	0x00000087	6	R/W	Pan/Scan Vertical Vector.
VID_OUT	0x00000090	4	R/W	Output of 4:2:2 display.
VID_MWV	0x0000009B	8	R/W	Mix Weight Video.
VID_MWS	0x0000009C	8	R/W	Mix Weight Still Picture.
VID_MWSV	0x0000009D	8	R/W	Mix Weight Still / Video.
VID_DIS	0x000000D4	7	R/W	Video configuration

Table 29 Video decoder registers (VID)

3 Asynchronous serial controller (ASC)

ASC_*n*_BaudRate ASC*n* baud rate generator

	31–16	15–0
0x00	Reserved	ReloadVal

Address: *ASCnBaseAddress* + 0x00
Type: Read/write
Reset value: 1

Description

The **ASC_*n*_BaudRate** register is the dual-function baud rate generator and reload value register. A read from this register returns the content of the 16-bit counter/accumulator; writing to it updates the 16-bit reload register.

If the **Run** bit of the control register is 1, then any value written in the **ASC_*n*_BaudRate** register is immediately copied to the timer. However, if the **Run** bit is 0 when the register is written, then the timer will not be reloaded until the first CPU clock cycle after the **Run** bit is 1.

The mode of operation of the baud rate generator depends on the setting of the **BaudMode** bit in the **ASC_*n*_Control** register. The 2 modes are described in the *Asynchronous Serial Controller* section of the STi5512 *Datasheet*.

Mode 0

When the **BaudMode** bit in the **ASC_*n*_Control** register is set to 0, the baud rate and the required reload value for a given baud rate can be determined by the following formulae:

$$BaudRate = \frac{f_{CPU}}{16 \times ASCBaudRate}$$

$$ASCBaudRate = \frac{f_{CPU}}{16 \times BaudRate}$$

where: *ASCBaudRate* represents the content of the **ASC_*n*_BaudRate** register, taken as an unsigned 16-bit integer, f_{CPU} is the frequency of the CPU.

Mode 0 shoud be used for all baud rates below 19.2K baud.

Table 30 lists commonly used baud rates with the required reload values and the approximate deviation errors for an example baud rate with a CPU clock of 60 MHz.

Baud rate	Reload value (exact)	Reload value (integer)	Reload value (hex)	Approx. deviation error
38.4 K	97.656	98	0062	0.35%
19.2 K	195.313	195	00C3	0.16%
9600	390.625	391	0091	0.1%
4800	781.250	781	030D	0.03%
2400	1562.500	1563	061B	0.03%
1200	3125.000	3125	0C35	0.00%
600	6250.000	6250	186A	0.00%

Table 30 Mode 0 Baud rates

Baud rate	Reload value (exact)	Reload value (integer)	Reload value (hex)	Approx. deviation error
300	12500.000	12500	30D4	0.00%
75	50000.000	50000	C350	0.00%

Table 30 Mode 0 Baud rates

Mode 1

When the **BaudMode** bit in the **ASC_*n*_Control** register is set to 1, the baud rate is given by::

$$BaudRate = \frac{ASCBaudRate \text{ x } f_{CPU}}{16 \text{ x } 2^{16}}$$

where: f_{CPU} is the CPU clock frequency and *ASCBaudRate* is the value written to the **ASC_*n*_BaudRate** register..

Baud rate	Reload value (exact)	Reload value (integer)	Reload value (hex)	Approx. deviation error
115200	2013.266	2013	07DD	0.01%
96000	1677.722	1678	068E	0.02%
38.4 K	671.089	671	029F	0.02%
19.2 K	335.544	336	0150	0.14%

Table 31 Mode 1 Baud rates

Mode 1 shoud be used for baud rates of 19.2 K and above as it has a lower deviation error than Mode 0 at higher frequencies.

ASC_*n*_Control ASC*n* control register

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0 C	Reserved	Baud Mode	Cts Enable	Fifo Enable	SC Enable	Rx Enable	Run	Loop Back	Parity Odd	Stop Bits		Mode		

Address: *ASCnBaseAddress* + 0x0C
Type: Read/write
Reset value: 0

Description

The **ASC_*n*_Control** register controls the operating mode of the UART ASC*n* and contains control bits for mode and error check selection, and status flags for error identification.

Programming the mode control field (**Mode**) to one of the reserved combinations may result in unpredictable behavior. Serial data transmission or reception is only possible when the baud rate generator run bit (**Run**) is set to 1. When the **Run** bit is set to 0, **TxD** will be 1. Setting the **Run** bit to 0 will immediately freeze the state of the transmitter and receiver. This should only be done when the ASC is idle.

Serial data transmission or reception is only possible when the baud rate generator **Run** bit is set to 1. A transmission is started by writing to the transmit buffer register **ASC_*n*_TxBuffer**.

Bitfield	Description
Mode	ASC mode control: Mode2: 000: RESERVED 001: 8-bit data 010: RESERVED 011: 7-bit data + parity 100: 9-bit data 101: 8-bit data + wake up bit 110: RESERVED 111: 8-bit data + parity
StopBits	Number of stop bits selection: 00: 0.5 stop bits 01: 1 stop bits 10: 1.5 stop bits 11: 2 stop bits
ParityOdd	Parity selection: 0: Even parity (parity bit set on odd number of ‘1’s in data) 1: Odd parity (parity bit set on even number of ‘1’s in data)
LoopBack	Loopback mode enable bit: 0: Standard transmit/receive mode 1: Loopback mode enabled
Run	Baudrate generator run bit: 0: Baudrate generator disabled (ASC inactive) 1: Baudrate generator enabled
RxEnable	Receiver enable bit: 0: Receiver disabled 1: Receiver enabled
SCEnable	SmartCard enable bit: 0: SmartCard mode disabled 1: SmartCard mode enabled
FifoEnable	FIFO enable bit: 0: FIFO disabled 1: FIFO enabled
CtsEnable	CTS enable bit: 0: CTS ignored 1: CTS enabled
BaudMode	Baud rate generation mode: 0: Baud counter decrements, ticks when it reaches 1 1: Baud counter added to itself, ticks when there’s a carry

ASC_*n*_GuardTime ASC*n* guard time

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
0x18	Reserved	GuardTime

Address: *ASCnBaseAddress* + 0x18
Type: Read/write
Reset value: 0

Description

The **ASC_*n*_GuardTime** register enables the user to delay the assertion of the interrupt **TxEmpty** by a programmable number of baud clock ticks. The value in the register is the number of baud clock ticks to delay assertion of **TxEmpty**. This value must be in the range 0 to 255.

ASC_*n*_IntEnable ASC*n* interrupt enable

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9	8	7	6	5	4	3	2	1	0
0x10	Reserved	RHF	TOI	TNE	OE	FE	PE	THE	TE	RBE

Address: *ASCnBaseAddress* + 0x10
Type: Read/write
Reset value: 0

Description

Bitfield	Description
RBE	**RxBufFullIE:** Receiver buffer full interrupt enable: 0: Receiver buffer full interrupt disable 1: Receiver buffer full interrupt enable
TE	**TxEmptyIE:** Transmitter empty interrupt enable: 0: Transmitter empty interrupt disable 1: Transmitter empty interrupt enable
THE	**TxHalfEmptyIE:** Transmitter buffer half empty interrupt enable: 0: Transmitter buffer half empty interrupt disable 1: Transmitter buffer half empty interrupt enable
PE	**ParityErrorIE:** Parity error interrupt enable: 0: Parity error interrupt disable 1: Parity error interrupt enable
FE	**FrameErrorIE:** Framing error interrupt enable: 0: Framing error interrupt disable 1: Framing error interrupt enable
OE	**OverrunErrorIE:** Overrun error interrupt enable: 0: Overrun error interrupt disable 1: Overrun error interrupt enable
TNE	**TimeoutNotEmptyIE:** Time-out when not empty interrupt enable: 0: Time-out when input FIFO or buffer not empty interrupt disable 1: Time-out when input FIFO or buffer not empty interrupt enable
TOI	**TimeoutIdleIE:** Time-out when the receiver FIFO is empty interrupt enable: 0: Time-out when the input FIFO or buffer is empty interrupt disable 1: Time-out when the input FIFO or buffer is empty interrupt enable
RHF	**RxHalfFullIE:** Receiver FIFO is half full interrupt enable: 0: Receiver FIFO is half full interrupt disable 1: Receiver FIFO is half full interrupt enable

ASC_*n*_Retries ASC*n* number of retries on transmission

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
0x28	Reserved	Number of Retries

Address: *ASCnBaseAddress* + 0x28
Type: Read/write
Reset value: 1

Description

The **ASC_*n*_Retries** register defines the number of transmissions attempted on a piece of data before the UART discards the data. If a transmission still fails after ‘Number of Retries’ the **Nacked** bit is set in the **ASC_*n*_Status** register where it can be read and acted on by software. This register does not have to be re-initialized after a Nack error.

ASC_*n*_RxBuffer ASC*n* receive buffer

	31–9	8–0
0x08	Reserved	RD

Address: *ASCnBaseAddress* + 0x08
Type: Read only
Reset value: 0

Description

Serial data reception is only possible when the baud rate generator **Run** bit in the **ASC_*n*_Control** register is set to 1.

Bitfield	Description
RD[6:0]	Receive buffer data **D0 to D6**
RD[7]	Receive buffer data **D7**, or parity error bit - depending on the operating mode (the setting of the **Mode** bit of the **ASCControl** register).
RD[8]	Receive buffer data **D8**, or parity error bit, or wake-up bit - depending on the operating mode (the setting of the **Mode** field of the **ASCControl** register) If the **Mode** field selects an 8-bit frame then this bit is undefined. Software should ignore this bit when reading 8-bit frames

ASC_*n*_RxReset ASC*n* receive FIFO reset

Address: *ASCnBaseAddress* + 0x24
Type: Write only

Description

Reset the receiver FIFO. The ‘registers’ **ASC_*n*_RxReset** have no actual storage associated with them. A write of any value to one of these registers resets the corresponding receiver FIFO.

ASC_*n*_Status ASC*n* interrupt status

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11	10	9	8	7	6	5	4	3	2	1	0
0x14	Reserved	NKD	TF	RHF	TOI	TNE	OE	FE	PE	THE	TE	RBF

Address: *ASCnBaseAddress* + 0x14
Type: Read only
Reset value: 3 (i.e. Rx buffer full and Tx buffer empty)

Description

Bitfield	Description
RBF	**RxBufFull:** Receiver FIFO not empty (FIFO operation) or buffer full (double-buffered operation): 0: Receiver FIFO is empty or buffer is not full 1: Receiver FIFO is not empty or buffer is full
TE	**TxEmpty:** Transmitter empty flag: 0: Transmitter is not empty 1: Transmitter is empty
THE	**TxHalfEmpty:** Transmitter FIFO at least half empty flag or buffer empty: 0: The FIFOs are enabled and the transmitter FIFO is more than half full (more than 8 characters) or the FIFOs are disabled and the transmit buffer is not empty. 1: The FIFOs are enabled and the transmitter FIFO is at least half empty (8 or less characters) or the FIFOs are disabled and the transmit buffer is empty
PE	**ParityError:** Input parity error flag: 0: No parity error 1: Parity error
FE	**FrameError:** Input frame error flag, i.e.stop bits not found: 0: No framing error 1: Framing error
OE	**OverrunError:** Overrun error flag: 0: No overrun error 1: Overrun error, i.e. data received when the input buffer is full
TNE	**TimeoutNotEmpty:** Time-out when the receiver FIFO or buffer is not empty: 0: No time-out or the receiver FIFO or buffer is empty 1: Time-out when the receiver FIFO or buffer is not empty
TOI	**TimeoutIdle:** Time-out when the receiver FIFO or buffer is empty: 0: No time-out or the receiver FIFO or buffer is not empty 1: Time-out when the receiver FIFO or buffer is empty
RHF	**RxHalfFull:** Receiver FIFO is half full: 0: The receiver FIFO contains 8 characters or less 1: The receiver FIFO contains more than 8 characters
TF	**TxFull:** Transmitter FIFO or buffer is full: 0: The FIFOs are enabled and the transmitter FIFO is empty or contains less than16 characters or the FIFOs are disabled and the transmit buffer is empty 1: The FIFOs are enabled and the transmitter FIFO contains 16 characters or the FIFOs are disabled and the transmit buffer is full
NKD	**Nacked:** Transmission failure acknowledgement by receiver: 0: Data transmitted successfully 1: Data transmission unsuccessful (data 'nacked' by SmartCard)

ASC_*n*_Timeout ASC*n* time out

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x1C	Reserved	Timeout

Address: *ASCnBaseAddress* + 0x1C
Type: Read/write
Reset value: 0

Description

The time-out period in baud rate ticks. The ASC contains an 8-bit time-out counter, which reloads from **ASC_*n*_Timeout** whenever one or more of the following is true:

- **ASC_*n*_RxBuffer** is read;
- the ASC is in the middle of receiving a character;
- **ASC_*n*_Timeout** is written to.

If none of these conditions hold the counter decrements towards 0 at every baud rate tick.

The **TimeoutNotEmpty** bit of the **ASC_*n*_Status** register is 1 when the input FIFO is not empty and the time-out counter is zero.The **TimeoutIdle** bit of the **ASC_*n*_Status** register is 1 when the input FIFO is empty and the time-out counter is zero.

When the software has emptied the input FIFO, the time-out counter will reset and start decrementing. If no more characters arrive, when the counter reaches zero the **TimeoutIdle** bit of the **ASC_*n*_Status** register will be set.

ASC_*n*_TxBuffer ASC*n* transmit buffer

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x04	Reserved	TD

Address: *ASCnBaseAddress* + 0x04
Type: Write only
Reset value: 0

Description

A transmission is started by writing to the transmit buffer register **ASC_*n*_TxBuffer**. Serial data transmission is only possible when the baud rate generator **Run** bit in the **ASC_*n*_Control** register is set to 1.

Data transmission is double-buffered or uses a FIFO, so a new character may be written to the transmit buffer register before the transmission of the previous character is complete. This allows characters to be sent back-to-back without gaps.

Bitfield	Description
TD[6:0]	Transmit buffer data **D0 to D6**
TD[7]	Transmit buffer data **D7**, or parity bit - depending on the operating mode (the setting of the **Mode** field of the **ASC-Control** register).
TD[8]	Transmit buffer data **D8**, or parity bit, or wake-up bit or undefined - depending on the operating mode (the setting of the **Mode** field of the **ASCControl** register). If the **Mode** field selects an 8-bit frame then this bit should be written as 0.

ASC_*n*_TxReset ASC*n* transmit FIFO reset

Address: *ASCnBaseAddress* + 0x20
Type: Write only

Description

Reset the transmitter FIFO. The ‘registers’ **ASC_*n*_TxReset** have no actual storage associated with them. A write of any value to one of these registers resets the corresponding transmitter FIFO.

4 Audio MPEG (AUD)

AUD_ADA Ancillary Data Buffer Size

	7	6	5	0
0x6C			AUD_ADA[5:0]	

Address: *AudioBaseAddress* + 0x6C
Type: Read only
Reset value: 0

Description

This register holds the number of bits available in the ancillary data buffer, **AUD_ANC [31:0]**. It is cleared by reading **AUD_ANC [31:24].**

AUD_ANC Ancillary Data Buffer

	7	0
0x06	AUD_ANC[7:0]	
0x07	AUD_ANC[15:8]	
0x08	AUD_ANC[23:16]	
0x09	AUD_ANC[31:24]	

Address: *AudioBaseAddress* + 0x06 to 0x09
Type: Read only
Reset value: Undefined

Description

The 4 8-bit ancillary data registers constitute a 32-bit FIFO which holds the ancillary data extracted from audio frames. The first bit of ancillary data received is stored in bit **AUD_ANC [0]**.

The extraction of ancillary data in **AUD_ANC** is started by enabling interrupt 7. An interrupt 7, i.e. **AUD_ITR [7]**, is generated when 32 bits have been written into **AUD_ANC**, i.e. when it is full.

When **AUD_ANC [31:24]** is read, **AUD_ADA** is cleared and the ancillary data buffer is reinitialized.

AUD_BBE Bit Buffer Enable

	7	0
0x70		AUD_BBE

Address: *AudioBaseAddress* + 0x70
Type: Read/write
Reset value: undefined

Description

This bit must be set in order for audio bit buffer data to be processed by the Audio Decoder. When reset, data may be input through the microcontroller interface via the **AUD_CDI** register, bypassing the audio bit buffer.

0: Bit buffer disabled, MCU input enabled

1: Bit buffer enabled, MCU input disabled

AUD_CDI Compressed Data Input

	7 … 0
0x18	AUD_CDI[7:0]

Address: *AudioBaseAddress* + 0x18
Type: Write only
Reset value: Undefined

Description

When **AUD_BBE** is reset, audio compressed data may be input via the microcontroller by writing the compressed data to this register. Note that as audio data enters the audio decoder directly, the system parser and bit buffer are bypassed. Thus, only elementary or PCM audio data may be entered, and these modes must be correctly programmed in the **AUD_ISS** register.

AUD_CRC CRC Error Concealment Mode

	7	1 … 0
0x2A		**AUD_CRC[1:0]**

Address: *AudioBaseAddress* + 0x2A
Type: Read/Write
Reset value: Undefined

Description

This register defines the action which will be taken upon detection of a CRC error in an input frame.

00 Disable CRC detection and error concealment
01 Mute on detection of CRC error
10 Illegal
11 Skip invalid frame

AUD_DEM De-Emphasis Mode

	7	1 … 0
0x46		AUD_DEM[1:0]

Address: *AudioBaseAddress* + 0x46
Type: Read only
Reset value: Undefined

Description

This register is set with the value of the *emphasis* field of the frame currently being decoded. If enabled, an interrupt 10, i.e. **AUD_ITR [10]**, is generated on a change in the value. The register is updated and the interrupt generated when the corresponding frame is at the PCM output stage.

00: None

01: 50/15 µs

10: Reserved.

11: TU-T J.17

AUD_DIF PCM Output Justification

	7	0
0x6F		AUD_DIF

Address: *AudioBaseAddress* + 0x6F
Type: Read/Write
Reset value: Undefined

Description

DIF stands for Data In Front. This bit selects whether the PCM output data is right or left justified with respect to the 32-clock frame when in 18-bit mode (i.e. when **AUD_P18** = 1). This bit is has no significance when in 16-bit mode, i.e. when **AUD_P18** = 0.

0: 8-bit PCM data is right justified, i.e. it occupies the last 18 cycles of each 32-cycle frame.

1: 18-bit PCM data is left justified, i.e. it occupies the first 18 cycles of each 32-cycle frame when **AUD_FOR** = 1, or the next 18 when **AUD_FOR** = 0.

AUD_DIV PCM Clock Divider

	7	5 0
0x6E		AUD_DIV[5:0]

Address: *AudioBaseAddress* + 0x6E
Type: Read/Write
Reset value: Undefined

Description

The number loaded into this register, in the range 0 to 63, defines the ratio of the frequency of the PCM bit clock, SCLK, to that of PCMCLK, according to the relationship:

$$f_{\mathrm{SCLK}} = \frac{f_{\mathrm{PCMCLK}}}{2 \times (\mathbf{AUD\_DIV} + 1)}$$

For example, **AUD_DIV** is loaded with 0, the frequency of SCLK is one half of the frequency of PCMCLK, while if **AUD_DIV** is loaded with 63, the frequency of SCLK is one 128th of the frequency of PCMCLK. The value of **AUD_DIV** = 16 is reserved. If this number is loaded, the divider is bypassed and the frequency of SCLK is equal to the frequency of PCMCLK.

AUD_DIV must be set up before the output of SCLK starts. This can be done by first disabling PCM outputs by de-asserting the **AUD_MUT** and **AUD_PLY** commands, and then writing to the **AUD_DIV** register. Once the register is set up, the **AUD_MUT** and/or **AUD_PLY** commands can be asserted. **AUD_DIV** cannot be changed “on the fly”.

AUD_ESC Elementary Stream Clock Reference

	7	0
0x0A	AUD_ESC[7:0]	
0x0B	AUD_ESC[15:8]	
0x0C	AUD_ESC[23:16]	
0x0D	AUD_ESC[31:24]	
0x0E	reserved	AUD_ESC[33:32]
0x0F	AUD_ESCX[7:0]	

Address: *AudioBaseAddress* + 0x0A to 0x0F
Type: Read only
Reset value: undefined

Description

The register contains the value of the last elementary stream clock reference (ESCR) which was detected in the audio bitstream. When a new ESCR is detected an interruption is signalled. See register description **AUD_INT**, **AUD_IMS**.

AUD_ESC[33] indicates the presence of the extension field **AUD_ESCX[7:0]** for 27 MHz system clocks.

AUD_EXT Decoding Mode Extension

	7	1 0
0x1F		AUD_EXT[1:0]

Address: *AudioBaseAddress* + 0x1F
Type: Read/write
Reset value: undefined

Description

This register enables “dual-mode” decoding and an improved mute.

00: Normal decoding

01: Decode dual-mode bitstream, where only right channel is decoded and is output on both output channels

10: Decode dual-mode bitstream, where only left channel is decoded and is output on both output channels

11: Perform “smooth” mute by changing attenuation from 0 to 63 on 32 consecutives samples.

AUD_FFL Free-Format Frame Length

	7 ... 0
0x14	AUD_FFL[7:0]
0x15	AUD_FFL[15:8]

Address: *AudioBaseAddress* + 0x14 to 0x15
Type: Read/write
Reset value: undefined

Description

When free-format decoding is used (*bitrate_index* = 0), the frame length, if known, can be loaded into this register, in units of bits. (In free-format, the frame length cannot be determined from bitstream parameters).

The length loaded into **AUD_FFL** is used in the internal synchronization algorithm. If the frame length is not known, **AUD_FFL** must be loaded with zero.

AUD_FOR PCM Output Format

	7 ... 1	0
0x19		AUD_FOR

Address: *AudioBaseAddress* + 0x19
Type: Read/write
Reset value: undefined

Description

This bit is used to select the I^2S-compatible PCM output format when in 18-bit left-justified mode (i.e. when **AUD_P18** = 1 and **AUD_DIF** = 1). In this mode the most-significant bit of the PCM data is output one cycle later than the change of LRCK. **AUD_FOR** has no significance when **AUD_P18** =0.

0: I^2S-compatible PCM output

1: Standard format (most-significant bit of data coincident with AOLRCK)

AUD_HDR Frame Header

	7 ... 0
0x5E	AUD_HDR[7:0]
0x5F	AUD_HDR[15:8]
0x60	AUD_HDR[23:16]
0x61	AUD_HDR[31:24]

Address: *AudioBaseAddress* + 0x5E to 0x61
Type: Read only
Reset value: undefined

Description

This 32-bit register contains the header of the frame currently being decoded.

This register is updated after interrupt 1 is enabled. An interrupt 1 is generated, i.e. **AUD_ITR[1]**, when a valid header has been received. The contents are retained until **AUD_HDR[31:24]** is read.

AUD_IDE Audio Stream ID Enable

	7	0
0x24		AUD_IDE

Address: *AudioBaseAddress* + 0x24
Type: Read/write
Reset value: undefined

Description

If this bit is reset, then the contents of **AUD_SID** are ignored. If it is set, then the register **AUD_SID** is taken into account. As they have opposite meanings, This bit and bit **PES_CF1.IAI** must always be forced to opposite values at the "same time" (beginning with **PES_CF1.IAI**).

0: Ignore **AUD_SID**

1: Use **AUD_SID**

AUD_IFT Input FIFO Threshold

	7	0
0x52	AUD_IFT[7:0]	

Address: *AudioBaseAddress* + 0x52
Type: Read/write
Reset value: undefined

Description

This value loaded into this register defines the input FIFO level at which an interrupt 12 can be generated, i.e. **AUD_ITR[12]**. The level is defined as a byte address, in the range 0 to 255. An interrupt can be generated each time the FIFO level is equal to **AUD_IFT**, regardless of whether it was approached from above or below. In addition, an interrupt 13, i.e **AUD_ITR[13]**, is generated whenever the FIFO is full.

Those interrupts AUD_ITR[13:12] are useful only when **AUD_BBE** is reset (i.e. Audio Bit Buffer disabled and data input directly into audio decoder through **AUD_CDI** register followed by this input FIFO. Note that **AUDREQ** refers to the input of data into the Bit Buffer (**AUD_BBE** is set). Thus controlling the direct input of audio data without Bit-Buffer, using **AUD_ITR[13:12]** is very constrained.

AUD_IMS Audio Interrupt Extension Mask

	7	1 0
0x5C		AUD_IMS[1:0]

Address: *AudioBaseAddress* + 0x5C
Type: Read/write
Reset value: undefined

Description

Interrupt status bits indicating new value for ESCR field received in MPEG-2 PES streams.
Setting **AUD_IMS[0]** must always be tied low. Setting **AUD_IMS[1]** enables ESCR interrupt.

AUD_ISS Input Stream Selection

	7	2 0
0x36		AUD_ISS[2:0]

Address: *AudioBaseAddress* + 0x36
Type: Read/write
Reset value: undefined

Description

This register defines the type of input bitstream expected by the audio decoder.

000: MPEG audio elementary stream

001: MPEG-1 packet stream

010: reserved

011: PCM data. In this mode, the audio decoder is bypassed and data is sent directly to the audio DAC interface. This bit must be set in CD-DA mode.

100 : MPEG-2 PES Streams

101 : Automatic detection MPEG-1/MPEG-2 packet streams

AUD_ITM Interrupt Mask Register

	7	6 ... 0
0x1C	AUD_ITM[7:0]	
0x1D		AUD_ITM[14:8]

Address: *AudioBaseAddress* + 0x1C to 0x1D
Type: Read/write
Reset value: 0. Also cleared on restart.

Description

A one in any bit position of this register will enable the corresponding bit of the **AUD_ITR** register.
In addition, setting certain bits of this register have additional actions, as specified below:

Setting **AUD_ITM[7]** enables the reading of ancillary data into **AUD_ANC**.

Setting **AUD_ITM[2]** enables the updating of the **AUD_PTS** register.

Setting **AUD_ITM[1]** enables the updating of the **AUD_HDR** register.

Setting **AUD_ITM[0]** enables the updating of the **AUD_SYS** register.

AUD_ITR Interrupt Status Request Register

	7	6 ... 0
0x1A	AUD_ITR[7:0]	
0x1B		AUD_ITR[14:8]

Address: *AudioBaseAddress* + 0x1A to 0x1B
Type: Read only
Reset value: 0. Also cleared on restart.

Description

An interrupt is signalled whenever one of the bits of **AUD_ITR** becomes set. This can only occur if the corresponding bit is set in the **AUD_ITM** register. The **AUD_ITR** register is cleared on reset (assertion of $\overline{\text{RESET}}$ pin or setting of **AUD_RES** register), or restart (setting the **AUD_RST** register). Also the most significant byte, and bit 5 of the least significant byte of **AUD_ITR** can be independently cleared by reading. Bits 0-2 and 7 are cleared by a different method, as indicated in the notes in the table below.

Bit	**Condition Signalled**	**Note**
14	First bit of new frame at PCM output	
13	Input FIFO full	
12	Input FIFO level = FIFO_THRES	
11	not used	
10	De-emphasis changed	
9	Sampling frequency changed	
8	PCM output buffer underflow	
7	Ancillary data register full	**AUD_ANC[31:24]** must be read in order to clear bit AUD_ITR[7] and to reinitialize the ancillary data buffer.
6	not used	
5	CRC error detected	
4	not used	
3	not used	
2	Valid PTS registered	**AUD_PTS[32]** must be read in order to clear bit **AUD_ITR[2]** and to reinitialize the **AUD_PTS** register.
1	Valid header registered	**AUD_HDR[31:24]** must be read in order to clear bit **AUD_ITR[1]** and to reinitialize the **AUD_HDR** register.
0	Change in synchronization status	**AUD_SYS** must be read in order to clear bit **AUD_ITR[0]** and to reinitialize the **AUD_SYS** register.

AUD_ITS Audio Interrupt Extension

0x5B	7	1 0
		AUD_ITS[1:0]

Address: *AudioBaseAddress* + 0x5B
Type: Read/write
Reset value: undefined

Description

Interrupt status bits indicating new values for ESCR and DTS fields received in MPEG-2 PES streams.

AUD_LAT Latency Selection

0x3C	7	0
		AUD_LAT

Address: *AudioBaseAddress* + 0x3C
Type: Read/write
Reset value: undefined

Description

This bit is used to select the latency mode of the decoder, and must be set to zero.

0: Low latency mode. The decoding latency is less than 5 ms.

1: Illegal

AUD_LCA Left Channel Attenuation

0x1E	7	5 0
		AUD_LCA[5:0]

Address: *AudioBaseAddress* + 0x1E
Type: Read/write
Reset value: undefined

Description

This register defines the left channel attenuation in steps of 2 dB. The minimum attenuation is 0 dB, the maximum is 2 x 63 = 126 dB.

AUD_LCK Sync Words Until Lock

0x28	7	1 0
		AUD_LCK[1:0]

Address: *AudioBaseAddress* + 0x28
Type: Read/write

Reset value: undefined

Description

This register defines how many valid synchronization words after the initial one must be found before locking audio frame synchronization.

When **AUD_SYE** is set to its default value, the audio decoder assumes that **AUD_LCK** is set to the value 3. Synchronization error concealment is still enabled when **AUD_LCK** has the value zero.

AUD_LRP LRCK Polarity

	7	0
0x11		AUD_LRP

Address: *AudioBaseAddress* + 0x11
Type: Read/write
Reset value: 0 after assertion of $\overline{\text{RESET}}$ pin only

Description

This bit is used to define the polarity of the output signal LRCK.

0: Left channel when LRCK = 1

1: Left channel when LRCK = 0

AUD_MUT Mute

	7	0
0x30		AUD_MUT

Address: *AudioBaseAddress* + 0x30
Type: Read/write
Reset value: 0 after assertion of $\overline{\text{RESET}}$ pin only

Description

See audio processor description for more details.

AUD_ORD PCM Output Bit Order

	7	0
0x38		AUD_ORD

Address: *AudioBaseAddress* + 0x38
Type: Read/write
Reset value: undefined

Description

This bit determines the order of PCM data output when in 16-bit mode (i.e. when **AUD_P18** = 0). It has no significance when **AUD_P18** = 1, when data is always output most-significant bit first.

0: Most-significant bit output first

1: Least-significant bit output first

AUD_P18 PCM Output Precision

	7	0
0x16		AUD_P18

Address: *AudioBaseAddress* + 0x16
Type: Read/write
Reset value: undefined

Description

This bit defines the PCM output precision.

0: 16-bit PCM data output

1: 18-bit PCM data output

Note that in 18-bit PCM data output, the 16 bits of the audio stream are on the MS bits of the 18 bits samples.

AUD_PLY Play

	7	0
0x2E		AUD_PLY

Address: *AudioBaseAddress* + 0x2E
Type: Read/write
Reset value: 0 after assertion of $\overline{\text{RESET}}$ pin only

Description

See audio processor section for more details.

AUD_PTS Presentation Time Stamp

	7	0
0x62	AUD_PTS[7:0]	
0x63	AUD_PTS[15:8]	
0x64	AUD_PTS[23:16]	
0x65	AUD_PTS[31:24]	
0x66		AUD_PTS[32]

Address: *AudioBaseAddress* + 0x62 to 0x66
Type: Read only
Reset value: undefined

Description

This 33-bit register contains the PTS associated with the frame currently being decoded.

This register is updated after interrupt 2 is enabled. An interrupt 2 is generated, i.e. **AUD_ITR[2**], when a valid PTS has been received. The contents are retained until **AUD_PTS[32]** is read.

AUD_RCA Right Channel Attenuation

	7	5	0
0x20		AUD_RCA[5:0]	

Address: *AudioBaseAddress* + 0x20
Type: Read/write
Reset value: undefined

Description

This register defines the right channel attenuation in steps of 2 dB. The minimum attenuation is 0 dB, the maximum is 2 x 63 = 126 dB.

AUD_RES Audio Decoder Software Reset

	7	0
0x40		AUD_RES

Address: *AudioBaseAddress* + 0x40
Type: Read/write
Reset value: 0 (command)

Description

Writing a 0 or 1 to this bit has an equivalent function to audio only hardware reset, except that the following registers are not cleared:

- AUD_SCP
- AUD_MUT
- AUD_PLY

This bit is reset automatically after being set.

AUD_REV Audio Decoder Revision

	7	0
0x6D	AUD_REV[7:0]	

Address: *AudioBaseAddress* + 0x6D
Type: Read only

Description

This register holds the version number of the audio decoder. Example version register read.

Register value:	1101	0011	
Mask:	0111	1111	
Version:	0101	0011	<=> 5.3

AUD_RST Restart

0x42	7 ...	0
		AUD_RST

Address: *AudioBaseAddress* + 0x42
Type: Read/write
Reset value: (command)

Description

When this bit is set, all data buffers are flushed, and then the **AUD_RST** bit is automatically reset. In addition the **AUD_ITR**, **AUD_ITM**, **AUD_IMS** and **AUD_IMT** registers are cleared.

AUD_SCM Sync Confirmation Mode

0x25	7 ...	0
		AUD_SCM

Address: *AudioBaseAddress* + 0x25
Type: Read/write
Reset value: undefined

Description

This bit selects one of two options in the packet synchronization algorithm.

0: After the first valid packet start code and stream ID are found, the packet length is used to locate the next start code and stream ID before synchronization is confirmed and audio decoding starts.

1: Synchronization is confirmed when the first valid packet start code and stream ID are found.

AUD_SCP SCLK Polarity

0x53	7 ...	0
		AUD_SCP

Address: *AudioBaseAddress* + 0x53
Type: Read/write
Reset value: 0 after assertion of $\overline{\text{RESET}}$ pin only

Description

This bit defines the polarity of the PCM bit clock output BCK.

0: The LRCK and PCMDATA outputs change on the falling edge of SCLK. The external DAC will sample LRCLK and PCMDATA on the rising edge of SCLK.

1: The LRCK and PCMDATA outputs change on the rising edge of SCLK. The external DAC will sample LRCK and PCMDATA on the falling edge of SCLK.

AUD_SEM Sync Error Concealment Mode

	7	1 0
0x2C		AUD_SEM[1:0]

Address: *AudioBaseAddress* + 0x2C
Type: Read/write
Reset value: undefined

Description

This register defines the action which will be taken upon detection of a synchronization error.

00: Ignore error

01: Mute on detection of synchronization error

10: Invalid.

11: Skip invalid frame

AUD_SFR Sampling Frequency

	7	1 0
0x44		AUD_SFR[1:0]

Address: *AudioBaseAddress* + 0x44
Type: Read only
Reset value: 0

Description

This register is loaded with the sampling frequency code extracted from the bit stream. The register is updated synchronously with the first sample out on the PCM data. If enabled, an interrupt 9 is generated, i.e. **AUD_ITR[9]**, at this time if the sampling frequency value has changed.

00: 44.1 kHz

01: 48 kHz

10: 32 kHz

11: reserved

AUD_SID Audio Stream ID

	7	4	0
0x22		AUD_SID[4:0]	

Address: *AudioBaseAddress* + 0x22
Type: Read/write
Reset value: undefined

Description

The value stored in this register is only taken into account if **AUD_IDE** is set. This register specifies the number (between 0 and 31) of the audio stream which is to be decoded. The stream number is defined in the field *stream_id* of the packet header. All other packets will be discarded. If **AUD_IDE** is reset, then all audio packets are decoded. Note that these bits are duplicated in register **PES_CF1.AUD_ID[4:0]** and should be set simultaneously.

AUD_SKP Skip Next Frame

	7	0
0x32		AUD_SKP

Address: *AudioBaseAddress* + 0x32
Type: Read/write
Reset value: 0 (command)

Description

When this bit is set, the next audio frame is skipped, and then the **AUD_SKP** bit is automatically reset.

AUD_SYE Sync Word Extension

	7	0
0x27	AUD_SYE[7:0]	

Address: *AudioBaseAddress* + 0x27
Type: Read/write
Reset value: undefined

Description

This register defines an extension to the frame synchronization word which can be used to increase the reliability of synchronization when the layer number, bit rate or sampling frequency are known. The three fields are defined in the following tables. Programming of this register is mandatory.

Bitfield	Definition
AUD_SYE[1:0]	00: 44.1 kHz 01: 48 kHz 10: 32 kHz 11: unused
AUD_SYE[5:2]	Value / LayerI / LayerII 0000: free format / free format 0001: 32 kbit/s / 32 kbit/s 0010: 64 kbits / 48 kbit/s 0011: 96 kbit/s / 56 kbit/s 0100: 128 kbit/s / 64 kbit/s 0101: 160 kbit/s / 80 kbit/s 0110: 192 kbit/s / 96 kbit/s 0111: 224 kbit/s / 112 kbit/s 1000: 256 kbit/s / 128 kbit/s 1001: 288 kbit/s / 160 kbit/s 1010: 320 kbit/s / 192 kbit/s 1011: 352 kbit/s / 224 kbit/s 1100: 384 kbit/s / 256 kbit/s 1101: 416 kbit/s / 320 kbit/s 1110: 448 kbit/s / 384 kbit/s 1111: This field not used / This field not used
AUD_SYE[7:6]	11: Layer I 10: Layer II 00: Not used

Examples :

1: Layer II, 48 kHz sampling frequency, variable bit rate: **AUD_SYE** = 10111101_2

2: Parameters unknown: **AUD_SYE** = 00111111_2

AUD_SYN Packet Sync Mode

	7 … 2	1 … 0
0x23		AUD_SYN[1:0]

Address: *AudioBaseAddress* + 0x23
Type: Read/write
Reset value: undefined

Description

This register defines the packet synchronization mode.

Bitfield	Definition
AUD_SYN	00: Synchronize only on 24-bit *packet_start_code_prefix*. (Multiplexed audio/video bitstream). 01: Synchronize both on 24-bit *packet_start_code_prefix* and first 3 bits of *stream_id*. (Multiplexed audio bitstream). 10: Synchronize both on 24-bit *packet_start_code_prefix* and all 8 bits of *stream_id*. (Audio bitstream with unique id).

AUD_SYS Synchronization Status

	7	1 0
0x26		AUD_SYS[1:0]

Address: *AudioBaseAddress* + 0x26
Type: Read only
Reset value: undefined

Description

This register is loaded with the synchronization status on every synchronization cycle. This status values are:

00: Unlocked , 01: Attempting to recover lost synchronization, 11: Locked

If the status changes an interrupt 0 is generated, i.e. **AUD_ITR[2]**. The status must then be read and the interrupt cleared by reading **AUD_SYS**.

5 Background color (BCK)

BCK_Y Background Color Y

	7 … 0
0x98	BCK_Y[7:0]

Address: *VideoBaseAddress* + 0x98
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register stores the value of the Y component of the background color mixable with the video or the still picture plane.

BCK_U Background Color U

	7 … 0
0x99	BCK_U[7:0]

Address: *VideoBaseAddress* + 0x99
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register stores the value of the U component of the background color mixable with the video or the still picture plane.

BCK_V Background Color V

	7 … 0
0x9A	BCK_V[7:0]

Address: *VideoBaseAddress* + 0x9A
Type: Read/write
Reset value: FF
Synchronization: VSYNC

Description

This register stores the value of the V component of the background color mixable with the video or the still picture plane.

6 Block move DMA (BMDMA)

BMDMAAbort Block move DMA abort

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x18	Abort																															

Address: *BMBaseAddress* + 0x18
Type: Write only

Description

Any write to this register aborts the current block move operation. The abort may take a few system clock cycles to complete depending on the access time of the memory and the **Active** bit in **BMDMAStatus** should be polled to check that the DMA has been aborted.

BMDMACount Block move DMA counts

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x08																	ByteCount															

Address: *BMBaseAddress* + 0x08
Type: Write only

Description

The number of bytes to block move. Writing to this register starts the block move.

BMDMADestAddress Block move DMA destination address

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x04	DestAddress																															

Address: *BMBaseAddress* + 0x04
Type: Write only

Description

The address of the base of the block move destination area.

BMDMAIntAck Block move DMA interrupt acknowledge

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x14	Ack																															

Address: *BMBaseAddress* + 0x14
Type: Write only

Description

Any write to this register acknowledges the interrupt and resets the **Interrupt** bit in the **BMDMAStatus** register.

BMDMAIntEn Block move DMA interrupt enable

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x0C		IntEn

Address: *BMBaseAddress* + 0x0C
Type: Read/write

Description

Interrupt enable; enabled when set to 1, masked when set to 0.

BMDMASrcAddress Block move DMA source address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x00	SrcAddress

Address: *BMBaseAddress* + 0x00
Type: Write only

Description

The address of the base of the block move source data.

BMDMAStatus Block move DMA status

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0
0x10	Reserved	Active	Interrupt

Address: *BMBaseAddress* + 0x10
Type: Read only

Description

Bitfield	Description
Active	Block move in progress when this bit is 1.
Interrupt	Interrupt pending when set to 1.

7 Cache control (CAC)

CAC_CacheControl0-1 Region 1 Cache control 0-1

	8	7	6	5	4	3	2	1	0
0x000	Block size	Cacheable7	Cacheable6	Cacheable5	Cacheable4	Cacheable3	Cacheable2	Cacheable1	Cacheable0
0x100	Reserved	Cacheable15	Cacheable14	Cacheable13	Cacheable12	Cacheable11	Cacheable10	Cacheable9	Cacheable8

Address: *CacheBaseAddress* + 0x000 (CacheControl0), 0x100 (CacheControl1)
Type: Read/write
Reset value: 0

Description

These two registers set the cacheability of 16 blocks in memory region 1. Each block is independently cacheable, i.e. bit *cacheable0*=1 sets block 0 cacheable and bit *cacheable0*=0 sets block 0 non-cacheable. The block size is set by bit 8 of CAC_CacheControl0 (when bit8=0 block size= 64 Kbyte; when bit8=1 block size= 512 Kbyte). The table below gives the block-start and block-end locations of each cacheable block, for both 64Kbit and 512Kbit block sizes.

Bitfield	**Block size 64Kbit (CacheControl0 bit 8 = 0)**		**Block size 512Kbit (CacheControl0 bit 8 = 1)**	
	Block start	**Block end**	**Block start**	**Block end**
Cacheable0	0xC0000000	0xC000FFFF	0xC0000000	0xC007FFFF
Cacheable1	0xC0010000	0xC001FFFF	0xC0080000	0xC00FFFFF
Cacheable2	0xC0020000	0xC002FFFF	0xC0100000	0xC017FFFF
Cacheable3	0xC0030000	0xC003FFFF	0xC0180000	0xC01FFFFF
Cacheable4	0xC0040000	0xC004FFFF	0xC0200000	0xC027FFFF
Cacheable5	0xC0050000	0xC005FFFF	0xC0280000	0xC02FFFFF
Cacheable6	0xC0060000	0xC006FFFF	0xC0300000	0xC037FFFF
Cacheable7	0xC0070000	0xC007FFFF	0xC0380000	0xC03FFFFF
Cacheable8	0xC0080000	0xC008FFFF	0xC0400000	0xC047FFFF
Cacheable9	0xC0090000	0xC009FFFF	0xC0480000	0xC04FFFFF
Cacheable10	0xC00A0000	0xC00AFFFF	0xC0500000	0xC057FFFF
Cacheable11	0xC00B0000	0xC00BFFFF	0xC0580000	0xC05FFFFF
Cacheable12	0xC00C0000	0xC00CFFFF	0xC0600000	0xC067FFFF
Cacheable13	0xC00D0000	0xC00DFFFF	0xC0680000	0xC06FFFFF
Cacheable14	0xC00E0000	0xC00EFFFF	0xC0700000	0xC077FFFF
Cacheable15	0xC00F0000	0xC00FFFFF	0xC0780000	0xC07FFFFF

CAC_EMIBank0-3 EMI Bank 0-3 cacheability

	7	6	5	4	3	2	1	0
0x300	Reserved				Cacheable3	Cacheable2	Cacheable1	Cacheable0

Address: *CacheBaseAddress* + 0x300
Type: Read/write
Reset value: 0

Description

This register sets the data cacheability of the 4 banks in region 3 cacheable. The table below gives the start and end locations of each bank. Register **CAC_EMIBlock0** sets the cacheability of the lower part of EMI Bank 0.

Bitfield	Bank start	Bank end	Remarks
Cacheable0	0x40000000	0x4007FFFF	Cacheable in 8 blocks by register CAC_EMIBlock0
	0x40080000	0x4FFFFFFF	
Cacheable1	0x50000000	0x5FFFFFFF	
Cacheable2	0x60000000	0x6FFFFFFF	
Cacheable3	0x70000000	0x7FFFFFFF	

CAC_EMIBlock0 EMI Block 0-7 cacheability

	7	6	5	4	3	2	1	0
0x200	Cacheable7	Cacheable6	Cacheable5	Cacheable4	Cacheable3	Cacheable2	Cacheable1	Cacheable0

Address: *CacheBaseAddress* + 0x200
Type: Read/write
Reset value: 0

Description

This register sets the data cacheability of the 8 blocks at the bottom of EMI Bank 0 in region 3 cacheable. The block start and end locations are shown in the table below. **CAC_EMIBank0-3** bit0 must be set to 1 to use this register.

Bitfield (1 for enabled)	Block start	Block end
Cacheable0	0x40000000	0x4000FFFF
Cacheable1	0x40010000	0x4001FFFF
Cacheable2	0x40020000	0x4002FFFF
Cacheable3	0x40030000	0x4003FFFF
Cacheable4	0x40040000	0x4004FFFF
Cacheable5	0x40050000	0x4005FFFF
Cacheable6	0x40060000	0x4006FFFF
Cacheable7	0x40070000	0x4007FFFF

CAC_DCacheNotSRAM DCache as a cache or SRAM

	7	6	5	4	3	2	1	0
0x400	Reserved							dcachenotsram

Address: *CacheBaseAddress* + 0x400
Type: write
Reset value: 0

Description

This register configures the DCACHE to be 2K SRAM; when bit0=0 it is 2K SRAM; when bit0=1 it is DCACHE. If attempts are made to change the mode from DCACHE to SRAM, the request will be masked until the DCACHE is ready to accept it (dcache_ready ='1') OR invalidate_dcache/flushing_dcache bits are off (no pending request to invalidate or flush the DCACHE).

CAC_InvalidateDCACHE Invalidate DCache

	7	6	5	4	3	2	1	0
0x500	Reserved							invalidatedcache

Address: *CacheBaseAddress* + 0x500
Type: write
Reset value: 0

Description

This register is used in conjunction with CAC_FlushingDCACHE and CAC_InvalidateICACHE to invalidate/flush theCACHEs. Writing 1 to these registers sets the function ON, writing 0 has no meaning, as the register is internally reset to 0 when the invalidate or flush sequence has completed. When they are accessed, the corresponding internal signals (invalidate_dcache, flushing_dcache and invalidate_icache) are set to 1. To check that the invalidate or flush sequence has completed the corresponding internal signals can be read from the CAC_STATUS register.

CAC_FlushingDCACHE Flushing DCache

	7	6	5	4	3	2	1	0
0x600	Reserved							FlushingDCache

Address: *CacheBaseAddress* + 0x600
Type: write
Reset value: 0

Description

This register is used in conjunction with CAC_InvalidateDCACHE and CAC_InvalidateICACHE to invalidate/flush the CACHEs. Writing 1 to these registers sets the function ON, writing 0 has no meaning, as the register is internally reset to 0 when the invalidate or flush sequence has completed.

CAC_InvalidateICACHE Invalidate ICache

	7	6	5	4	3	2	1	0
0x800	Reserved							InvalidateIcache

Address: *CacheBaseAddress* + 0x800
Type: write
Reset value: 0

Description

This register is used in conjunction with CAC_InvalidateDCACHE and CAC_FlushingDCACHE to invalidate/flush the CACHEs. Writing 1 to these registers sets the function ON, writing 0 has no meaning, as the register is internally reset to 0 when the invalidate or flush sequence has completed.

CAC_EnableICACHE Enable ICache

	7	6	5	4	3	2	1	0
0x700	Reserved							EnableICache

Address: *CacheBaseAddress* + 0x700
Type: write
Reset value: 0

Description

When bit0 of register CAC_EnableICACHE is 1 then ICACHE is ON; when 0 then ICACHE is OFF. If attempts are made to change the mode from ICACHE ON to ICACHE OFF, the request will be masked until the ICACHE is ready to accept it (icache_ready = 1) OR invalidate_icache bit is off (no pending request to invalidate the ICACHE).

CAC_STATUS Internal register status

	7	6	5	4	3	2	1	0
0x90 0		ICache Ready	DCache Ready	Flushing DCache	Invalidating ICache	Invalidating DCache	Enable ICache	DCache NotSRAM

Address: *CacheBaseAddress* + 0x900
Type: Read only
Reset value: 0

Description

This register is read only, and shows the current state of the caches. It is described in the table below

Bitfield	Description
DCacheNotSRAM	Data cache or SRAM: 0=SRAM, 1=Data cache
EnableICache	Instruction cache: 0=Disabled, 1=Enabled
InvalidatingDCache	Invalidating data cache: 0=Not invalidating, 1=Invalidating
InvalidatingICache	Invalidating instruction cache: 0=Not invalidating, 1=Invalidating
FlushingDCache	Flushing instruction cache: 0=Not flushing, 1=Flushing
DCacheReady	Data cache ready: 0=Busy performing an operation, 1= Ready to perform another operation
ICacheReady	Instruction cache ready: 0=Busy performing an operation, 1=Ready to perform another operation

CAC_CacheControlLock Cache control lock

	7	6	5	4	3	2	1	0
0xA00								Lock

Address: *CacheBaseAddress* + 0xA00
Type: Read/write
Reset value: 0

Description

This register is used to lock the cache configuration. When bit0=1 the cacheability control, CAC_DCacheNotSRAM , CAC_EnableICache and CAC_CacheControlLock registers are all locked. Reset of this flag is only performed by a hardware reset.

8 Configuration and control (CFG)

CFG_CCF Chip Configuration

	7	6	5	4	3	2	1	0
0x01	EAI	EOU	PBO	EC3	EC2	ECK	EDI	EVI

Address: *VideoBaseAddress* + 0x01
Type: Read/write
Reset value: 0
Synchronization: Edge triggered

Description

Bitfield	Description
EAI	*Enable audio interface.* When this bit is reset the audio interface is put into its high impedance state and the internal PCMCLK is disabled. This bit must be set for normal operation and for reduced power mode (if the display interface is used). It is reset in low power mode.
EOU	*Enable overflow/underflow errors.* When this bit is reset overflow and underflow errors are not treated internally. This bit must be set for normal operation.
PBO	*Prevent bit buffer overflow.* When this bit is set, bit buffer overflow (and thus the loss of data) is prevented by disabling the transfer of data from the compressed data FIFO to the bit buffer whenever the bit buffer level reaches the threshold defined in the **VID_VBT** or **VID_ABT** register.
EC3	*Enable clock 3.* When this bit is reset, the internal clock3 (clock3 = memory clock = SDRAM clock/4) is disabled. This bit must be set for normal operation and for reduced power mode. It is reset in low power mode.
EC2	*Enable clock 2.* When this bit is reset, the internal clock2 (clock2 = decoder clock = SDRAM clock/2) is disabled. This bit must be set for normal operation, and reset in reduced and low power modes.
ECK	*Enable clocks.* When this bit is reset, all internal clocks are disabled. This bit must be set for normal operation and for reduced power mode. It is reset in low power mode.
EDI	*Enable SDRAM interface.* When this bit is reset the SDRAM interface is put into its high impedance state. This bit must be set for normal operation and for reduced power mode. It is reset in low power mode.
EVI	*Enable video interface.* When this bit is reset the video interface is put into its high impedance state and the internal PIXCLK disabled. This bit must be set for normal operation and for reduced power mode (if the display interface is used). It is reset in low power mode.

CFG_CDR Compressed Data Input

	7 … 0
0x44	CFG_CDR[7:0]

Address: *VideoBaseAddress* + 0x44
Type: Write only
Reset value: Undefined
Synchronization: None

Description

This register is a compressed data input register. It is used to input compressed data using a different path from the usual DMA path. Depending on the value of **CFG_GCF.CDR[1:0]** the data stored in this register will enter either the audio, video or sub-picture compressed data FIFO.

CFG_DRC DRAM configuration

	7	6	5	4	3	2	1	0
0x38	MY64	Reserved	MRS	Reserved	P[1:0]		ERQ	SDR

Address: *VideoBaseAddress* + 0x38
Type: Read/write
Reset value: 0
Synchronization: None

Description

Bitfield	Description
SDR	Synchronous DRAM mode This bit must always be written as 1.
ERQ	Enable Processes Requests This bit when reset disables all processes requests to the local memory controller. This bit must be set for normal operation.
P[1:0]	Clk3 phase to MemClk The following procedure has to be executed to make sure the SDRAM interface is properly initialized: P[1:0] = 0, write four 32-bit words to SDRAM, read back the four words from SDRAM, if they do not read back correctly, increment the value of P[1:0] until the read is correct.
MRS	Mode Register Set When this bit is set the special power-on-reset sequence and mode register programming is carried out. When this bit is reset the procedure is inhibited. Mode used is: burst length = 4 and CAS latency = 3.
MY64	Memory format 0: Memory mode is 16Mb SDRAM (1 or 2 units) 1: Memory mode is 64Mb SDRAM

CFG_GCF General Configuration

	7	6	5	4	3	2	1	0
0x3A	PXD	A3Rq	A3DI	CDR		SCK	A3M	

Address: *VideoBaseAddress* + 0x3A
Type: Read/write
Reset value: 0
Synchronization: None

Description

Bitfield	Description
PXD	*Add one PIXCLK delay.* When this bit is set the active video is delayed by one PIXCLK cycle with respect to $\overline{\text{HSYNC}}$. This is to allow its horizontal position to be defined more precisely than is possible with **VID_XDO** and **VID_XDS**, which have a resolution of 2 PIXCLK cycles. Changing the value of PXD also has the effect of inverting the phasing of the Y/C output samples with respect to $\overline{\text{HSYNC}}$.
A3Rq	*External AC3 request polarity.* External polarity is reversed if set to 1.
A3DI	*External AC3 data strobe mode.* This bit, when set, allows MSB data being strobed on the second rising edge of SCLK following a transition of LRCLK. Otherwise, the first rising edge is used.
CDR[1:0]	Compressed data register: 00: Automatic strobe generation when writing to CD FIFOs 10: Sub-picture 01: Audio 11: Video When one internal strobe is chosen, it has to be driven by writing to register **CFG_CDR**.
SCK	*Audio strobe clock select.* This bit, when set, selects clock2 as serial strobe period. Otherwise, clock3 is selected. Note clock2 = SDRAM clock/2, clock3 = SDRAM clock/4
A3M	*AC3 / MPEG audio decoder Select.* This bit, when set, configures the audio pins to use the external AC3 audio decoder. When reset, the configuration is done for the internal MPEG audio decoder.

CFG_MCF Memory Refresh Interval

	7	6 ... 0
0x00		RFI[6:0]

Address: *VideoBaseAddress* + 0x00
Type: Read/write
Reset value: 0
Synchronization: none

Description

This register defines the SDRAM refresh interval in units of 32 SDRAM clock periods. For example if 2048 rows must be refreshed every 32 ms, with an SDRAM clock of 108 MHz, the following value must be stored in **MCF.RFI[6:0]**:

$(32 \times 10^{-3}/2048) \times (108 \times 10^{6}/32) = 52$

If the register is set to 0 then refresh is disabled.

9 Clock generator (CKG)

CKG_AUX Clock Generator for the Auxiliary Clock Divider

	7	6	5	4	3 - 0
0xDC			ENA	DV2	P0[3:0]
0xDD		PR[9:3]			
0xDE		PR[2:0]			Q[10:7]
0xDF		Q[6:0]			

Address: *VideoBaseAddress* + 0xDC to 0xDF
Type: Read/write
Reset value: 0

Description

This register controls the fractional divider for the auxiliary clock. Writing to the highest address (DF) latches the new configuration. The bits have the significance given in the following table.

Bitfield	Address	Description
ENA	0xDC	*Enable*. Controls whether or not the fractional divider is enabled or disabled for low power. 0: fractional divider is disabled, 1: fractional divider is enabled
DV2	0xDC	*Divide-by-2*. The output of the fractional divider is divided by two. 0: not divided by 2, 1: divided by 2.
P0[3:0]	0xDC	P_0: Defines the value of P_0 as described in the STi5512 datasheet. These bits should be programmed to a value of P_0-1.
PR[9:3]	0xDD	P_r: Contains the value of P_r as described in the STi5512 datasheet.
PR[2:0]	0xDE	
Q[10:7]	0xDE	*Q*: Contains the value of Q as described in the STi5512 datasheet. These bits should be programmed to Q (i.e. one’s compliment).
Q[6:0]	0xDF	

CKG_CFG Clock Generator Configuration

	7	6 - 0
0x31		CKG_CFG[6:0]

Address: *VideoBaseAddress* + 0x31
Type: Read/write
Reset value: 0

Description

Controls the input/output modes for MEMCLK and PCMCLK and signal selection for MPEG decoders clocks and test modes. After reset, clock pin PCMCLK is in input mode and clock MEMCLK is in output mode.

The table below shows the significance of the bits.

Bitfield	Description
CFG[6]	*Configuration 6.* Controls the input/output state of the **MEMCLK** pin. 0: the **MEMCLK** pin is an output from MEMCLKo fractional divider (see bits **CKG_MEM.ENA**/**DV2**) 1: the **MEMCLK** can be input externally (bypassing PLL) using the **A_C_REQ** pin.
CFG[5]	*Configuration 5.* Controls the input/output state of the **PCMCLK** pin. 0: the **PCMCLK** pin is an input 1: the **PCMCLK** pin is an output from PCMCLKo fractional divider (see bits **CKG_PCM.ENA**/**DV2**)
CFG[3:2]	Configuration 3 and 2. Controls the AUX multiplexor. 00: Selects the clock generated by its fractional divider (see bits CKG_AUX.ENA/DV2). 01: Select a fractional divider output for test purpose. 10: Select 27 MHz clock. 11: Select the PLL clock f_{PLL} - TEST MODE.

CKG_MCK Clock Generator MEMCLK Divider

	7	6	5	4	3	0
0xD8		DIV	ENA	DV2	P0[3:0]	
0xD9		PR[9:3]				
0xDA		PR[2:0]			Q[10:7]	
0xDB		Q[6:0]				

Address: *VideoBaseAddress* + 0xD8 to 0xDB
Type: Read/write
Reset value: 0

Description

This register controls the fractional divider for the memory clock (see the STi5512 datasheet). Writing to the highest address (DB) latches the new configuration. The bits have the significance given in the table below.

Bitfield	Address	Description
DIV	0xD8	*Divide*. Controls whether or not the fractional or the hardwired divide by 2 is enabled 0: Fixed divide by 2 1: Fractional divider is enabled
ENA	0xD8	*Enable*. Controls whether or not the fractional divider is enabled or disabled for low power 0: Fractional divider is disabled 1: Fractional divider is enabled
DV2	0xD8	*Divide-by-2*. The output of the fractional divider is divided by two 0: Not divided by 2 1: Divided by 2
P0[3:0]	0xD8	P_0: Defines the value of P_0 as described in the STi5512 datasheet. These bits should be programmed to a value of P_0-1.
PR[9:3]	0xD9	P_r: Contains the value of P_r as described in the STi5512 datasheet.
PR[2:0]	0xDA	
Q[10:7]	0xDA	*Q*: Contains the value of Q as described in the STi5512 datasheet. These bits should be programmed to Q (i.e. one's compliment).
Q[6:0]	0xDB	

CKG_PCM Clock Generator PCMCLK Divider

	7	6	5	4	3			0
0xD0			ENA	DV2	P0[3:0]			
0xD1		PR[9:3]						
0xD2		PR[2:0]			Q[10:7]			
0xD3		Q[6:0]						

Address: *VideoBaseAddress* + 0xD0 to 0xD3
Type: Read/write
Reset value: 0

Description

This register controls the fractional divider for the PCM clock (see the STi5510 datasheet). Writing to the highest address (D3) latches the new configuration. The bits have the significance shown in the following table.

Bitfield	Address	Description
ENA	0xD0	*Enable*. Controls whether or not the fractional divider is enabled or disabled for low power. 0: Fractional divider is disabled 1: Fractional divider is enabled
DV2	0xD0	*Divide-by-2*. The output of the fractional divider is divided by two. 0: Not divided by 2 1: Divided by 2
P0[3:0]	0xD0	P_0: Defines the value of P_0 as described in the STi5512 data sheet. These bits should be programmed to a value of P_0-1.
PR[9:3]	0xD1	P_r: Contains the value of P_r as described in the STi5512 data sheet.
PR[2:0]	0xD2	
Q[10:7]	0xD2	*Q*: Contains the value of Q as described in the STi5512 data sheet. These bits should be programmed to Q (i.e. one's compliment).
Q[6:0]	0xD3	

CKG_PLL Clock Generator PLL Parameters

	7	6	5	4	3			0
0x30	PDM	REF[1:0]		N	M[3:0]			

Address: *VideoBaseAddress* + 0x30
Type: Read/write
Reset value: 10000000_2, i.e. PLL in power down mode.

Description

Controls the reference frequency input, pre-divider and multiplication factor of the PLL.

The bits have the following significance:

Bitfield	Description
PDM	*Power down mode:* When set, the PLL is in power down mode.
REF[1:0]	*Clock Generator Reference:* Control the reference frequency input multiplexer to the clock generator PLL. 00: Power down mode (no clock: grounded) 01: Selects external PCMCLK (if selected by **CKG_CGF.CFG[5]**) 10: Selects external 27 MHz PIXCLK 11: Selects external 27 MHz PIXCLK
N	*Divide-by-N+1:* When set, the reference input to the PLL is divided by two. This bit corresponds to *N* in the STi5512 data sheet.
M[3:0]	*PLL Multiplication Factor, M+7:* These bits control the reference frequency input to the PLL of the clock generator. The actual multiplication factor is M + 7. This field corresponds to *M* in the STi5512 data sheet formulae.

10 Digital encoder (DEN)

10.1 Register descriptions

DEN_CFG0 Configuration0

	7	6	5	4	3	2	1	0
0x0	std1	std0	sync2	sync1	sync0	polh	polv	freerun

Address: DencBaseAddress + 0x0
Type: R/W
Reset value: 10010010

Description

Bitfield	Description
std[1:0]	Standard 0 0: PAL BDGHI 0 1: PAL N (see bit set-up) 1 0: NTSC M [1] [2] 1 1: PAL M
sync[2:1:0]	Configuration 0 0 0: ODDEV-only based SLAVE mode (frame locked) 0 0 1: F based SLAVE mode (frame locked) 0 1 0: ODDEV + HSYNC based SLAVE mode (line locked) 0 1 1: F+ H based SLAVE mode (line locked) 1 0 0: VSYNC-only based SLAVE mode (frame locked) [3] 1 0 1: VSYNC + HSYNC based SLAVE mode (line locked) 1 1 0: MASTER mode 1 1 1: AUTOTEST mode (color bar pattern)
polh	Synchro: active edge of HSYNC selection (when input) or polarity of HSYNC (when output) 0: HSYNC is a negative pulse (128 Tckref wide) or falling edge is active 1: HSYNC is a positive pulse (128 Tckref wide) or rising edge is active
polv	Synchro: active edge of ODDEV/VSYNC selection (when input) 0: Falling edge of ODDEV flags start of field1 (odd field) or VSYNC is active low 1: Rising edge of ODDEV flags start of field1 (odd field) or VSYNC is active high
freerun	Bit “freerun” is taken into account in ODDEV-only, VSYNC-only or F-based slave modes. It is irrelevant to other synchronization modes. Synchro: active edge of ODDEV/VSYNC selection (when input) 0: Disabled 1: Enabled

1. The standard on hardware reset is NTSC; any standard modification automatically selects the right parameters for correct subcarrier generation.
2. If bit ‘secam’ from register DEN_CFG7 is set, ‘std1’ and ‘std0’ bits are not taken into account.
3. In VSYNC-only based slave mode (sync[2:0]=”100”), HSYNC is nevertheless needed as an input.

DEN_CFG1 Configuration1

	7	6	5	4	3	2	1	0
0x01	blkli	flt1	flt0	syncok	coki	setup	cc2	cc1

Address: DencBaseAddress + 0x01
Type: R/W
Reset value: 01000100

Description

Bitfield	Description
cc[2:1]	00:[2] No closed caption/extended data encoding 01: Closed caption/extended data encoding enabled in field 1 (odd) 10: Closed caption/extended data encoding enabled in field 2 (even) 11: Closed caption/extended data encoding enabled in both fields
setup	Pedestal enable 0:[2] Blanking level and black level are identical on all lines (ex: Arg. PAL-N, Japan NTSC-M, PAL-BDGHI) 1: Black level is 7.5 IRE above blanking level on all lines outside VBI (ex: Paraguayan and Uruguayan PAL-N). In all cases, gain factor is adjusted to obtain the required chrominance levels.
coki	Color killer 0:[2] Color ON 1: Color suppressed on CVBS output signal (CVBS=YS) but color still present on C output For color suppression on chroma DAC 'C', see register DEN_CFG5.bkdac2.
syncok	Sync signal availability (analog & digital) for input synchronization loss with free-run inactive freerun=0 0:[2] No synchro output signals 1: Output synchros available on YS, CVBS and, when applicable, HSYNC (if output port), ODDEV (if output port),: i.e same behavior as free-run except that video outputs are blanked in the active portion of the line
flt[1:0]	U/V Chroma filter bandwidth selection. Typical applications for 3dB bandwidth are given for each flt bit configuration. 00: f_{-3dB}=1.1MHz low def NTSC filter 01: f_{-3dB}=1.3MHz low def PAL filter 10: [2] f_{-3dB}=1.6MHz high def. NTSC filter (ATSC compliant) & PAL M/N (ITU-R 624.4 compliant) 11: f_{-3dB}=1.9MHz high def. PAL filter: Rec 624 - 4 for PAL BDG/I compliant
blkli [1]	Vertical Blanking Interval selection for active video lines area. 0: [2] ('partial blanking') Only the following lines inside the vertical interval are blanked: NTSC-M: lines [1..9], [263(half)..272] (525-SMPTE), PAL-M: lines [523..6], [260(half)..269] (525-CCIR) other PAL: lines [623(half)..5], [311..318] (625-CCIR) This mode preserves embedded VBI data within the incoming YCrCb, e.g. Teletext (lines [7..22] and [320..335]), Wide Screen signalling (full line 23), Video Programing Service (line16), etc.). 1: ('full blanking') All lines inside VBI are blanked NTSC-M: lines [1..19], [263(half)..282] (525-SMPTE), PAL-M: lines [523..16], [260(half)..279] (525-CCIR) other PAL: lines [623(half)..22], [311..335] (625-CCIR)

1. **blkli** must be set to '0' when closed captions are to be encoded on the following lines:
   in 525/60 system: before line 20(SMPTE) or before line 283(SMPTE)
   in 625/50 system: before line 23(CCIR) or before line 336(CCIR)
2. DEFAULT mode when denc_nrst pin is active (LOW level)

DEN_CFG2 Configuration2

	7	6	5	4	3	2	1	0
0x02	nintrl	enrst	bursten	set4:4:4	selrst	rstosc_buf	valrst1	valrst0

Address: *DencBaseAddress* + 0x02
Type: R/W
Reset value: 00100000

Description

Bitfield	Description
valrst[0:1] [1]	00:[2] Automatic reset of the oscillator every line 01: Automatic reset of the oscillator every 2^{nd} field 10: Automatic reset of the oscillator every 4^{th} field 11: Automatic reset of the oscillator every 8^{th} field
rstosc_buf [3]	Software phase reset of DDFS (Direct Digital Frequency Synthesizer) buffer 0:[2] No reset 1: When a 0-to-1 transition occurs, either the hard-wired default phase value, or the value loaded in the DEN_PDFS1/2 register (according to bit 'selrst'), is put to phase buffer. This value is loaded into accumulator (phase of sub-carrier) when bits **ph_rst_osc** from DEN_CFG8 are programmed, or when the standard changes or softreset occurs.
selrst	Selects reset values for Direct Digital Frequency Synthesizer 0:[2] Hardware reset values for subcarrier oscillator phase (see DEN_PDFS1/2 register descriptions for values) 1: Loaded reset values selected (see contents of the DEN_PDFS1/2 registers)
set4:4:4	Selects the 4:4:2 or 4:4:4 video input from the mixing unit 0:[2] YCrCb (4:2:2) input : The presence of OSD on the DENC output can be selected or deselected by OSD block header field S. 1: YCrCb (4:4:4) input : In this mode OSD is always present on the DENC output.
bursten	Chrominance burst control 0: Burst is turned off on CVBS, chrominance output is not affected 1:[2] Burst is enabled
enrst	Cyclic update of DDFS phase 0:[2] No cyclic subcarrier phase reset 1: Cyclic subcarrier phase reset depending of valrst1 and valrst0 (see below)
nintrl [4]	Non-interlaced mode select 0:[2] Interlaced mode (625/50 or 525/60 system) 1: Non-interlaced mode(2x312/50 or 2x262/60 system)

1. **valrst[1:0]** is taken into account only if bit 'enrst' is set. Resetting the oscillator means here forcing the value of the phase accumulator to its nominal value to avoid accumulating errors due to the finite number of bits used internally. The value to which the accumulator is reset is either the hard-wired default phase value or the value loaded in the DEN_PDFS1/2 registers (according to bit 'selrst'), to which a 0^0, 90^0, 180^0, or 270^0 correction is applied according to the field and line on which the reset is performed.Note: If SECAM is performed the oscillator is reset every line.
2. Default mode when **denc_nrst** pin is active (LOW level)
3. **rstosc_buf** is automatically set back to '0' after the buffer is loaded
4. **nintrl** update is internally taken into account at the beginning of the next frame. In SECAM mode, only the interlaced mode is available.

DEN_CFG3 Configuration3

	7	6	5	4	3	2	1	0
0x03	entrap	trap_4.43	encgms	ck_in_phase	del2	del1	del0	enwss

Address: DencBaseAddress + 0x03
Type: R/W
Reset value: 00000000

Description

Bitfield	Description
enwss	WSS encoding enable 0:[4] Disabled 1: Enabled
del[2:0]	Delay on luma path with reference to chroma path on 4:2:2 inputs 0 1 0: + 2 pixel delay on luma 0 0 1: + 1 pixel delay on luma 0 0 0: [4] + 0 pixel delay on luma 1 1 1: - 1 pixel delay on luma 1 1 0: - 2 pixel delay on luma Other configurations: + 0 pixel delay on luma
ck_in_phase [1]	Choice of active edge of **denc_ref_ck** (master clock) that samples incoming YCrCb data. This bit can be used to analyze the cause of application synchronization problems. 0:[4] **denc_ref_ck** falling edge 1: **denc_ref_ck** rising edge
encgms [2]	CGMS encoding enable 0:[4] Disabled 1: Enabled
trap_4.43	Enable trap filter: Trap_4.43 is taken into account only if bit **entrap** is set 0:[4] Select the trap filter centered around 3.58 MHz 1: Select the trap filter centered around 4.43 MHz
entrap [3]	Enable trap filter 0:[4] Trap filter disabled 1: Trap filter enabled

1. This bit is typically used when synchronization problems appear in application.
2. When encgms is set to 1, Closed-Captions/Extended Data Services should not be programmed on lines 20 and 283 (525/60, SMPTE line number convention).
3. When SECAM is performed trap filter is always enabled (entrap ='1').
4. Default mode when **denc_nrst** pin is active (LOW level)

DEN_CFG4 Configuration4

	7	6	5	4	3	2	1	0
0x04	syncin_ad1	syncin_ad0	syncout_ad1	syncout_ad0	aline	del4_[2:0]		

Address: *DencBaseAddress* + 0x04
Type: R/W
Reset value: 00000000

Description

Bitfield	Description
del4[2:0]	Delay on luma path w.r.t chroma path on YUV and RGB outputs when 4:4:4 input is used 0 1 0: + 2 pixel delay on luma 0 0 1: + 1 pixel delay on luma 0 0 0: [4] + 0 pixel delay on luma 1 1 1: - 1 pixel delay on luma 1 1 0: - 2 pixel delay on luma Other configurations: + 0 pixel delay on luma
aline	Video active line duration control 0:[1] Full digital video line encoding (720 pixels - 1440 clock cycles) 1: Active line duration follows ITU-R/SMPTE ‘analog’ standard requirements
syncout_ad[1:0]	Adjustment of outgoing sync signals. Used to ensure correct interpretation of incoming video samples as Y, Cr or Cb when the encoder is master and supplies sync signals 00:[1] Nominal 01: +1 ckref 10: +2 ckref 11: +3 ckref
syncin_ad[1:0]	Adjustment of incoming sync signals. Used to ensure correct interpretation of incoming video samples as Y, Cr or Cb when the encoder is slaved to incoming sync signals (incl. ‘F/H’ flags stripped off ITU-R656/D1 data) 00:[1] Nominal 01: +1 ckref 10: +2 ckref 11: +3 ckref

1. Default mode when **denc_nrst** pin is active (LOW level)

DEN_CFG5 Configuration5

	7	6	5	4	3	2	1	0
0x05	selrst_inc	bkdac1	bkdac2	bkdac3	bkdac4	bkdac5	bkdac6	dacinv

Address: DencBaseAddress + 0x05
Type: R/W
Reset value: 00000000

Description

Bitfield	Description
dacinv	‘Inverts’ DAC codes to compensate for an inverting output stage in the application 0:[1] Non inverted DAC inputs (outputs) 1: Inverted DAC inputs (outputs)
bkdacN	Blanking of DACs (N = 1,2,3,4,5 or 6) 0:[1] DAC N in normal operation 1: DAC N input code forced to black level (if RGB, UV or C) or blanking level (if Y or CVBS) depending on conf_out bits of the DEN_CFG8 register.
selrst_inc	Choice of Digital Frequency Synthesizer increment after soft-reset or when ph_rst_mode = ‘01’ (see the DEN_CFG8 register) 0:[1] Hard wired value (depending on TV standard) 1: Soft (value from DEN_IDFS1/2/3 registers)

1. Default mode when **denc_nrst** pin is active (LOW level)

DEN_CFG6 Configuration6

	7	6	5	4	3	2	1	0
0x06	softreset	jump	dec_ninc	free_jump	cfc1	cfc2	ttx_en	maxdyn

Address: DencBaseAddress + 0x06
Type: R/W
Reset value: 00010000

Description

Bitfield	Description	
max_dyn[1]	Max dynamic magnitude allowed on YCrCb inputs for encoding	
	0:[5]	10hex to EBhex for Y, 10hex to E0hex for chrominance (Cr,Cb)
	1:	01hex to FEhex for Y, Cr and Cb
ttx_en	Teletext enable bit	
	0:[5]	Disabled
	1:	Enabled
cfc[1:0]	Color frequency control via CFC line	
	0 0:[5]	Update mode disabled (update is carried out by loading DEN_IDFS1/2/3 registers)
	0 1:	Update of increment for DDFS just after serial loading via CFC
	1 0:	Update of increment for DDFS on next active edge of HSYNC
	1 1:	Update of increment for DDFS just before next color burst
jump: [2] dec_ninc: free_jump (bits[6:5:4])	Update mode	
	0 0 0:	Normal mode (no line skip/insert capability) ITU-R (CCIR): 313/312 or 263/262 non-interlaced: 312/312 or 262/262
	0 x 1: [5]	Manual mode for line insert (“dec_ninc”=0) or skip (“dec_ninc”=1) capability. Both fields of all the frames following the writing of this value are modified according to “lref” and “ltar” bits of DEN_LJPM1/2/3 registers (by default, “lref”=0 and “ltar”=1 giving the normal mode above).
	1 0 0:	Automatic line insert mode. The 2nd field of the frame following the writing of this value is increased. Line insertion is done after line 245 in 525/60 and after line 290 in 625/50. “lref” and “ltar” are ignored.[3]
	1 1 0	Automatic line skip mode. The 2nd field of the frame following the writing of this value is decreased. Line suppression is done after line 245 in 525/60 and after line 290 in 625/50. “lref” and “ltar” are ignored.[3]
	1 x 0:	DO NOT USE!
softreset [4]	Software reset	
	0:[5]	No reset
	1:	Software reset

1. EAV and SAV words are replaced by blanking values before being fed to the luminance and chrominance processing.
2. Bit jump is automatically reset after use.
3. Two lines are skipped (inserted) in 525/60 and four lines in 625/50 standards.
4. Bit **softreset** is automatically reset after internal reset generation. Software reset is active for 4 CKREF periods. When softreset is activated, all the device is reset as with hardware reset except for the first nine user registers (DEN_CFG0 to DEN_CFG8).
5. Default mode when denc_nrst pin is active (LOW level)

DEN_CFG7 Configuration7

	7	6	5	4	3	2	1	0
0x07	secam	gen_secam	inv_phi_secam	not used	setupYUV	uv_lev	envps	sqpix

Address: DencBaseAddress + 0x07
Type: R/W
Reset value: 00000000

Description

Bitfield	Description
sqpix	Square pixel mode enable: This mode is not available in secam standard. 0:[1] Disable 1: Enable
envps	VPS encoding enable 0:[1] Disable 1: Enable
uv_lev	UV output level 0:[1] Same peak to peak amplitude for both U and V (default value - 0.7 Vpp for 100/0/100/0 color bar pattern) 1: U and V outputs as defined by ITU-R624-4
setupYUV	Pedestal enable on YUV outputs: This bit is significant only if register DEN_CFG8 bit conf_out(1:0) = ‘01’ (YUV outputs). If this is the case, the Y output on dac5_g_y has a 7.5 IRE pedestal. Furthermore, if uv_lev = ‘1’ then the U and V levels also depend on the value of setupYUV. 0:[1] Disable 1: Enable
inv_phi_secam	Inversion of sub-carrier phase 0:[1] 0, 0, π,... in odd fields and π, π, 0 in even fields 1: π, π, 0 in odd fields and 0, 0, π,... in even fields
gen_secam /secam	Where the bit order is secam - gen_secam: 00[1]: Standard selected by std1 and std0 bits of DEN_CFG0 (PAL or NTSC) 01: Genlock slave mode using external sync extraction 10: Secam standard, the sub-carrier phase sequence start-point is line1 11: Secam standard, the sub-carrier phase sequence start-point is line23 If master mode is performed in the secam standard, bit “secam” must be set before sync2, sync1 and sync0 bits of DEN_CFG0. If DEN_CFG0 is not programmed, then a soft reset must be performed after programming secam. In secam, the trap filter should always be enabled (see DEN_CFG3 register).

1. Default value

DEN_CFG8 Configuration8

	7	6	5	4	3	2	1	0
0x08	ph_rst_mode1	ph_rst_mode0	conf_out1	conf_out0	blk_all	ttx_notmv	xxx	xxx

Address: DencBaseAddress + 0x08
Type: R/W
Reset value: 00100000

Description

<table>
<tr><th>Bitfield</th><th>Description</th></tr>
<tr><td>ttx_notmv</td><td>WARNING! After reset, the default value of this bit is zero, however, for devices using teletext, this bit must then be reprogrammed to 1. This is to avoid the occurrence of teletext glitches in Secam mode and loss of teletext data in all modes.

Priority of ancillary data on a VBI line. Note, higher priority data overwrites lower priority data.
0[1] Priority is: CGMS > Closed caption > Macrovision[2] > WSS > VPS > Teletext
1 Priority is: Teletext > CGMS > Closed caption > WSS > VPS > Macrovision</td></tr>
<tr><td>blk_all</td><td>Blanking of all video lines
0[1] Disabled
1 Enabled (all inputs ignored - 80hex instead of Cr and Cb and 10hex instead of Y and Y4)</td></tr>
<tr><td>conf_out[1:0]</td><td>DENC output configuration
<table>
<tr><th></th><th></th><th>dac1</th><th>dac2</th><th>dac3</th><th>dac4</th><th>dac5</th><th>dac6</th></tr>
<tr><td>0</td><td>0</td><td>Y</td><td>C</td><td>CVBS</td><td>C</td><td>Y</td><td>CVBS</td></tr>
<tr><td>0</td><td>1</td><td>Y</td><td>C</td><td>CVBS</td><td>V</td><td>Y</td><td>U</td></tr>
<tr><td>1</td><td>x[1]</td><td>Y</td><td>C</td><td>CVBS</td><td>R</td><td>G</td><td>B</td></tr>
</table>
If conf_out(1:0) = ‘01’ and register DEN_CFG2 bit set_4:4:4 = ‘1’, then the dac5 output Y comes from the Y4 input.</td></tr>
<tr><td>ph_rst_mode[1:0]</td><td>Sub-carrier phase reset mode.
In modes “01” and “10”, this bit is automatically reset to “00” after oscillator reset.
00 [1] Disabled
01 Enabled - phase is updated with value from phase buffer register (see DEN_CFG2 bit rstosc_buf) on the beginning of the next video line. Increment is updated with hard or soft value depending on selreg_inc value (see DEN_CFG5)
10 Enabled - phase is updated with value from the DEN_IDFS1/2/3 registers on the next increment updating by cfc (depending on cfc loading moment and DEN_CFG6 cfc(1:0) bits.
11 Enabled - phase is reset after detecting of rst bit on cfc line, up to 9 denc_ref_ck after loading of cfc’s LSB.</td></tr>
</table>

1. Default value
2. If there is no Macrovision programmed on the line, then VBI line blanking (when register DEN_CFG1 bit blki=1) will overwrite VPS or teletext, and no VPS or teletext data will be encoded.

DEN_STA Status (read only)

	7	6	5	4	3	2	1	0
0x09	hok	atfr	buf2_free	buf1_free	fieldct2	fieldct1	fieldct0	jump

Address: DencBaseAddress + 0x09
Type: R0
Reset value: Undefined

Description

Bitfield	Description
jump	Indicates whether a frame length modification has been programmed at '1' from programming of bit 'jump' to end of frame(s) concerned default = 0 Refer to DEN_CFG6 and DEN_LJPM1/2/3 registers
fieldct[2:0]	Digital field identification number 000 Indicates field 1 ... 111 Indicates field 8 NOTE: fieldct[0] also represents the odd/even information (odd='0', even='1')
buf1_free	Closed caption registers access condition for field 1 Same as buf2_free but concerns field 1. Reset value:= 1 (access authorized)[2]
buf2_free	Closed caption registers access condition for field 2 Closed caption data for field 2 is buffered before being output on the relevant TV line; buf2_free is reset if the buffer is temporarily unavailable. If the microcontroller can guarantee that the DEN_CCF2 registers are never written more than once between two frame reference signals, then bit 'buf2_free' will always be true (set). Otherwise, closed caption field2 registers (DEN_CCF2) access might be temporarily forbidden by resetting bit 'buf2_free' until the next field2 closed caption line occurs. Note that this bit is false (reset) when 2 pairs of data bytes are awaiting to be encoded, and is set back immediately after one of these pairs has been encoded (so at that time, encoding of the last pair of bytes is still pending) Reset value:= 1 (access authorized)[2]
atfr	Frame synchronization flag 0[2] Encoder not synchronized 1 In slave mode: encoder synchronized
hok[1]	Hamming decoding of frame sync flag embedded within ITU-R656 / D1 compliant YCrCb streams 0 Consecutive errors 1[2] A single or no error

1. Signal quality detector is issued from Hamming decoding of EAV,SAV from YCrCb
2. DEFAULT mode when denc_nrst pin is active (LOW level)

DEN_IDFS1/2/3 Increment for digital frequency synthesizer

These three registers contain the 24-bit increment used by the DDFS, **ONLY IF** bit ‘selrst_inc’ (register DEN_CFG5) equals ‘1’. They generate the phase of the subcarrier, i.e. the address that is supplied to the sine ROM. It, therefore, customizes the synthesized subcarrier frequency: 1 LSB ~ 1.6 Hz

To validate use of these registers instead of the hard-wired values:

- Load the registers with the required value
- Set bit ‘selrst_inc’ to 1 (register DEN_CFG5)
- Perform a software reset (register DEN_CFG6)

Note *The values loaded in DEN_IDFS1/2/3 are taken into account after a software reset, and* **ONLY IF** *bit* ‘selrst_inc*’=’1’ (*register DEN_CFG5*)*
These registers are never reset and must be explicitly written, to ensure that they contain sensible information.

On hardware or software reset the DDFS is initialized with a hardwired increment, independent of DEN_IDFS1/2/3. These hardwired values cannot be read out of the DENC.

	d23	d22	d21	d20	d19	d18	d17	d16
0x0A								
	d15	d14	d13	d12	d11	d10	d9	d8
0x0B								
	d7	d6	d5	d4	d3	d2	d1	d0
0x0C								

Address: DencBaseAddress+ 0x0A(DEN_IDFS1), 0x0B (DEN_IDFS2), 0x0C(DEN_IDFS3)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description		
DEN_IDFS1/2/3	Value	Frequency synthesized	Ref.Clock
	d(23:0): 21F07C hexa for NTSC M	f=3.5795452 MHz	27 MHz
	d(23:0): 2A098B hexa for PAL BGHIN	f=4.43361875MHz	27 MHz
	d(23:0): 21F694 hexa for PAL N	f=3.5820558 MHz	27 MHz
	d(23:0): 21E6F0 hexa for PAL M	f=3.57561149 MHz	27 MHz
	d(23:0): 255554 hexa for NTSC M square pixel	f=3.5795434 MHz	24.545454 MHz
	d(23:0): 26798C hexa for PAL BGHIN square pixel	f=4.43361867 MHz	29.5 MHz
	d(23:0): 1F15C0 hexa for PAL N square pixel	f=3.58205605 MHz	29.5 MHz
	d(23:0): 254AD4 hexa for PAL M square pixel	f=3.57561082 MHz	24.545454 MHz

DEN_PDFS1/2 Phase_dfs

These registers are never reset and must be explicitly written to.

If bit 'selrst'=0 (e.g. after a hardware reset) the phase offset used every time the DDFS is reinitialized is a hard-wired value. The hard-wired values *cannot be read out of the DENC*. These are:

D9C000hex for PAL BDGHI, N, M,
1FC000hex for NTSC-M,
000000hex (blue lines)
43C000hex (red lines) for SECAM

When secam = '0' (DEN_CFG8)

Static phase offset for digital frequency synthesizer (10 bits only)

	-	-	-	-	-	-	o23	o22
0x0D								
	o21	o20	o19	o18	o17	o16	o15	o14
0x0E								

Address: DencBaseAddress + 0x0D (DEN_PDFS1), 0x0E (DEN_PDFS2)
Type: R/W
Reset value: Undefined

Under certain circumstances (detailed below), these registers contain the 10 MSBs of the value with which the phase accumulator of the DDFS is initialized after a 0-to-1 transition of bit 'rstosc' (DEN_CFG2), or after a standard change, or when cyclic phase readjustment has been programmed (see bits valrst[1:0] of DEN_CFG2). The 14 remaining LSBs loaded into the accumulator in these cases are all '0's (defining the phase reset value with a 0.35^{o} accuracy).

To validate use of these registers instead of the hard-wired values:

- Load the registers with the required value
- Set bit 'selrst' to 1 (DEN_CFG2)
- Perform a software reset (DEN_CFG6), or set DEN_CFG6 bit rstosc_buf to "1". This puts the soft-phase value into a tampon register and sets register DEN_CFG8 bit ph_rst_mode[1:0] to put this value into an accumulator at the beginning of the next line.

When secam = '1' (DEN_CFG8)

Static phase offset for digital frequency synthesizer for two SECAM sub-carriers (8 MSB only) (blue lines - DEN_PDFS1 *and red lines -* DEN_PDFS2*)*

	b21	b20	b19	b18	b17	b16	b15	b14
0x0D								
	r21	r20	r19	r18	r17	r16	r15	r14
0x0E								

Address: DencBaseAddress + 0x0D (DEN_PDFS1), 0x0E (DEN_PDFS2),
Type: R/W
Reset value: Undefined

These registers contain the 8 bits (21 to14) of the value with which the phase accumulator of the DDFS is initialized on **every line** in SECAM mode. The phase is calculated with 1.4° accuracy as:

(Blue lines) DEN_PDFS1 * 16384 (dec), or DEN_PDFS1 * 4000 (hex)

(Red lines) (256 + DEN_PDFS1 + DEN_PDFS2) * 16384 (dec), or (100 + DEN_PDFS1 + DEN_PDFS2) * 4000 (hex)

To validate use of these registers instead of the hard-wired values, follow the PAL and NTSC mode.

DEN_WSS1/2 WSS_BIT[15:0]: WSS data registers

	15	14	13	12	11	10	9	8
0x0F	wss15	wss14	wss13	wss12	wss11	wss10	wss9	wss8
	7	6	5	4	3	2	1	0
0x10	wss7	wss6	wss5	wss4	wss3	wss2	wss1	wss0

Address: DencBaseAddress + 0x0F(DEN_WSS1), 0x10 (DEN_WSS2)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description
wss[0]	0: No subtitles within teletext 1: Subtitles within teletext
wss[2:1]	0 0: No open subtitles 0 1: Subtitles in active image area 1 0: Subtitles out of image area 1 1: reserved
wss[7:3]	reserved = 0
wss[11:8] [1]	1 0 0 0: full format 4:3 0 0 0 1: box 14:9 center 0 0 1 0: box 14:9 top 1 0 1 1: box 16:9 center 0 1 0 0: box 16:9 top 1 1 0 1: > box 16:9 center 1 1 1 0: full format 4:3 (shoot and protect 14:9 center) 0 1 1 1: full format 16:9 (anamorphic)
wss[12]	0: Camera mode 1: Film mode
wss[15:13]	reserved = 0

1. wss11 is an odd parity bit

DEN_DAC13 Dac1 and dac3 multiplying factors

	7	6	5	4	3	2	1	0
0x11	dac1_mult[3:0]				dac3_mult[3:0]			

Address: DencBaseAddress + 0x11
Type: R/W
Reset value: 10001000

Description

Bitfield	Description
dac3_mult[3:0]	multiplying factor on dac3_cvbs digital signal before the D/A converters with 3.125% step 0 0 0 0: 75.00% dac3_cvbs value compared to default 0 0 0 1: 78.125% dac3_cvbs value compared to default 0 0 1 0: 81.25% dac3_cvbs value compared to default 0 0 1 1: 84.375% dac3_cvbs value compared to default .. .. 1 0 0 0: [1] 100% dac3_cvbs value compared to default .. .. 1 1 1 1: 121.875% dac3_cvbs value compared to default
dac1_mult[3:0]	multiplying factor on dac1_y digital signal before the D/A converters with 3.125% step 0 0 0 0: 75.000% dac1_y value compared to default 0 0 0 1: 78.125% dac1_y value compared to default 0 0 1 0: 81.250% dac1_y value compared to default 0 0 1 1: 84.375% dac1_y value compared to default .. .. 1 0 0 0: [1] 100% dac1_y value compared to default .. .. 1 1 1 1: 121.875% dac1_y value compared to default

1. Default value

DEN_DAC45 DAC4 and DAC5 multiplying factors

	7	6	5	4	3	2	1	0
0x12	dac4_mult[3:0]				dac5_mult[3:0]			

Address: DencBaseAddress + 0x12
Type: R/W
Reset value: 10001000

Description

Bitfield	Description		
dac5_mult[3:0]	Multiplying factor on dac5_g_y digital signal before the D/A converters: dac5_g_y value (in% of default value)		
	Value	if G(conf_out=1x)	if Y(conf_out=0x)
	0 0 0 0:	80.49%	75.000%
	0 0 0 1:	82.93%	78.125%
	0 0 1 0:	85.37%	81.250%
	0 0 1 1:	87.81%	84.375%
	.. ..	...	...
	1 0 0 0: [1]	100%	100%
	.. ..		
	1 1 1 1:	117.07%	121.875%
dac4_mult[3:0]	Multiplying factor on dac4_r_v_c digital signal before the D/A converters: dac4_r_v_c value (in% of default value)		
	Value	if R (conf_out=1x)	if V or C (conf_out=0)
	0 0 0 0:	80.49%	75.000%
	0 0 0 1:	82.93%	78.125%
	0 0 1 0:	85.37%	81.250%
	0 0 1 1:	87.81%	84.375%
	.. ..	...	...
	1 0 0 0: [1]	100%	100%
	.. ..		
	1 1 1 1:	117.07%	121.875%

1. DEFAULT mode when denc_nrst pin is active (LOW level)

DEN_DAC6C DAC6 and C multiplying factors

	7	6	5	4	3	2	1	0
0x13	dac6_mult[3:0]				c_mult[3:0]			

Address: DencBaseAddress + 0x13
Type: R/W
Reset value: 10000000

Description

Bitfield	Description
c_mult[3:0]	Multiplying factor on C digital output (before D/A convertors) and on color part of CVBS signal: Value Factor value (c_mult) 0 0 0 0: 1.000000 (1.000000 dec) 0 0 0 1: 1.000001 (1.015625 dec) 0 0 1 0: 1.000010 (1.031250 dec) 0 0 1 1: 1.000011 (1.046875 dec) .. .. 1 1 1 1: 1.001111 (1.234375 dec)
dac6_mult[3:0]	Multiplying factor on dac6_b_cvbs digital signal before the D/A converters: dac6_b_cvbs value (in% of default value) Value If B(conf_out=1x) If CVBS (conf_out=01) 0 0 0 0: 80.49% 75.000% 0 0 0 1: 82.93% 78.125% 0 0 1 0: 85.37% 81.250% 0 0 1 1: 87.81% 84.375% .. .. 1 0 0 0: [1] 100% 100% 1 1 1 1: 117.07% 121.875%

Default peak to peak amplitude of U and V outputs corresponds to 70% of default Y or CVBS peak to peak amplitude if 100/0/100/0 color bar pattern is inputted. In other words, when Iref is set to deliver 1Vpp for CVBS on DAC3 for example (and dac3_mult = “1000”), when switched to U, DAC6 delivers 0.7 Vpp. If bit ‘uv_lev’ from register DEN_CFG7 is set, default peak to peak amplitude is 86% (80% if setupYUV=’1’) for V output and 67% (57% if setupYUV=’1’) for U output, of default Y or CVBS peak to peak amplitude if 100/0/100/0 color bar pattern is inputted, according to ITU-R 624-4 definition of UV signals.

Default peak to peak amplitude of RGB outputs corresponds to 70% of default Y or CVBS peak to peak amplitude if 100/0/75/0 color bar pattern is inputted. In other words, when Iref is set to deliver 1Vpp for CVBS on DAC3 for example (and dac3_mult = “1000”), when switched to B, DAC6 delivers 0.7 Vpp.

DEN_LJPM1/2/3 CLIG_I_REG = ltarg[8:0] and lref[8:0]

	7	6	5	4	3	2	1	0
0x15	ltarg8	ltarg7	ltarg6	ltarg5	ltarg4	ltarg3	ltarg2	ltarg1
0x16	ltarg0	lref8	lref7	lref6	lref5	lref4	lref3	lref2
0x17	lref1	lref0	-	-	-	-	-	-

Address: DencBaseAddress+ 0x15(DEN_LJPM1), 0x16(DEN_LJPM2), 0x17(DEN_LJPM3)
Type: R/W
Reset value: Undefined

These registers can be used to jump from a reference line (end of that line) to a target line of the SAME FIELD. However, not all lines can be skipped or repeated without problems. This functionality should be USED WITH CAUTION.

Bitfield	Description
lref[8:0]	Binary format of the reference line from which a jump is required. Default value: lref[8:0]:= 000000000
ltarg[8:0]	Binary format of the target line. Default value: ltarg[8:0]:= 000000001

DEN_CID CHIPID (read only): DENC version identification number

	7	6	5	4	3	2	1	0
0x18	CHIPID							

Address: DencBaseAddress + 0x18
Type: RO
Reset value: Undefined

Description

Bitfield	Description
CHIPID	1001 0000 (for V9, 1000 0000 for V8)

DEN_VPS1 VPS: VPS Data registers

	7	6	5	4	3	2	1	0
0x19	s1	s0	xxx	xxx	cni3	cni2	cni1	cni0
0x1A	np7	np6	d4	d3	d2	d1	d0	m3
0x1B	m2	m1	m0	h4	h3	h2	h1	h0
0x1C	min4	min3	min3	min2	min1	min0	c3	c2
0x1D	c1	c0	np5	np4	np3	np2	np1	np0
0x1E	pt7	pt6	pt5	pt4	pt3	pt2	pt1	pt0

Address: DencBaseAddress + 0x19(DEN_VPS1), 0x1A(DEN_VPS2), 0x1B(DEN_VPS3), 0x1C(DEN_VPS4), 0x1D(DEN_VPS5), 0x1E(DEN_VPS6)

Type: R/W

Reset value: Undefined

Description

Bitfield	Description
pt[7:0]	Program type, binary
np[5:0]	Network or program CNI (Country and Network Identification)
c[3:0]	Country, binary
min[5:0]	Minute, binary
h [4:0]	Hour, binary
m[3:0]	Month, binary
d[4:0]	Day, binary
np[7:6]	Network or program CNI
cni[3:0]	4 bits of CNI Reserved for enhancement of VPS
S[1:0]	Sounds 0 0: don’t know 0 1: mono 10: stereo 1 1: dual sound

DEN_CGMS1/2/3 CGMS_BIT[1:20]: CGMS data registers (20 bits only)

	7	6	5	4	3	2	1	0
0x1F	-	-	-	-	b1	b2	b3	b4
0x20	b5	b6	b7	b8	b9	b10	b11	b12
0x21	b13	b14	b15	b16	b17	b18	b19	b20

Address: DencBaseAddress + 0x1F (DEN_CGMS1), 0x20 (DEN_CGMS2), 0x21 (DEN_CGMS3)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description
b15 - b20	CRC (not internally computed)
b11 - b14	Word2
b7 - b10	Word1
b4 - b6	Word0B
b1 - b3	Word0A

DEN_TTX1-4/M TTX__[1:4]_DEF Teletext block definition

	7	6	5	4	3	2	1	0
0x22	fp_ttxt	ttxdel2	ttxdel1	ttxdel0	ttx_l6	ttx_l7	ttx_l8	ttx_l9
0x23	ttx_l10	ttx_l11	ttx_l12	ttx_l13	ttx_l14	ttx_l15	ttx_l16	ttx_l17
0x24	ttx_l18	ttx_l19	ttx_l20	ttx_l21	ttx_l22	ttx_l23	ttx_l318	ttx_l319
0x25	ttx_l320	ttx_l321	ttx_l322	ttx_l323	ttx_l324	ttx_l325	ttx_l326	ttx_l327
0x26	ttx_l328	ttx_l329	ttx_l330	ttx_l331	ttx_l332	ttx_l333	ttx_l334	ttx_l335

Address: DencBaseAddress + 0x22(DEN_TTX1), 0x23(DEN_TTX2), 0x24(DEN_TTX3), 0x25(DEN_TTX4), 0x26(DEN_TTXM)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description
ttx_lX	Each of these bits enables teletext on line X (ITU-R line numbering and 625 line systems). Teletext line selections for other standards refer to the table below:
ttxdel[2:0]	Teletext data latency: The encoder clocks the first Teletext data sample on the (2+txdl[2:0])th rising edge of the master clock, following the rising edge of TTXS (teletext synchronization signal, supplied by the encoder). 1 0 0: default value
fp_ttxt	Full page teletext mode enable 0 0:[1] Disabled 0 1: Enabled

1. Default mode when denc_nrst pin is active (low level).

TTX__[1:4]_DEF bitfield	PAL BDGHIN line (CCIR 625 line numbering)	PAL M line (CCIR 525 line numbering)	NTSC M line (SMPTE 525 line numbering)
ttx_l6	6	6	9
ttx_l7	7	7	10
ttx_l8	8	8	11
ttx_l9	9	9	12
ttx_l10	10	10	13
ttx_l11	11	11	14
ttx_l12	12	12	15
ttx_l13	13	13	16
ttx_l14	14	14	17
ttx_l15	15	15	18
ttx_l16	16	16	19
ttx_l17	17	17	20
ttx_l18	18	18	21
ttx_l19	19	19	22
ttx_l20	20	20	23
ttx_l21	21	21	24
ttx_l22	22	22	25
ttx_l23	23	23	26
ttx_l318	318	268	271
ttx_l319	319	269	272
ttx_l320	320	270	273
ttx_l321	321	271	274
ttx_l322	322	272	275
ttx_l323	323	273	276
ttx_l324	324	274	277
ttx_l325	325	275	278
ttx_l326	326	276	279
ttx_l327	327	277	280

Table 32 Teletext block definition for PAL and NTSC standards

TTX__[1:4]_DEF bitfield	PAL BDGHIN line (CCIR 625 line numbering)	PAL M line (CCIR 525 line numbering)	NTSC M line (SMPTE 525 line numbering)
ttx_l328	328	278	281
ttx_l329	329	279	282
ttx_l330	330	280	283
ttx_l331	331	281	284
ttx_l332	332	282	285
ttx_l333	333	283	286
ttx_l334	334	284	287
ttx_l335	335	285	288

Table 32 Teletext block definition for PAL and NTSC standards

DEN_CCF1 CCCF1: closed caption characters/extended data for field 1

	7	6	5	4	3	2	1	0
0x27	opc11	c117	c116	c115	c114	c113	c112	c111
0x28	opc12	c127	c126	c125	c124	c123	c122	c121

Address: DencBaseAddress + 0x27 (DEN_CCF11), 0x28 (DEN_CCF12)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description
c11[7:1]	First byte to encode in field1
opc11	Odd-parity bit of US-ASCII 7-bit character c11[7:1]
c12[7:1]	Second byte to encode in field1
opc12	Odd-parity bit of US-ASCII 7-bit character c12[7:1]

- DEFAULT value: none, but closed captions enabling without loading these registers will issue a null character.
- DEN_CCF1 registers are never reset.

DEN_CCF2 CCCF2: closed caption characters/extended data for field 2

	7	6	5	4	3	2	1	0
0x29	opc21	c217	c216	c215	c214	c213	c212	c211
0x2A	opc22	c227	c226	c225	c224	c223	c222	c221

Address: DencBaseAddress + 0x29 (DEN_CCF21), 0x2A (DEN_CCF22)
Type: R/W
Reset value: Undefined

Description

Bitfield	Description
c21[7:1]	First byte to encode in field2
opc21	Odd-parity bit of US-ASCII 7-bit character c21[7:1]
c12[7:1]	Second byte to encode in field2
opc22	Odd-parity bit of US-ASCII 7-bit character c22(7:1)

- DEFAULT value: none, but closed captions enabling without loading these registers will issue a null character.
- DEN_CCF2 registers are never reset.

DEN_CLF1 CCLIF1: closed caption/extended data line insertion for field 1

This register programs the TV line number of the closed caption/extended data encoded in field 1:

	7	6	5	4	3	2	1	0
0x2B	xx	xx	xx	l14	l13	l12	l11	l10

Address: DencBaseAddress + 0x2B
Type: R/W
Reset value: 00001111

525/60 system: (525-SMPTE line number convention)

Only lines 10 to 22 should be used for closed caption or extended data services (lines 1 to 9 contain the vertical sync pulses with equalizing pulses).

Bitfield	Description	
l1[4:0]	0 0 0 0 0:	No line selected for closed caption encoding
	0 0 0 x x:	Do not use these codes
	....	
	i code line (i+6)	(SMPTE) selected for encoding
	....	
	11111:	line 37 (SMPTE) selected

625/50 system: (625-CCIR/ITU-R line number convention)

Only lines 7 to 23 should be used for closed caption or extended data services.

Bitfield	Description	
l1[4:0]	0 0 0 0 0:	No line selected for closed caption encoding
	....	
	i code line (i+6)	(CCIR) selected for encoding (i>0)
	....	
	11111:	line 37 (CCIR) selected

DEFAULT value = 01111 line 21 (525/60, 525-SMPTE line number convention), which corresponds to line 21 in l625/50 system,(625-CCIR line number convention)

DEN_CLF2 CCLIF2: closed caption/extended data line insertion for field 2

This register programs the TV line number of the closed caption/extended data encoded in field 1

	7	6	5	4	3	2	1	0
0x2C	xx	xx	xx	l24	l23	l22	l21	l20

Address: DencBaseAddress + 0x2C
Type: R/W
Reset value: 00001111

525/60 system: (525-SMPTE line number convention)

Only lines 273 to 284 should be used for closed caption or extended data services (preceding lines contain the vertical sync pulses with equalizing pulses), although it is possible to program over a wider range.

Bitfield	Description	
l2[4:0]	0 0 0 0 0:	No line selected for closed caption encoding
	0 0 0 x x:	Do not use these codes
	i line (269 +i)	(SMPTE) selected for encoding
	....	
	01111:	Line 284 (SMPTE) selected for encoding
	11111:	Line 289 (SMPTE)

Note *If cgms is allowed on lines 20 and 283 (525/60, 525-SMPTE line number convention), closed captions should not be programmed on these lines.*

625/50 system: (625-CCIR line number convention)

Only lines 319 to 336 should be used for closed caption or extended data services (preceding lines contain the vertical sync pulses with equalizing pulses), although it is possible to program over a wider range.

Bitfield	Description	
l2[4:0]	0 0 0 0 0:	No line selected for closed caption encoding
	i line (318 +i)	(CCIR) selected for encoding
	....	
	10010:	Line 336 (CCIR) selected for encoding
	11111:	Line 349 (CCIR)

DEFAULT value:= 01111 line 284 (525/60, 525-SMPTE line number convention) this value also corresponds to line 333 in 625/50 system, (625-CCIR line number convention)

DEN_REG 45...63 Reserved

DEN_REG_64 TTX_CONF

	7	6	5	4	3	2	1	0
0x40	ttxt100IRE	ttxt_abcd[1:0]		xxx	xxx	xxx	xxx	xxx

Address: DencBaseAddress + 0x40
Type: R/W
Reset value: 01010000

Description

Bitfield	Description
ttxt_abcd[1:0]	Teletext standard selection 0 0 Teletext A 0 1 Teletext B Teletext B in 625 line systems is known as World System Teletext 1 0 [1] Teletext C Teletext C in 525 line systems is known as NABTS 1 1 Teletext D
ttxt100IRE	Teletext waveform amplitude 0[1] 70 IRE 1 100 IRE

1. DEFAULT mode when denc_nrst pin is active (LOW level)

Note Do not change the value of bits 4 to 0.

DEN_REG_65 DAC2MULT&TTXS

	7	6	5	4	3	2	1	0
0x41	dac2_mult[3:0]				ttx_mask_off	xxx	bcs_en_4	bcs_en_2

Address: DencBaseAddress + 0x41
Type: R/W
Reset value: 10000010

Description

Bitfield	Description
bcs_en_2	Brightness, Contrast and Saturation control by registers DEN_REG_69 to DEN_REG_71 on 4:2:2 input 0:[1] Disable 1: Enable
bcs_en_4	Brightness, Contrast and Saturation control by registers DEN_REG_69 to DEN_REG_71 on 4:4:4 input 0: Disable 1:[1] Enable
ttx_mask_off	Masking of 2 bits in Teletext framing code, depending on selected teletext standard 0:[1] Enable 1: Disable
dac2_mult[3:0]	Multiplying factor on dac1_c digital signal before the D/A converters with 3.125% step Bitvalue dac2_c value compared to default 0 0 0 0: 75.000% 0 0 0 1: 78.125% 0 0 1 0: 81.250% 0 0 1 1: 84.375% ... ... 1 0 0 0:[1] 100% ... 1 1 1 1: 121.875%

1. DEFAULT mode when denc_nrst pin is active (LOW level)

DEN_REG_69 Brightness

	7	6	5	4	3	2	1	0
0x45	b[7:0]							

Address: DencBaseAddress + 0x45
Type: R/W
Reset value: 10000000

The register contents are used by the following formula to adjust the luminance intensity of the display video image:

$$Y_{out} = Y_{in} + b - 128$$

Where Yin is 8-bit input luminance and Yout is the result of 'Brightness' operation (still on 8 bits).

This value is saturated at 235 (16) or 254 (1) according to DEN_CFG6 bit "maxdyn", b: brightness (unsigned value with center at 128, default 128)

DEN_REG_70 Contrast

	7	6	5	4	3	2	1	0
0x46	c[7:0]							

Address: DencBaseAddress + 0x46
Type: R/W
Reset value: 00000000

The register contents are used by the following formula to adjust the relative difference between the display image higher and lower intensity luminance values:

$$Y_{out} = \frac{(Y_{in} - 128)(c + 128)}{128} + 128$$

Where, Yin is 8-bit input luminance, Yout is the result of 'Contrast' operation (still on 8 bits)

This value is saturated at 235 (16) or 254 (1) according to DEN_CFG6 bit 'maxdyn', c: contrast (2's complement value from -128 to 127, default 0)

DEN_REG_71 Saturation

	7	6	5	4	3	2	1	0
0x47	s[7:0]							

Address: DencBaseAddress + 0x47
Type: R/W
Reset value: 10000000

The register contents are used by the following formula to adjust the color intensity of the displayed video image:

$$Crout = \frac{s(Crin - 128)}{128} + 128$$

$$Cbout = \frac{s(Cbin - 128)}{128} + 128$$

Where Crin and Cbin are the 8-bit input chroma, Crout and Cbout are the result of 'Saturation' operation (still on 8 bits)

This value is saturated at 240 (16) or 254 (1) according to DEN_CFG6 bit 'maxdyn', s: saturation value (unsigned value with centre at 128, default 128).

11 External memory interface (EMI)

EMI_ConfigData0 EMI configuration data register 0

SDRAM/DRAM format

		15	14	13	12	11 10	9 8	7 6 5	4 3	2 1 0
Bank0	0x00			DataDrive Delay	BusRelease Time	Sub Banks	SubBank Size	ShiftAmount	PortSize	DeviceType
Bank1	0x10			DataDrive Delay	BusRelease Time	Sub Banks	SubBank Size	ShiftAmount	PortSize	DeviceType
Bank2	0x20			DataDrive Delay	BusRelease Time	Sub Banks	SubBank Size	ShiftAmount	PortSize	DeviceType
Bank3	0x30			DataDrive Delay	BusRelease Time	Sub Banks	SubBank Size	ShiftAmount	PortSize	DeviceType

SRAM/peripheral format

		15 14 13	12 11	10 9	8 7	6 5	4 3	2 1 0
Bank0	0x00	DataDriveDelay	BusRelease Time	CSActive	OEActive	BEActive	PortSize	DeviceType
Bank1	0x10	DataDriveDelay	BusRelease Time	CSActive	OEActive	BEActive	PortSize	DeviceType
Bank2	0x20	DataDriveDelay	BusRelease Time	CSActive	OEActive	BEActive	PortSize	DeviceType
Bank3	0x30	DataDriveDelay	BusRelease Time	CSActive	OEActive	BEActive	PortSize	DeviceType

Address: *EMIBaseAddress* + 0x00, 0x10, 0x20 and 0x30
Type: Read/write
Reset value: See Table 35: *EMI_ConfigData0 register format for SRAM/peripherals on page 92*.

Description

EMI configuration data register 0 for banks 0 to 3. There is one of these configuration registers for each bank. For safe configuration, each of the four banks should be configured after reset and then have their configuration locked by writing to the **EMI_ConfigLock** register before any access to an external bank is made.

The format of the configuration registers for a bank depends on the memory device type, i.e. whether the bank is configured for SDRAM/DRAM or for SRAM and peripherals. The type is defined by the **DeviceType** field in this register.

Bitfield	Function
Reserved	Write 0
DataDriveDelay	Data drive delay; 0-1 phases
BusReleaseTime	Bus release time; 0 means 1 cycle, 1 means 2 cycles
SubBanks	00: 1 sub-bank, 01: 2 sub-banks, 10:reserved, 11: reserved
SubBankSize	00: 256 k words, 01: 1 M words, 10: 4 M words, 11: 16 M words
ShiftAmount	Column address width (0 means 7 bits, 1 means 8 bits etc.)
PortSize	00: reserved, 01: 32 bit, 10: 16 bit, 11: 8 bit
DeviceType	000 = DRAM. Sets the format of the configuration data registers for the bank.

Table 33 EMI_ConfigData0 register format for DRAM

Bitfield	Function
Reserved	Write 0.
DataDriveDelay	Data drive delay; 0-1 phases
BusReleaseTime	Bus release time; 0 means 1 cycle, 1 means 2 cycles
SubBanks	00: 1 sub-bank, 01: 2 sub-banks, 10: reserved, 11: reserved
SubBankSize	00: 16 Mbit, 01: 32 Mbit, 10: 64 Mbit, 11: 128 Mbit
ShiftAmount	Column address width (0 means 7 bits, 1 means 8 bits etc.)
PortSize	00: reserved, 01: 32 bit, 10: 16 bit, 11: 8 bit
DeviceType	010 = SDRAM. Sets the format of the configuration data registers for the bank.

Table 34 EMI_ConfigData0 register format for SDRAM

Bitfield	Function	Reset
DataDriveDelay	Data drive delay; 0-7 phases	5
BusReleaseTime	Bus release time; 0-3 cycles	2
CSactive	00: Inactive, 01: Active during read accesses only, 10: Active during write accesses only, 11: Active during read and write accesses	01
OEactive	00: Inactive, 01: Active during read accesses only, 10: Active during write accesses only, 11: Active during read and write accesses	01
BEactive	00: Inactive, 01: Active during read accesses only, 10: Active during write accesses only, 11: Active during read and write accesses	00
Portsize	00 = reserved, 01 = 32 bit, 10 = 16 bit, 11 = 8 bit	Value on BootSource0-1
DeviceType	001 = SRAM/peripheral. Sets the configuration registers fromat for the bank.	1

Table 35 EMI_ConfigData0 register format for SRAM/peripherals

EMI_ConfigData1 EMI configuration data register 1

SDRAM/DRAM format

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bank0	0x04	RASbits[22:7]															
Bank1	0x14	RASbits[22:7]															
Bank2	0x24	RASbits[22:7]															
Bank3	0x34	RASbits[22:7]															

SRAM/peripheral format

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bank0	0x04	AccessTimeRead				CSe1Time Read		CSe2Time Read		OEe1Time Read		OEe2Time Read		BEe1Time Read		BEe2Time Read	
Bank1	0x14	AccessTimeRead				CSe1Time Read		CSe2Time Read		OEe1Time Read		OEe2Time Read		BEe1Time Read		BEe2Time Read	
Bank2	0x24	AccessTimeRead				CSe1Time Read		CSe2Time Read		OEe1Time Read		OEe2Time Read		BEe1Time Read		BEe2Time Read	
Bank3	0x34	AccessTimeRead				CSe1Time Read		CSe2Time Read		OEe1Time Read		OEe2Time Read		BEe1Time Read		BEe2Time Read	

Address: *EMIBaseAddress* + 0x04, 0x14, 0x24 and 0x34
Type: Read/write
Reset value: See Table 37 .

Description

EMI configuration data register 1 for banks 0 to 3. There is one of these configuration registers for each bank. For safe configuration, each of the four banks should be configured after reset and then have their configuration locked by writing to the **EMI_ConfigLock** register before any access to an external bank is made.

The format of the configuration registers for a bank depends on the memory device type, i.e. whether the bank is configured for SDRAM/DRAM or for SRAM/peripherals. The type is defined by the **DeviceType** field in register **EMI_ConfigData0**.

Bitfield	Description
RASbits[22:7]	Page address mask for address bits 22:7

Table 36 EMI_ConfigData1 register format for DRAM

Bitfield	Description	Reset
AccessTimeRead	2 cycles + 0-15 cycles	8
CSe1TimeRead	Delay of falling edge of CS in3 phases after start of access cycle	0
CSe2TimeRead	Delay from rising edge of CS in phases before end of access cycle	0
OEe1TimeRead	Delay of falling edge of OE in phases after start of access cycle	0
OEe2TimeRead	Delay from rising edge of OE in phases before end of access cycle	0
BEe1TimeRead	Delay of falling edge of BE in phases after start of access cycle	0
BEe2TimeRead	Delay from rising edge of BE in phases before end of access cycle	0

Table 37 EMI_ConfigData1 register format for SRAM

EMI_ConfigData2 EMI configuration data register 2

DRAM format

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bank0	0x08		RASbits[29:23]							PrechargeTime			RefreshInterval				Refresh RASedge
Bank1	0x18		RASbits[29:23]							PrechargeTime			RefreshInterval				Refresh RASedge
Bank2	0x28		RASbits[29:23]							PrechargeTime			RefreshInterval				Refresh RASedge
Bank3	0x38		RASbits[29:23]							PrechargeTime			RefreshInterval				Refresh RASedge

SDRAM format

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bank0	0x08		RASBits[29:23]							PrechargeTime			RefreshInterval				
Bank1	0x18		RASBits[29:23]							PrechargeTime			RefreshInterval				
Bank2	0x28		RASBits[29:23]							PrechargeTime			RefreshInterval				
Bank3	0x38		RASBits[29:23]							PrechargeTime			RefreshInterval				

SRAM/peripheral format

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bank0	0x08	AccessTimeWrite				CSe1Time Write		CSe2Time Write		OEe1Time Write		OEe2Time Write		BEe1Time Write		BEe2Time Write	
Bank1	0x18	AccessTimeWrite				CSe1Time Write		CSe2Time Write		OEe1Time Write		OEe2Time Write		BEe1Time Write		BEe2Time Write	
Bank2	0x28	AccessTimeWrite				CSe1Time Write		CSe2Time Write		OEe1Time Write		OEe2Time Write		BEe1Time Write		BEe2Time Write	
Bank3	0x38	AccessTimeWrite				CSe1Time Write		CSe2Time Write		OEe1Time Write		OEe2Time Write		BEe1Time Write		BEe2Time Write	

Address: *EMIBaseAddress* + 0x08, 0x18, 0x28 and 0x38
Type: Read/write
Reset value: See Table 38 .

Description

EMI configuration data register 2 for banks 0 to 3. There is one of these configuration registers for each bank. For safe configuration, each of the four banks should be configured after reset and then have their configuration locked by writing to the **EMI_ConfigLock** register before any access to an external bank is made.

The format of the configuration registers for a bank depends on the memory device type, i.e. whether the bank is configured for DRAM (Table 38), SDRAM (Table 39) or SRAM and peripherals (Table 40). The type is defined by the **DeviceType** field in register **EMI_ConfigData0**.

Bitfield	Function
Reserved	Write 0
RASBits[29:23]	Page address mask for address bits 29:23
PrechargeTime	1 to 8 cycles
RefreshInterval	1 to 16 periods of 128 cycles
RefreshRASedge	0 means 1 cycle or 1 means 2 cycles after start of refresh

Table 38 EMI_ConfigData2 register format for DRAM

Bitfield	Function
Reserved	Write 0
RASBits[29:23]	Page address mask for address bits 29:23
PrechargeTime	1 to 8 cycles
RefreshInterval	1 to 16 periods of 128 cycles
Reserved	Write 0

Table 39 EMI_ConfigData2 register format for SDRAM

Bitfield	Function
AccessTimeWrite	2 cycles + 0-15 cycles
CSe1TimeWrite	Delay of falling edge of CS in phases after start of access cycle
CSe2TimeWrite	Delay from rising edge of CS in phases before end of access cycle
OEe1TimeWrite	Delay of falling edge of OE in phases after start of access cycle
OEe2TimeWrite	Delay from rising edge of OE in phases before end of access cycle
BEe1TimeWrite	Delay of falling edge of BE in phases after start of access cycle
BEe2TimeWrite	Delay from rising edge of BE in phases before end of access cycle

Table 40 EMI_ConfigData2 register format for SRAM

EMI_ConfigData3 EMI configuration data register 0

DRAM format

		15 14 13 12	11	10 9	8	7 6	5 4	3 2	1	0
Bank0	0x0C		RAStime	CAStime	RASe1Time	RASe2Time	CASe1Time	CASe2Time	Multi byte	Latch Point
Bank1	0x1C		RAStime	CAStime	RASe1Time	RASe2Time	CASe1Time	CASe2Time	Multi byte	Latch Point
Bank2	0x2C		RAStime	CAStime	RASe1Time	RASe2Time	CASe1Time	CASe2Time	Multi byte	Latch Point
Bank3	0x3C		RAStime	CAStime	RASe1Time	RASe2Time	CASe1Time	CASe2Time	Multi byte	Latch Point

SDRAM format

		15	14	13-10	9-8	7-6	5-4	3	2-1	0
Bank0	0x0C		Mode set delay	Refresh time 1 to 16 cycles	Activate to read	Activate to write	CAS latency	SDRAM banks	Write recovery time	
Bank1	0x1C		Mode set delay	Refresh time 1 to 16 cycles	Activate to read	Activate to write	CAS latency	SDRAM banks	Write recovery time	
Bank2	0x2C		Mode set delay	Refresh time 1 to 16 cycles	Activate to read	Activate to write	CAS latency	SDRAM banks	Write recovery time	
Bank3	0x3C		Mode set delay	Refresh time 1 to 16 cycles	Activate to read	Activate to write	CAS latency	SDRAM banks	Write recovery time	

SRAM/peripheral format

		15-1	0
Bank0	0x0C		LatchPoint
Bank1	0x1C		LatchPoint
Bank2	0x2C		LatchPoint
Bank3	0x3C		LatchPoint

Address: *EMIBaseAddress* + 0x0C, 0x1C, 0x2C and 0x3C
Type: Read/write
Reset value: 0

Description

EMI configuration data register 3 for banks 0 to 3. There is one of these configuration registers for each bank. For safe configuration, each of the four banks should be configured after reset and then have their configuration locked by writing to the **EMI_ConfigLock** register before any access to an external bank is made.

The format of the configuration registers for a bank depends on the memory device type, i.e. whether the bank is configured for SDRAM/DRAM or for SRAM and peripherals. The type is defined by the **DeviceType** field in register **EMI_ConfigData0**.

Bitfield	Function
Reserved	Write 0
RAStime	1 or 2 cycles
CAStime	2 cycles + 0-3 cycles
RASe1Time	Falling edge of RAS. 1-2 phases after start of **RASTime**
RASe2Time	Rising edge of RAS. 0-3 phases before end of **CASTime**
CASe1Time	Falling edge of CAS. 1-4 phases after start of **CASTime**
CASe2Time	Rising edge of CAS. 0-3 phases before end of **CASTime**
Multibyte	When set, enables byte addressing using CAS strobes (byte mode)
LatchPoint	0: end of **CASTime** 1: 1 cycle before end of **CASTime**

Table 41 EMI_ConfigData3 register format for DRAM

Bitfield	Function
Reserved	Write 0
ModeSetDelay	3 or 4 cycles
RefreshTime	1 to 16 cycles
ActivateToRead	1 to 4 cycles
ActivateToWrite	1 or 4 cycles
CASlatency	1 to 4 cycles
SDRAM banks	The number of internal memory banks in SDRAM organisation 0 = 2 SDRAM banks, 1 = 4 SDRAM banks
WriteRecoveryTime	00 = reserved, 01 = 1 cycle, 10 = 2 cycles, 11 = 3 cycles)
Reserved	Write 0

Table 42 EMI_ConfigData3 register format for SDRAM

Bitfield	Function	Reset
Reserved	Write 0	
LatchPoint	0: end of access cycle 1: 1 cycle before end of access cycle 2: 2 cycles before end of access cycle	0

Table 43 EMI_ConfigData3 register format for SRAM and peripherals

EMI_ConfigLockBank3-0 EMI configuration lock

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x40																Lock0
0x44																Lock1
0x48																Lock2
0x4C																Lock3

Address: *EMIBaseAddress* + 0x40 to 0x4C
Type: Write only
Reset value: 0

Description

EMI configuration write protection locks. When the EMI for a bank has been configured, the configuration should be locked by setting the corresponding lock register to 1. When **Lock***n* is set, **EMI_ConfigData3:0Bank***n* become read only.

EMI_ConfigPadlogic EMI configuration of pad logic

	15	14	13	12	11	10	9 8 7	6	5	4	3	2	1	0
0x70	1284 slave/ master				SDRAM clock tristate	SDRAM clock enable	Sub bank size	Bank3 subde- code	Mem Wait Config		Bank3 Addr ShiftEn	Bank2 Addr ShiftEn	Bank1 Addr ShiftEn	Bank0 Addr ShiftEn

Address: *EMIBaseAddress* + 0x70
Type: Read/write
Reset value: 0

Description

The **EMI_ConfigPadlogic** register is a read/write register which configures the bank address shift enables and configures the **MemWait**. It also provides (via bit 15) slave/master mode selection for the IEEE 1284 interface.

Bitfield	Function
Bank0AddrShiftEn	0 = shift disabled, 1= shift enabled for Bank 0
Bank1AddrShiftEn	0 = shift disabled, 1= shift enabled for Bank 1
Bank2AddrShiftEn	0 = shift disabled, 1= shift enabled for Bank 2
Bank3AddrShiftEn	0 = shift disabled, 1= shift enabled for Bank 3
MemWaitConfig	0 = asynchronous MemWait, 1 = synchronous MemWait
Bank3subdecode	0 = disabled, 1 = enabled Note : if enabled, subdecode strobe is notMemCAS3.
Subbanksize	000 = 4Mbit, 001 = 8Mbit, 010 = 16Mbit, 011 = 32Mbit,100 = 64Mbit,101 = 1Gbit
SDRAMclockenable	1 = enable (notMemCAS1 used for SDRAM clock) 0 = disable (notMemCAS1 used for DRAM CAS strobe) Program to 1 if using SDRAM.
SDRAMclocktristate	1 = SDRAM clock is not tristate with MemReq 0 = SDRAM clock is tristated Note : This MUST be programmed to 1 if using SDRAM as it also controls SDRAM strobe phase control.
1284 slave/master mode selection	0 = slave mode (default mode, as STi5510), 1 = master mode

EMI_ConfigStatus EMI configuration status

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x50	Reserved								Bank3 Config Lock	Bank2 Config Lock	Bank1 Config Lock	Bank0 Config Lock	Bank3 Config	Bank2 Config	Bank1 Config	Bank0 Config

Address: *EMIBaseAddress* + 0x50
Type: Read only
Reset value: 0

Description

The **ConfigStatus** register is a read only register which contains information on whether the **ConfigData** registers have been written to and locked.

Bitfield	Function
Reserved	Write 0
Bank3ConfigLock	Bank 3 configuration has been locked.
Bank2ConfigLock	Bank 2 configuration has been locked.
Bank1ConfigLock	Bank 1 configuration has been locked.
Bank0ConfigLock	Bank 0 configuration has been locked.
Bank3Config	When set all the four configuration registers for bank3 have been written at least once.
Bank2Config	When set all the four configuration registers for bank2 have been written at least once.
Bank1Config	When set all the four configuration registers for bank1 have been written at least once.
Bank0Config	When set all the four configuration registers for bank0 have been written at least once.

EMI_SDRAMModeReg0 SDRAM mode register

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x58	Reserved									Latency mode			Burst type	Burst length		

Address: *EMIBaseAddress* + 0x58
Type: W
Reset value: 0

Description

SDRAM mode register for single SDRAM bank, or SDRAM bank 0 when there are two SDRAM banks.

Bitfield	Descritpion
Burst length	Default 010
Burst type	Default 0
Latency mode	Default 010

EMI_SDRAMModeReg1 (SDRAM mode)

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x5C	Reserved									Latency mode			Burst type	Burst length		

Address: *EMIBaseAddress* + 0x5C
Type: W
Reset value: 0

Description

SDRAM mode register for SDRAM bank 1 when there are two SDRAM banks.

Bitfield	Descritpion
Burst length	Default 010
Burst type	Default 0
Latency mode	Default 010

EMI_DRAMInitialize DRAM/SDRAM initialize

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x60																DRAMInit

Address: *EMIBaseAddress* + 0x60
Type: Write only

Description

A write to this address initializes any DRAM or SDRAM in the system and enables refreshes to take place.

The **EMI_DRAMInitialize** register should only be written to when there is DRAM or SDRAM in the system.

12 IEEE 1284 port (PC parallel port)

All enables are active high unless otherwise stated.

1284Checksum 1284 check sum

	7	6	5	4	3	2	1	0
0x28	Checksum							

Address: *1284BaseAddress* + 0x28
Type: Read only
Reset value: 0

Description

The **1284Checksum** register contains a checksum of all bytes transmitted or received by the IEEE 1284. A read from this register causes it to be reset. The checksum is calculated by the accumulative bitwise XOR of each bit in the byte passing through the IEEE 1284 with the previous checksum value.

This register is not defined in transport modes for a single cycle pulse width.

This function is not part of the *IEEE 1284 Standard*, and is an addition to allow rapid checksum calculation in a specific application.

1284Control 1284 control

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x08								Override Host LogicH	Reset	ExtBus Direction	HwInput RLE expan			Mode		

Address: *1284BaseAddress* + 0x08
Type: Write only
Reset value: 0

Description

The **1284Control** register controls the operating mode of the IEEE 1284 port.

Setting the **Reset** bit forces all machines back to the idle status, discarding any stored data. Note, this may cause protocol errors and loss of data. This is a functional synchronous reset, and returns the module to the initialization state in the enabled mode. This only resets the IEEE 1284 module, and not the DMA engines.

If any IEEE 1284 mode or the transport stream mode is disabled during a transaction, the mode is disabled next time the controlling state machine reaches idle, after completing any ongoing transactions. In transport stream modes, the current package is completed before returning to idle, in IEEE 1284 mode it waits until returning to compatible mode.

If the hardware enables (bits 3 and 5) are changed during a transaction, the change takes effect next time the action associated with that transaction occurs.

Note, when operating in IEEE 1284 mode, the point at which the IEEE 1284 port returns to idle is controlled by the host and therefore may be an unbounded period of time.

Bitfield	Function
OverrideHostLogicH	When set to 1, the **1284HostLogicH** input signal is forced high, so when operating in any IEEE 1284 mode the IEEE 1284 module always assumes that the input signals from the host are valid.
Reset	When set to 1, the IEEE 1284 port is reset, and any stored data is discarded.
ExtBusDirection	Enable external bus direction control.
HwInputRLEexpan	Enable hardware input RLE expansion.
Reserved	Write 0.
Mode	IEEE 1284 port operating mode: 000: software mode 001: IEEE 1284 mode 010: transport stream mode A 011: transport stream mode B

1284DataIn 1284 data input

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x20								Control/ Address	Control/ Data[7]	Data[6:0]						

Address: *1284BaseAddress* + 0x20
Type: Read only
Reset value: 0

Description

When in any IEEE 1284 mode, a read from the **1284DataIn** register reads the input data token stored in the IEEE 1284 module. In other modes it reflects the data currently on the input data pins (**1284Data0-7**).

This data is valid and available only when the **InputDataReady** bit is set high in the **1284Status** register, see 1284Status on page 111. If data is available, then reading this register removes the data token from the IEEE 1284 port and allows the next access sequence to proceed.

Reading from the register during a DMA transfer will interrupt that transfer sequence, and may invalidate the DMA transfer.

Bit 8 is valid only in ECP mode and indicates byte packet type, in all other modes it is undefined. The type of an incoming data package is mode dependent.

Bitfield	Function
Control/Address	Byte packet type: 0: data 1: control packet in ECP mode
Control/Data[7]	Data currently on the data pin (**1284Data7**) - in any other mode. Control packet type - in ECP mode, where 1 indicates channel number packet, 0 indicates RLE count packet.
Data[6:0]	Input data token stored in the IEEE 1284 module - in any IEEE 1284 mode. Data currently on the data pins (**1284Data0-6**) - in any other mode.

1284DataOut 1284 data output

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x24								Control/ Address	Control/ Data[7]	Data[6:0]						

Address: *1284BaseAddress* + 0x24
Type: Write only
Reset value: 0

Description

A write to the **1284DataOut** register writes a data token to the IEEE 1284 module.

If a write to this register occurs in IEEE 1284 mode or transport stream mode, and the **OutputDataReady** bit of the **1284Status** register is set high, then a data token is transferred to the IEEE 1284 port and the access sequence started. In other modes it reflects the value on the data pins if the data bus is being driven.

Bit 8 is valid in ECP mode only and controls the type of access triggered.

Writing to this register during a DMA access will interrupt the access sequence and may invalidate the DMA transfer.

Bitfield	**Function**
Control	Byte packet type: 0: data 1: control packet in ECP mode
Control/Data[7]	Data currently on the data pin (**1284Data7**) - in any other mode. Control packet type - in ECP mode, where 1 indicates channel number packet, 0 indicates RLE count packet.
Data[6:0]	Output data token stored in the IEEE 1284 module - in any IEEE 1284 mode or transport mode. Data currently on the data pins (**1284Data0-6**) - in any other mode.

1284DmaAddress 1284 DMA address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x40	DmaAddress

Address: *1284BaseAddress* + 0x40
Type: Write only
Reset value: 0

Description

This defines the byte address from which the DMA starts. Following the completion of a DMA access this register points to the next location in memory.

This register is undefined following system reset.

1284DmaControl 1284 DMA control

	7	6	5	4	3	2	1	0
0x48						DmaReset	DmaStall	DmaDirection

Address: *1284BaseAddress* + 0x48

Type: Read/write
Reset value: 0

Description

The **1284DmaControl** register controls the DMA transfer.

The direction of the DMA access, either from memory to the IEEE 1284 port or vice versa is specified by the **DmaDirection** bit.

Setting the **DmaReset** bit terminates the DMA transfer. Buffered incoming data is written to memory. Stored outgoing data is lost. The **1284DmaCount** register shows the number of bytes successfully transferred before reset occurred. When reset is complete, the **DmaReset** bit is set to zero.

If the DMA engine is reset while a DMA output is occurring and the byte transfer is in progress on the IEEE 1284 port, the byte transfer may be corrupted, or the host left in a position of expecting data to be transferred. This byte is not included in the DMA count.

DMA reset is only expected to be used to clear the DMA engines in exceptional conditions such as, errors, at which point the interface is stalled, or by stalling the DMA engines for long enough for all buffered tokens to be removed.

Setting the **DmaStall** bit stops the DMA transfer. The **1284DmaCount** register shows the total number of bytes remaining to be transferred. Resetting the **DmaStall** bit allows the DMA transfer to continue.

Bitfield	**Function**
DmaReset	Terminates the DMA transfer
DmaStall	Stalls the DMA transfer
DmaDirection	Direction of the DMA access: 0: from IEEE 1284 port to memory 1: from memory to IEEE 1284 port

1284DmaCount 1284 DMA count

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x44	DmaCount															

Address: *1284BaseAddress* + 0x44
Type: Read/write
Reset value: Undefined

Description

In the event of a DMA error, or other exception, the **1284DmaCount** register contains the number of bytes which remain to be transferred.

Writing to this register starts a new DMA sequence, starting from the address given in the **1284DmaAddress** register, for the number of bytes written to this location. Reading from this register gives the number of bytes remaining to be transferred. This value is constant only when the DMA engine has been stalled or reset. A value of zero in this register indicates that the DMA transfer has completed transfers to or from memory. If a zero is written to this register, a memory access may occur, but no data is transferred.

This register is undefined following system reset.

1284DmaToken 1284 DMA token

	7	6	5	4	3	2	1	0
0x30							TokenFromDma	TokenToDma

Address: *1284BaseAddress* + 0x30
Type: Read/write
Reset value: 0

Description

This register allows the DMA engines to be used when driving the IEEE 1284 port directly in software mode. The register is valid only when software mode is enabled, writes to this register in other modes are undefined.

A read from this register indicates whether the token has been successfully transferred to the DMA engine. If the bit is high then the memory system has not yet accepted the token, if the bit is zero it indicates that it has accepted the token.

The token transfer will only occur if the direction of the DMA engine corresponds to the token transfer direction.

The DMA engine may assemble each byte token in word packets before writing the token to memory.*

Writing a 1 to bit 0 transfers a data token to the DMA engine from the data pins. Writing a 1 to bit 1 transfers a data token to the data pins from the DMA engine.

Bitfield	Function
TokenFromDma	Data token transfer from the DMA engine to the data pins.
TokenToDma	Data token transfer from the data pins to the DMA engine.

1284IntAck 1284 interrupt acknowledge

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x54							Reset Ack	Error Ack	Request Ack	Direc ChangeAck	Mode ChangeAck	PinInt Ack	Dma Error Ack	Dma Com-plete Ack	Input Avail Ack	Output Avail Ack

Address: *1284BaseAddress* + 0x54
Type: Write only
Reset value: 0

Description

The **1284IntAck** register is write only. Writing a ‘1’ to a bit in this register explicitly clears the associated active interrupt.

The bits with their functions marked as ‘Not applicable’ in the table below reference interrupts which are implicitly cleared by completing the action associated with the interrupt. An explicit reset will clear these bits, but the interrupt will be immediately re-asserted if the triggering condition is still true.

The IEEE 1284 input and output interrupts are cleared when the associated data token is transferred. The DMA error and DMA complete interrupts can be cleared by resetting/restarting the DMA engine. A IEEE 1284 request is cleared by outputting a data token.

Bitfield	Function
ResetAck	When set, the associated interrupt is cleared.
ErrorAck	When set, the associated interrupt is cleared.
RequestAck	Not applicable
DirecChangeAck	When set, the associated interrupt is cleared.
ModeChangeAck	When set, the associated interrupt is cleared.
PinIntAck	Not applicable
DmaErrorAck	Not applicable
DmaCompleteAck	Not applicable
InputAvailAck	Not applicable
OutputAvailAck	Not applicable

1284IntEnable 1284 interrupt enable

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x4C							Reset En	Error En	Request En	Direc Change En	Mode Change En	PinInt En	Dma Error En	Dma Complete En	Input Avail En	Output Avail En

Address: *1284BaseAddress* + 0x4C
Type: Read/write
Reset value: 0

Description

The **1284IntEnable** register determines whether an interrupt is enabled.

If the bit relating to the interrupt is set, then if that event occurs, an interrupt is generated.

A DMA error occurs if a non-data packet (an RLE count, channel number or an address value) is passed during a DMA transfer. The DMA sequence stalls at this point. The DMA engine must then be reset to flush valid buffered incoming bytes to memory. The erroneous data token can be accessed directly and removed from the IEEE 1284 module by reading the 1284DataIn register (see1284DataIn on page 102). Outgoing data tokens are not checked.

The DMA access can also be stalled by a number of events such as mode and direction changes, protocol errors and IEEE 1284 requests. These events can be monitored and treated as a DMA error if the events are seen, by explicitly resetting the DMA engines, which flushes buffered valid bytes to memory, leaving the engines in the same state as a DMA error following a reset.

Bitfield	Function
ResetEn	When set, an interrupt is generated if the host system resets the IEEE 1284 port.
ErrorEn	When set, an interrupt is generated if a protocol error is detected by the IEEE 1284 port.
RequestEn	When set, an interrupt is generated if a device id request is made.
DirecChangeEn	When set, an interrupt is generated if the transfer direction of the IEEE 1284 port changes. The current transfer direction can be read from the **1284Status** register.
ModeChangeEn	When set, an interrupt is generated if the IEEE 1284 port changes mode. The operational modes are specified in the **1284Status** register. Additional information on the direction of the bidirectional modes is also available in the **1284Status** register or can be interpreted from bits 1:0 of the **1284IntStatus** register.

Bitfield	Function
PinIntEn	When set, an interrupt is generated when the enabled (**1284PinInEnable** register) IEEE 1284 input pins fail to match the pattern in the **1284PinInValue** register. The value of the input pins can be read from the **1284PinIn** register, see 1284PinIn on page 108.
DmaErrorEn	When set, an interrupt is generated if a non-data packet (an RLE count, channel number or address value) is passed by the IEEE 1284 port during a DMA transfer.
DmaCompleteEn	When set, an interrupt is generated when a DMA transfer is completed and all tokens have been transferred to/from the IEEE 1284 port.
InputAvailEn	When set, and interrupt is generated when a IEEE 1284 input byte is available.
OutputAvailEn	When set, an interrupt is generated when the IEEE 1284 output is clear and available.

1284IntStatus 1284 interrupt status

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x50							Reset	Error	Request	Direc Change	Mode Change	PinInt	Dma Error	Dma Complete	Input Avail	Output Avail

Address: *1284BaseAddress* + 0x50
Type: Read only
Reset value: 0

Description

The 1284IntStatus register gives the identity of the event which caused the interrupt. This register may also be read to monitor the status of masked interrupts.

Bitfield	Function
Reset	When set, indicates reset interrupt was generated.
Error	When set, indicates error interrupt was generated.
Request	When set, indicates request interrupt was generated.
DirecChange	When set, indicates direction change interrupt was generated.
ModeChange	When set, indicates mode change interrupt was generated.
PinInt	When set, indicates input pin interrupt was generated.
DmaError	When set, indicates DMA error interrupt was generated.
DmaComplete	When set, indicates DMA complete interrupt was generated.
InputAvail	When set, indicates IEEE 1284 input byte available interrupt was generated.
OutputAvail	When set, indicates IEEE 1284 output clear and available interrupt was generated.

1284ModeEnable 1284 mode enable

	7	6	5	4	3	2	1	0
0x00			EnRLE	EnECP		EnDevID		EnByte

Address: *1284BaseAddress* + 0x00
Type: Write only
Reset value: 0

Description

The 1284ModeEnable register bits are a direct mask of the IEEE 1284 extensibility request values. If a bit corresponding to the mode is set low, the peripheral is refused access to enter the mode when operating in IEEE 1284 mode. If all the bits are disabled the device operates in the nibble and compatible modes.

If the register is modified, the change takes effect at the next IEEE 1284 negotiation transaction.

This register is only valid when IEEE 1284 mode is enabled.

Bitfield	Function	Reset value
Reserved.	Write 0.	0
EnRLE	Enable RLE.	0
EnECP	Enable ECP.	0
Reserved.	Write 0.	0
EnDevID	Enable device identification.	0
Reserved.	Write 0.	0
EnByte	Enable byte.	0

1284PacketSize 1284 packet size

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x2C					PacketSize											

Address: *1284BaseAddress* + 0x2C
Type: Write only
Reset value: 0

Description

The **1284PacketSize** register contains the packet size during a transport stream transfer. This register is valid only when transport mode is enabled.

1284PinIn 1284 pin input

	7	6	5	4	3	2	1	0
0x10				HostLogicH	notSelectIn	notInit	notAutoFd	notStrobe

Address: *1284BaseAddress* + 0x10
Type: Read only
Reset value: 0

Description

The **1284PinIn** register reflects the current status of the input pins in all modes. The value read is the value on the pins when the request is granted.

Bitfield	Function
HostLogicH	**1284HostLogicH** pin status
notSelectIn	**1284notSelectIn** pin status
notInit	**1284notInit** pin status
notAutoFd	**1284notAutoFd** pin status
notStrobe	**1284notStrobe** pin status

1284PinInEnable 1284 pin input enable

	7	6	5	4	3	2	1	0
0x14				HostLogicH IntEn	notSelectIn IntEn	notInit IntEn	notAutoFd IntEn	notStrobe IntEn

Address: *1284BaseAddress* + 0x14
Type: Read/write
Reset value: 0

Description

This register enables generation of an interrupt based on the values contained in the **1284PinInValue** register, see 1284PinInValue on page 109.

Bitfield	Function
HostLogicHIntEn	When set, it enables generation of an interrupt if the associated value given in the **1284PinInValue** register does not match the **1284HostLogicH** input pin setting.
notSelectInIntEn	When set, it enables generation of an interrupt if the associated value given in the **1284PinInValue** register does not match the **1284notSelect** input pin setting.
notInitIntEn	When set, it enables generation of an interrupt if the associated value given in the **1284PinInValue** register does not match the **1284notInit** input pin setting.
notAutoFdIntEn	When set, it enables generation of an interrupt if the associated value given in the **1284PinInValue** register does not match the **1284notAutoFd** input pin setting.
notStrobeIntEn	When set, it enables generation of an interrupt if the associated value given in the **1284PinInValue** register does not match the **1284notStrobe** input pin setting.

1284PinInValue 1284 pin input value

	7	6	5	4	3	2	1	0
0x18				HostLogicH IntVal	notSelectIn IntVal	notInit IntVal	notAutoFd IntVal	notStrobe IntVal

Address: *1284BaseAddress* + 0x18
Type: Read/write
Reset value: 0

Description

This register holds the value against which the input pins are compared. Any difference in the associated bits results in an interrupt being generated if the corresponding bit in the enable register (see 1284PinInEnable on page 109) is set.

The compare function is level sensitive, and any change in input must be held for longer than the interrupt response time to be seen as a valid interrupt.

Following system reset this register is undefined.

Bitfield	Function
HostLogicHIntVal	Value to which the **1284HostLogicH** input pin setting is compared.
notSelectInIntVal	Value to which the **1284notSelect** input pin setting is compared.
notInitIntVal	Value to which the **1284notInit** input pin setting is compared.
notAutoFdIntVal	Value to which the **1284notAutoFd** input pin setting is compared.
notStrobeIntVal	Value to which the **1284notStrobe** input pin setting is compared.

1284PinOut 1284 pin output

	7	6	5	4	3	2	1	0
0x1C		DataBus Enable	Periph LogicH	Busy	notAck	Perror	Select	notFault

Address: *1284BaseAddress* + 0x1C
Type: Read/write
Reset value: 0

Description

The bus direction pin signals are only under the control of this register when not being controlled by the IEEE 1284 or transport mode state machine. For details of the pin control in the transport and IEEE 1284 modes refer below.

All pins are under user control when the device is operating in software mode.

A read from this register gives the current value of the output pin/bus direction. If the pin is not under user control this may not be the value written to this register, but will reflect the current value on the output pins.

If **DataBusEnable** is set low, the data bus is high impedance and may be driven by the host.

Bitfield	Function
DataBusEnable	**1284Out** output pin setting.
PeriphLogicH	**1284PeriphLogicH** output pin setting.
Busy	**1284Busy** output pin setting.
notAck	**1284notAck** output pin setting.
Perror	**1284PError** output pin setting.
Select	**1284Select** output pin setting.
notFault	**1284notFault** output pin setting.

1284PulseWidth 1284 pulse width

	7	6	5	4	3	2	1	0
0x04	ClockCycles							

Address: *1284BaseAddress* + 0x04
Type: Write only
Reset value: Undefined

Description

In IEEE 1284 mode, the **1284PulseWidth** register specifies the time period (Tp[1]) in number of system clock cycles. For the STi5512 running at 40 MHz this value must be 20 system clock cycles minimum in order to comply with the IEEE 1284 minimum time period of 500 ns. This register must only be written when transport and IEEE 1284 modes are disabled.

In transport stream mode, this register specifies the minimum period between byte clock (**TSByteClk**) edge transitions. For details see the transport mode description.

A write to this register takes effect at the next byte transfer.

1. Tp: Defined in the IEEE 1284 spec as the Minimum setup or pulse width for IEEE 1284 handshakes.

1284Status 1284 status

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0C								Data Transfer Dir	OpMode				Enable RLEext	1284 Request	Input Data Ready	Output Data Ready

Address: *1284BaseAddress* + 0x0C
Type: Read only
Reset value: 0

Description

The **1284Status** register gives the current status of the IEEE 1284 module.

Bit three is valid in ECP mode. It indicates that RLE expansion has been enabled. If the hardware expansion is not enabled then software is required to expand the byte stream.

Following system reset the IEEE 1284 module starts in software mode.

A **1284Request** is cleared when the next data token is transferred to the IEEE 1284 module.

Bitfield	**Function**
DataTransferDir	Data transfer direction 0: from host to IEEE 1284 module 1: from IEEE 1284 module to host
OpMode	Operational mode. Valid values are as follows: 0000: EEE 1284 mode: initialization 0001: IEEE 1284 mode: compatible 0010: IEEE 1284 mode: negotiation 0011: IEEE 1284 mode: nibble 0100: IEEE 1284 mode: byte 0101: IEEE 1284 mode: ECP 0111: IEEE 1284 mode: terminate 1000: software mode - peripheral control of IEEE 1284 port 1001: transport stream mode A 1010: transport stream mode B
EnableRLEext	RLE extensions enabled - in ECP mode
1284Request	Device id request.
InputDataReady	Input byte available. Data from the host is available in the IEEE 1284 module input buffer.
OutputDataReady	Output clear and available. The IEEE 1284 module is ready to output data to the host.

13 Interrupt controller (INC)

INC_Clear_Exec Clear a bit of the Exec register

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x108																Clr									Clr	Clr	Clr	Clr	Clr	Clr	Clr	Clr

Address: *IntControllerBase* + 0x108
Type: Write only

Description

INC_Clear_Exec allows bits of **Exec** to be cleared individually. Writing a ‘1’ in this register resets the corresponding bit in the **Exec** register, a ‘0’ leaves the bit unchanged.

INC_Clear_Mask Clear a bit of the interrupt enable mask

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xC8																Clr									Clr	Clr	Clr	Clr	Clr	Clr	Clr	Clr

Address: *IntControllerBase* + 0xC8
Type: Write only

Description

INC_Clear_Mask allows bits of **Mask** to be cleared individually. Writing a ‘1’ in this register resets the corresponding bit in the **Mask** register, a ‘0’ leaves the bit unchanged.

INC_Clear_Pending Clear a bit of the Pending register

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x88																Clr									Clr	Clr	Clr	Clr	Clr	Clr	Clr	Clr

Address: *IntControllerBase* + 0x88
Type: Write only

Description

INC_Clear_Pending allows bits of **Pending** to be cleared individually. Writing a ‘1’ in this register resets the corresponding bit in the **Pending** register, a ‘0’ leaves the bit unchanged.

INC_Exec Interrupts executing

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x100																									IntEx7	IntEx6	IntEx5	IntEx4	IntEx3	IntEx2	IntEx1	IntEx0

Address: *IntControllerBase* + 0x100
Type: Read/write
Reset value: 0

Description

The **Exec** register keeps track of the currently executing and pre-empted interrupts. A bit is set when the CPU starts running code for that interrupt. The highest priority interrupt bit is reset once the interrupt handler executes a return from interrupt (*iret*).

The **Exec** register is mapped onto two additional addresses **INC_Set_Exec** and **INC_Clear_Exec** so that bits can be set or cleared individually.

INC_HandlerWptr*n* Interrupt handler work space pointer

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

	31–2	1	0
0x00	HandlerWptr0[31:2]		P
0x04	HandlerWptr1[31:2]		P
0x08	HandlerWptr2[31:2]		P
0x0C	HandlerWptr3[31:2]		P
0x10	HandlerWptr4[31:2]		P
0x14	HandlerWptr5[31:2]		P
0x18	HandlerWptr6[31:2]		P
0x1C	HandlerWptr7[31:2]		P

Address: *IntControllerBase* + 0x00 to 0x1C
Type: Read/write
Reset value: Undefined

Description

The **INC_HandlerWptr** registers (1 per interrupt level) each contain a pointer to the work space of the corresponding interrupt handler. The base of the work space is 32-bit word aligned, so the two least significant bits of the 32-bit address are always zero, and are not held. Each register also contains a priority bit **P** which determines whether the interrupt is at a higher or lower priority than the high priority process queue.

Before the interrupt is enabled by writing a 1 in the **Mask** register, the software must ensure that there is a valid **INC_HandlerWptr** in the register.

Bit field	**Function**
HandlerWptr[31:2]	The 30 most significant bits of the address of the work space of the interrupt handler.
P	Sets the priority of the interrupt. If this bit is set to 0, the interrupt is a higher priority than the high priority process queue; if this bit is 1, the interrupt is a lower priority than the high priority process queue. o: high priority 1: low priority

INC_Mask Interrupt enable mask

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xC0																GloEn									IntEn7	IntEn6	IntEn5	IntEn4	IntEn3	IntEn2	IntEn1	IntEn0

Address: *IntControllerBase* + 0xC0
Type: Read/write
Reset value: 0

Description

An interrupt mask register is provided in the interrupt controller to selectively enable or disable external interrupts. This mask register also includes a global interrupt disable bit to disable all external interrupts whatever the state of the individual interrupt mask bits.

To complement this the interrupt controller also includes an interrupt pending register which contains a pending flag for each interrupt channel. The **Mask** register performs a masking function on the **Pending** register to give control over what is allowed to interrupt the CPU while retaining the ability to continually monitor external interrupts.

On start-up, the **Mask** register is initialized to zeros, so all interrupts are disabled, both globally and individually. When a 1 is written to the **GlobalEnable** bit, the individual interrupt channels are still disabled. To enable an interrupt channel, a 1 must also be written to the corresponding **InterruptEnable** bit.

Bit field	Function
IntEn[7:0]	When set to 1, the corresponding interrupt is enabled. When 0, interrupt is disabled.
GloEn	When set to 1, the setting of the interrupt is determined by the specific **InterruptEnable** bit. When 0, all interrupts are disabled.

The **Mask** register is mapped onto two additional addresses **Set_Mask** and **Clear_Mask** so that bits can be set or cleared individually.

INC_Pending Interrupt pending

	7	6	5	4	3	2	1	0
0x80	PendInt7	PendInt6	PendInt5	PendInt4	PendInt3	PendInt2	PendInt1	PendInt0

Address: *IntControllerBase* + 0x80
Type: Read/write
Reset value: 0

Description

The **Pending** register contains one bit per interrupt level with each bit controlled by the corresponding interrupt. A read can be used to examine the state of the interrupt controller while a write can be used to explicitly trigger an interrupt.

A bit is set when the triggering condition for an interrupt is met. All bits are independent so that several bits can be set in the same cycle. Once a bit is set, a further triggering condition will have no effect. The triggering condition is independent of the **Mask** register.

The highest priority interrupt bit is reset once the interrupt controller has made an interrupt request to the CPU.

The interrupt controller receives external interrupt requests and makes an interrupt request to the CPU when it has a pending interrupt request of higher priority than the currently executing interrupt handler.

If the software needs to write or clear some bits of the **Pending** register, the interrupts should be masked (by writing or clearing the **Mask** register) before writing or clearing the **Pending** register. The interrupts can then be unmasked.

The **Pending** register is mapped onto two additional addresses **Set_Pending** and **Clear_Pending** so that bits can be set or cleared individually.

INC_Set_Exec Set a bit of the Exec register

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x104																Set									Set	Set	Set	Set	Set	Set	Set	Set

Address: *IntControllerBase* + 0x104
Type: Write only

Description

INC_Set_Exec allows bits of **Exec** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **Exec** register, a ‘0’ leaves the bit unchanged.

INC_Set_Mask Set a bit of the interrupt enable mask

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xC4																Set									Set	Set	Set	Set	Set	Set	Set	Set

Address: *IntControllerBase* + 0xC4
Type: Write only

Description

INC_Set_Mask allows bits of **Mask** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **Mask** register, a ‘0’ leaves the bit unchanged.

INC_Set_Pending Set a bit of the Pending register

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x84																Set									Set	Set	Set	Set	Set	Set	Set	Set

Address: *IntControllerBase* + 0x84
Type: Write only

Description

INC_Set_Pending allows bits of **Pending** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **Pending** register, a ‘0’ leaves the bit unchanged.

INC_TriggerMode Interrupt trigger mode

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3	2 1 0
0x40		Trigger0
0x44		Trigger1
0x48		Trigger2
0x4C		Trigger3
0x50		Trigger4
0x54		Trigger5
0x58		Trigger6
0x5C		Trigger7

Address: *IntControllerBase* + 0x40 to 0x5C
Type: Read/write
Reset value: Undefined

Description

These registers control the triggering conditions of the interrupts. Each interrupt channel can be programmed to trigger on rising or falling edges or high or low levels on the incoming interrupt signal, as shown in the table below.

Trigger2:0	**Interrupt triggers on**
000	No trigger mode
001	High level - triggered while input high
010	Low level - triggered while input low
011	Rising edge - low to high transition
100	Falling edge - high to low transition
101	Any edge - triggered on rising and falling edges
110	No trigger mode
111	No trigger mode

Level triggering is different from edge triggering in that if the input is held at the triggering level, a continuous stream of interrupts is generated.

14 Interrupt level controller (INL)

INL_InputInterrupts Input interrupt status

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x7C										IntIn22	IntIn21	IntIn20	IntIn19	IntIn18	IntIn17	IntIn16	IntIn15	IntIn14	IntIn13	IntIn12	IntIn11	IntIn10	IntIn9	IntIn8	IntIn7	IntIn6	IntIn5	IntIn4	IntIn3	IntIn2	IntIn1	IntIn0

Address: *InterruptLevelBase* + 0x7C
Type: Read only
Reset value: Undefined

Description

The **InputInterrupts** register contains a vector which shows the state of each input interrupt signal. This allows the interrupt handler to determine the source of the interrupt, even when several sources are mapped onto the same level.

Bit 0 of the read data corresponds to **Interrupt0**, bit 1 corresponds to **Interrupt1** in sequence up to the maximum interrupt. Each bit is 1 when the corresponding interrupt signal is high and 0 when the signal is low.

The STi5512 data sheet gives the assignment of interrupt signals and therefore the bits in this register to the peripherals and external pins.

INL_Int*n*Priority Interrupt priority

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00																														Int0Pri		
...																...																
0x58																														Int22Pri		

Address: *InterruptLevelBase* + 0x00 to 0x58
Type: Read/write
Reset value: Undefined

Description

The priority assigned to each of the input interrupts is programmable via the **Int*n*Priority** registers. There is one register for each source of interrupts. Each register has 3 bits and is word aligned. The value in the register is the priority of the interrupt, as shown in the table below.

Int*n*Pri[2:0]	**Assert output interrupt**
000	0 (lowest priority)
001	1
010	2
011	3
100	4
101	5
110	6

Int*n*Pri[2:0]	Assert output interrupt
111	7 (highest priority)

Table 44 Output interrupt coding

14.1 Low power controller support registers

The interrupt level controller has additional registers to support the low power controller. The external interrupts can be used to provide a wake-up from power-down mode. These registers do not affect the active level or enable interrupts at the interrupt controller for any of the external interrupts.

INL_SelectnotInv External interrupt sense for low power controller

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x60																													SelectnotInv	SelectnotInv	SelectnotInv	SelectnotInv

Address: *InterruptLevelBase* + 0x60
Type: R/W

Bit field	Function
SelectnotInv	External interrupt sense for low power controller. 0: Active low. Low level on the corresponding external interrupt pin will wake up the low power controller. 1: Active high. High level on the corresponding external interrupt pin will wake up the low power controller.

Description

Each of the four external interrupts can be programmed to be not inverting or inverting, depending on whether the interrupt is active high or active low. Bit 0 corresponds to pin Interrupt0. After reset all the bits of this register are set, i.e. the signals are assumed to be active high.

INL_ExtIntEnable Enable external interrupt for low power controller

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x64																													ExtIntEnable	ExtIntEnable	ExtIntEnable	ExtIntEnable

Address: *InterruptLevelBase* + 0x64
Type: R/W

Bit field	Function
ExtIntEnable	Enable external interrupt for low power controller. 0: Disable external interrupt from causing the wake up from low power mode. 1: Enable external interrupt to cause the wake up from low power mode.

This register enables each of the four external interrupts to wake up the chip when it is in low power mode. Bit 0 corresponds to pin **Interrupt0**. After reset, this register is set to zero, i.e. all the interrupts are disabled from waking up the device.

15 IEEE1394 link layer interface (LLI)

LLI_LLIControl LLI control register

	7	6	5	4	3	2	1	0
0x00				SernotPar	AVOutnotIn	LocalnotTSClk	AVnot1284	AVnotTS

Address: *LLIBaseAddress* + 0x00
Type: Read/write
Reset value: 0

Description

This register has various control bits for the Link Layer Interface.

Bit field	**Function**
AVnotTS	Select the PTI input source: 0: Select the stream from the TSIn pins. 1: Select the stream from the IEEE 1394 pins if in AV mode or from the IEEE 1284 port module if in 1284 mode.
AVnot1284	Select AV or 1284 mode for the IEEE 1394 pins: 0: Select 1284 mode. 1: Select AV mode.
LocalnotTSClk	Select the output byte clock source: 0: Select **TSInByteClk.** 1: Select the locally generated byte clock**.**
AVOutnotIn	Select the direction of the IEEE 1394 pins when operating in AV mode: 0: IEEE 1394 pins input to the PTI. 1: IEEE 1394 pins outputs from the PTI.
SernotPar	Select serial or parallel mode for the TSIn input stream: 0: TSIn pins are in parallel mode. 1: TSIn pins are in serial mode. The local byte clock is selected for byte input to the PTI.

LLI_LLIByteClockRatio LLI byte clock ratio

	7	6	5	4	3	2	1	0
0x04					DivRatio			

Address: *LLIBaseAddress* + 0x04
Type: Read/write
Reset value: 0

Description

This register defines the speed of the local byte clock. To obtain of the local byte clock, the frequency of the system clock is divided by 4 more than the value in this register, as shown in the table below.

DivRatio	**Division**
0	Divide by 4.

Table 45 DivRatio encoding

DivRatio	Division
1	Divide by 5.
2	Divide by 6.
3	Divide by 7.
4	Divide by 8.
5	Divide by 9.
6	Divide by 10.
7	Divide by 11.
8	Divide by 12.

Table 45 DivRatio encoding

LLI_LLISyncLock LLI sync lock

	7	6	5	4	3	2	1	0
0x14					SyncLock			

Address: *LLIBaseAddress* + 0x14
Type: Write only
Reset value: 0

Description

The number of successive correct Sync bytes found before the Sync detect is locked.

LLI_LLISyncDrop LLI sync drop value

	7	6	5	4	3	2	1	0
0x16					SyncDrop			

Address: *LLIBaseAddress* + 0x16
Type: Write only
Reset value: 0

Description

The number of successive erroneous Sync bytes found before the lock is abandoned.

LLI_LLISyncPeriod LLI sync period

	7	6	5	4	3	2	1	0
0x18	SyncPeriod							

Address: *LLIBaseAddress* + 0x18
Type: Write only
Reset value: 188

Description

The expected number of valid byte clock cycles between Sync bytes. On reset, this will default to 188.

16 Low power module (LPM)

LPM_Timer Low power timer

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x400	LPTimer[31:0]
0x404	LPTimer[63:32]

Address: *LPCBaseAddress* + 0x400 and 0x404
Type: Read/write
Reset value: Undefined

Description

The **LPTimer** registers are the least significant and most significant words of the low power timer register. These enable the least significant or most significant word to be written independently without affecting other words.

When either word of the register is written, the low power timer is stopped and the new value in **LPTimer** is available to be written to the low power timer.

LPM_TimerStart Low power timer start

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x408		TimerStart

Address: *LPCBaseAddress* + 0x408
Type: Write only

Description

A write of any value to the **LPTimerStart** register starts the low power timer counter. The counter is stopped and the **LPTimerStart** register reset if either word of **LPTimer** is written.

Setting the **LPTimerStart** register to zero does not stop the timer.

LPM_Alarm Low power alarm

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
0x410	LPAlarm[31:0]	
0x414		LPAlarm[39:32]

Address: *LPCBaseAddress* + 0x410 and 0x414
Type: Read/write
Reset value: Undefined

Description

The **LPAlarm** registers are the least significant and most significant words of the low power alarm. They are used to program the alarm register.

LPM_AlarmStart Low power alarm start

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x418																																Alarm Start

Address: *LPCBaseAddress* + 0x418
Type: Write only

Description

Any write to the **LPAlarmStart** register starts the low power alarm counter. The counter is stopped and the **LPStart** register reset if either word of the **LPTimer** register is written.

LPM_SysPll System clock PLL

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x420																															LPSysPll	

Address: *LPCBaseAddress* + 0x420
Type: Read/write
Reset value: Undefined

Description

The **LPSysPll** register controls the System Clock PLL operation when low power mode is entered. The coding of the LPSysyPll value is shown in the table below.

Bitfield	Function
LPSysPll	00: PLL off 01: PLL reference on and power on 10: reference on and power on 11: PLL on

LPM_SysRatio System clock ratio

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x500																											SysRatio					

Address: *LPCBaseAddress* + 0x500
Type: Read only
Reset value: Undefined

Description

The **SysRatio** register is a read only register which reflects the frequency of the ST20 system clock (clock used by the CPU) selected by the **SpeedSelect0-1** pins.

It contains the binary code of the state of the **SpeedSelect0-1** pins respectively.

Bitfield	Function
SysRatio	0: times one mode 1: 60 2: 39.9 3: 49.875

LPM_WdEnable Watchdog enable

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x510		Enable

Address: *LPCBaseAddress* + 0x510
Type: Read/write
Reset value: Undefined

Description

Setting the **WdEnable** register enables the low power alarm counter to be used as a watchdog timer.

Bitfield	Function
Enable	0: Alarm 1: Watchdog

LPM_WdFlag Watchdog flag

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x514		WdFlag

Address: *LPCBaseAddress* + 0x514
Type: Read only
Reset value: Undefined

Description

The **WdFlag** register is set when a watchdog reset occurs.

Bitfield	Function
WdFlag	0: No watchdog reset has occurred. 1: A watchdog reset has occurred.

17 On-screen display (OSD)

OSD_ACT Active Signal

	7	6	5	4	3	2	1	0
0x3E		OAM	OAD[5:0]					

Address: *VideoBaseAddress* + 0x3E
Type: Read/write
Reset value: 0

Description

Bitfield	Function
OAD	*OSD Active signal delay.* These bits are used to define the delay of the OSD active signal corresponding to output OSD pixels.
OAM	*OSD Active signal mode.* When this bit is set, OSD active signal is an input. When it is reset, OSD active signal is an output. Note that OSD active signal must never be driven when VID_CTL.EVI=1 and VID_OSD.OAM=0.

OSD_BDW OSD Boundary Weight

	7	6	5	4	3	2	1	0
0x92			OSD_BDW[5:0]					

Address: *VideoBaseAddress* + 0x92
Type: Read/write
Reset value: 0

Description

These bits represent the value of mix_weight at the horizontal border of an OSD region or at a horizontal border of a transparent region within an OSD region. The OSD_BDW value is used in OSD filter formulae and has therefore no meaning when the OSD is in normal mode.

OSD_BOT OSD Bottom Field Pointer

	7	6	5	4	3	2	1	0
1st cycle		OBP[14:8]						
2nd cycle	OBP[7:0]							

Address: *VideoBaseAddress* + 0x2B
Type: Serial read/write
Reset value: 0
Synchronization: VSYNC bottom

Description

This register is written or read in two cycles:

- first cycle: OBP[14:8]
- second cycle: OBP[7:0]

The circularity is reset by a hardware reset or a bottom field VSYNC. The register holds the start address, in units of 256 bytes of the current OSD specification buffer for the bottom field. This specification will be decoded during bottom fields when OSD is enabled.

OSD_CFG OSD configuration

	7	6	5	4	3	2	1	0
0x91				LLG	Reserved	ENA	NOR	FIL

Address: VideoBaseAddress + 0x91
Type: RW
Reset value: 0

Description

Bitfield	Description
FIL	*OSD Filter mode*: to activate one of the two filter modes bit NOR must be set. 0: Anti-flicker mode active 1: Anti-flutter mode active
NOR	*OSD Normal mode:* 0: OSD normal mode active, filter modes specified in OSD CFG.FIL will NOT be active 1: Filter mode specified in OSD_CFG.FIL is active
ENA	*OSD Enable.* 0: OSD disabled, 1: OSD enabled
LLG	Linked list granularity: 0: OSD linked-list pointer can point to any word in the lower 32Mbits of the 64Mbit SDRAM area 1: OSD linked-list pointer can point to any *alternate* word in *all* of the 64Mbit SDRAM area. This effectively doubles the available area by halving the placement precision.

OSD_TOP OSD Top Field Pointer

	7	5	0
1st cycle		OTP[14:8]	
2nd cycle	OTP[7:0]		

Address: *VideoBaseAddress* + 0x2A
Type: Serial read/write
Reset value: 0
Synchronization: VSYNC top

Description

This register is written or read in two cycles:

- first cycle: OTP[14:8]
- second cycle: OTP[7:0]

The circularity is reset by a hardware reset or a top VSYNC. The register holds the start address, in units of 256 bytes, of the current OSD specification buffer for the top field. This specification will be decoded during top fields when OSD is enabled.

18 PES parser (PES)

PES_CF1 PES Audio Decoding Control

	7	6	5	4 ... 0
0x40	SDT		IAI	AUD_ID[4:0]

Address: *VideoBaseAddress* + 0x40
Type: Read/write
Reset value: 0
Synchronization: None

Description

This register controls the audio stream decoding and some miscellaneous parser functions.

Bitfield	Description
AUD_ID	Audio Stream id.
IAI	Ignore Audio Stream id.
SDT	Store DTS not PTS.

PES_CF2 PES Video Parser Control

	7 6	5	4	3 ... 0
0x41	MOD[1:0]	SS	IVI	VID_ID[3:0]

Address: *VideoBaseAddress* + 0x41
Type: Read/write
Reset value: 0
Synchronization: None

Description

This register controls the video stream parser.

Bitfield	Description
VID_ID[3:0]	Video Stream i.d.
IVI	Ignore Video Stream i.d.
SS	System Stream. This bit, when set, indicates that the stream sent to the parser is a system stream. To decode a pure video or audio stream this bit should be set to zero.
MOD[1:0]	00: Automatic configuration. The parser will configure itself to decode the incoming stream. 01: MPEG-1 system stream with one data strobe input format 10: MPEG-2 PES separate data strobes input format (standard mode). 11: MPEG-2 whole PES video and whole PES audio one after the other with single data strobe input format.

PES_TM1 DSM Trick Mode

	7	0
0x42	PES_TM1[7:0]	

Address: *VideoBaseAddress* + 0x42
Type: Read only
Reset value: 0
Synchronization: None

Description

This register stores the DSM trick mode bits. This register may be used only if the ES rate flag of the bit stream is set to zero.

PES_TM2 PES Parser Status

	7	1	0
0x43		M2	DSA

Address: *VideoBaseAddress* + 0x43
Type: Read only
Reset value: 2
Synchronization: None

Description

Bitfield	Definition
M2	*MPEG-2 not MPEG-1.* This bit indicates, in automatic mode, if the current stream being decoded is an MPEG-2 or an MPEG-1 stream.
DSA	DSM association flag. This bit, when set indicates that the picture header present in the start code detector is associated to DSM values present in the **PES_TM1** register.

PES_TS PES Time Stamps

	7	1	0
0x49	PES_TS [7:0]		
0x4A	PES_TS [15:8]		
0x4B	PES_TS [23:16]		
0x4C	PES_TS [31:24]		
0x4D		TSA	PES_TS [32]

Address: *VideoBaseAddress* + 0x49 to 0x4D
Type: Read only
Reset value: 0
Synchronization: None

Description

Bitfield	Definition
TSA	Time stamp association. When this bit is set it indicates that the picture to be decoded next (from which the header is available in the start code detector) has an associated time stamp available in PES_TS.

19 Parallel input/output (PIO)

Each eight-bit PIO port has a set of eight-bit registers. Each of the eight bits of each register refers to the corresponding pin in the corresponding port.

PIO_Clear_P*n*C2:0 Clear bits of P*n*C2:0

		7	6	5	4	3	2	1	0
Clear_P*n*C0	0x28	Clear_PC0[7:0]							
Clear_P*n*C1	0x38	Clear_PC1[7:0]							
Clear_P*n*C2	0x48	Clear_PC2[7:0]							

Address: *PIO*n*BaseAddress* + 0x28, 0x38 and 0x48
Type: Write only

Description

Clear_P*n*C2:0 allow the bits of registers **P*n*C2:0** to be cleared individually. Writing a ‘1’ in one of these register clears the corresponding bit in the corresponding **P*n*C2:0** register, while a ‘0’ leaves the bit unchanged.

PIO_Clear_P*n*Comp Clear bits of P*n*Comp

	7	6	5	4	3	2	1	0
0x58	Clear_PComp[7:0]							

Address: *PIO*n*BaseAddress* + 0x58
Type: Write only

Description

Clear_P*n*Comp allows bits of **P*n*Comp** to be cleared individually. Writing a ‘1’ in this register clears the corresponding bit in the **P*n*Comp** register, while a ‘0’ leaves the bit unchanged.

PIO_Clear_P*n*Mask Clear bits of P*n*Mask

	7	6	5	4	3	2	1	0
0x68	Clear_PMask[7:0]							

Address: *PIO*n*BaseAddress* + 0x68
Type: Write only

Description

Clear_P*n*Mask allows bits of **P*n*Mask** to be cleared individually. Writing a ‘1’ in this register clears the corresponding bit in the **P*n*Mask** register, while a ‘0’ leaves the bit unchanged.

PIO_Clear_P*n*Out Clear bits of P*n*Out

	7	6	5	4	3	2	1	0
0x08	Clear_POut[7:0]							

Address: *PIOnBaseAddress* + 0x08
Type: Write only

Description

Clear_P*n*Out allows bits of **P*n*Out** to be cleared individually. Writing a ‘1’ in this register clears the corresponding bit in the **P*n*Out** register, while a ‘0’ leaves the bit unchanged.

PIO_P*n*C2:0 PIO configuration

		7	6	5	4	3	2	1	0
P*n*C0	0x20	ConfigData0[7:0]							
P*n*C1	0x30	ConfigData1[7:0]							
P*n*C2	0x40	ConfigData2[7:0]							

Address: *PIOnBaseAddress* + 0x20, 0x30 and 0x40
Type: Read/write
Reset value: 0

Description

There are three configuration registers (**P*n*C0**, **P*n*C1** and **P*n*C2**) for each port, which are used to configure the PIO port pins. Each pin can be configured as an input, output, bidirectional, or alternative function pin (if any), with options for the output driver configuration.

Three bits, one bit from each of the three registers, configure the corresponding bit of the port. The configuration of the corresponding I/O pin for each valid bit setting is given in the table below.

P*n*C2[*y*]	**P*n*C1[*y*]**	**P*n*C0[*y*]**	**Bit *y* configuration**	**Bit *y* output**
0	0	0	Input	Hi-Z with weak pull-up
0	0	1	Bidirectional	Open drain.
0	1	0	Output	Push-pull.
0	1	1	Bidirectional	Open drain.
1	0	0	Input	Hi-Z.
1	0	1	Input	Hi-Z.
1	1	0	Alternative function output	Push-pull.
1	1	1	Alternative function bidirectional	Open drain.

The **P*n*C** registers are each mapped onto two additional addresses **Set_P*n*C** and **Clear_P*n*C** so that bits can be set or cleared individually.

PIO_P*n*Comp PIO input comparison

	7	6	5	4	3	2	1	0
0x50	PComp[7:0]							

Address: *PIO*n*BaseAddress* + 0x50
Type: Read/write
Reset value: 0

Description

The input compare register **P***n***Comp** can be used to cause an interrupt if the input value differs from a fixed value.

The input data from the PIO ports pins will be compared with the value held in **P***n***Comp**. If any of the input bits is different from the corresponding bit in the **P***n***Comp** register and the corresponding bit position in **P***n***Mask** is set to 1, then the internal interrupt signal for the port will be set to 1.

The compare function is sensitive to changes in levels on the pins. For the comparison to be seen as a valid interrupt by an interrupt handler, the change in state on the input pin must be longer in duration than the interrupt response time.

The compare function is operational in all configurations for each PIO bit, including the alternative function modes.

The **P***n***Out** register is mapped onto two additional addresses **Set_P***n***Out** and **Clear_P***n***Out** so that bits can be set or cleared individually.

PIO_P*n*In PIO input

	7	6	5	4	3	2	1	0
0x10	PIn[7:0]							

Address: *PIO*n*BaseAddress* + 0x10
Type: Read only
Reset value: 0

Description

The data read from this register will give the logic level present on the input pins of the port at the start of the read cycle to this register. Each bit reflects the input value of the corresponding bit of the port. The read data will be the last value written to the register regardless of the pin configuration selected.

PIO_P*n*Mask PIO input comparison mask

	7	6	5	4	3	2	1	0
0x60	PMask[7:0]							

Address: *PIO*n*BaseAddress* + 0x60
Type: Read/write
Reset value: 0

Description

When a bit is set to 1, the compare function for the internal interrupt for the port is enabled for that bit. If the respective bit (7 to 0) of the input is different from the corresponding bit in the **P***n***Comp** register, then an interrupt is generated.

The **P*n*Mask** register is mapped onto two additional addresses **Set_P*n*Mask** and **Clear_P*n*Mask** so that bits can be set or cleared individually.

PIO_P*n*Out PIO output

	7	6	5	4	3	2	1	0
0x00	POut[7:0]							

Address: *PIO*n*BaseAddress* + 0x00
Type: Read/write
Reset value: 0

Description

This register holds output data for the port. Each bit defines the output value of the corresponding bit of the port.

The **P*n*Out** register is mapped onto two additional addresses **Set_P*n*Out** and **Clear_P*n*Out** so that bits can be set or cleared individually.

PIO_Set_P*n*C2:0 Set bits of P*n*C2:0

		7	6	5	4	3	2	1	0
Set_P*n*C0	0x24	Set_PC0[7:0]							
Set_P*n*C1	0x34	Set_PC1[7:0]							
Set_P*n*C2	0x44	Set_PC2[7:0]							

Address: *PIO*n*BaseAddress* + 0x24, 0x34 and 0x44
Type: Write only

Description

Set_P*n*C2:0 allow the bits of registers **P*n*C2:0** to be set individually. Writing a ‘1’ in one of these registers sets the corresponding bit in the corresponding **P*n*C2:0** register, while a ‘0’ leaves the bit unchanged.

PIO_Set_P*n*Comp Set bits of P*n*Comp

	7	6	5	4	3	2	1	0
0x54	Set_PComp[7:0]							

Address: *PIO*n*BaseAddress* + 0x54
Type: Write only

Description

Set_P*n*Comp allows bits of **P*n*Comp** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **P*n*Comp** register, while a ‘0’ leaves the bit unchanged.

PIO_Set_P*n*Mask Set bits of P*n*Mask

	7	6	5	4	3	2	1	0
0x64	Set_PMask[7:0]							

Address: *PIOnBaseAddress* + 0x64
Type: Write only

Description

Set_P*n*Mask allows bits of **P*n*Mask** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **P*n*Mask** register, while a ‘0’ leaves the bit unchanged.

PIO_Set_P*n*Out Set bits of P*n*Out

	7	6	5	4	3	2	1	0
0x04	Set_POut[7:0]							

Address: *PIOnBaseAddress* + 0x04
Type: Write only

Description

Set_P*n*Out allows bits of **P*n*Out** to be set individually. Writing a ‘1’ in this register sets the corresponding bit in the **P*n*Out** register, while a ‘0’ leaves the bit unchanged.

20 Programmable transport interface (PTI)

20.1 PTI DMA registers

PTI_DMA0Status DMA channel 0 status

	7	6	5	4	3	2	1	0
0x1040							DMA0Overflow	Reserved

Address: *PTIBaseAddress* + 0x1040
Type: Read/write
Reset value: Undefined

Description

The **DMA0Status** register shows whether DMA channel 0 overflowed. This is only used when debugging TC code. The TC code normally is designed to read the **DMA0Overflow** bit and signal this condition to the ST20 software via one of the interrupt status bits. The interrupt bit is also used as a handshake that the ST20 software has acknowledged the condition. Data is discarded by DMA channel 0 if the buffer it is writing overflows.

Bitfield	Description
DMA0Overflow	If ‘1’, the channel 0 circular buffer overflowed. Reset by writing ‘1’ to this bit.
Reserved	Write 0.

PTI_DMA*n*Base DMA buffer base address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x1000	DMA0Base
0x1004	DMA1Base
0x1008	DMA2Base
0x100C	DMA3Base

Address: *PTIBaseAddress* + 0x1000 to 0x100C
Type: Write only

Description

Each of these registers holds the base address of the DMA buffer for the corresponding DMA channel. This address must be aligned to a 16-byte boundary, so bits 3:0 must be written as 0. Bits 31:30 of this address must be the same as bits 31:30 of the corresponding **DMA*n*Top** address register. The DMA channel 0 register would be set up for each PID in the PID data structure in the PTI data memory.

PTI_DMA*n*Burst DMA byte or word transfers

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x1044	Reserved	DMA1Burst
0x1048	Reserved	DMA2Burst
0x104C	Reserved	DMA3Burst

Address: *PTIBaseAddress* + 0x1044 to 0x104C
Type: Write only

Description

The **DMA1-3Burst** registers determine the size of the DMA transfer that occurs each time the corresponding **notCDReq1-3** signal is low.

'0' means write one word at a time. '1' means write one byte at a time.

PTI_DMA*n*Holdoff DMA hold off time

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5	4 3 2 1 0
0x1054		DMA1Holdoff
0x1058		DMA2Holdoff
0x105C		DMA3Holdoff

Address: *PTIBaseAddress* + 0x1054 to 0x105C
Type: Write only

Description

The **DMA1-3Holdoff** registers are used to specify the delay time between the end of a burst of data being transferred and re-sampling the **notCDReq1-3** signals before another transfer is started on a DMA channel. The time is in units of byte clock cycles in the range 0 to 31 cycles.

PTI_DMA*n*Read DMA buffer read address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x1030	DMA0Read
0x1034	DMA1Read
0x1038	DMA2Read
0x103C	DMA3Read

Address: *PTIBaseAddress* + 0x1030 to 0x103C
Type: Read/write except **DMA0Read** which is write only
Reset value: Undefined

Description

Each of these registers holds the read address within the DMA buffer for the corresponding DMA channel. The address in the register is always a pointer to the next byte to be read except when the buffer is empty. The pointer must remain between the addresses defined by the base and top register. The channel 0 register would be initialized for each PID in the PID data structure in the PTI data shared memory, and an updated read pointer would be written into the same data structure.

The read pointer is initialized to be equal to the write pointer. As data is read from the buffer the read pointer will be updated by the hardware.

For channels 1 to 3, the read pointer is automatically wrapped round, and cannot overtake the write pointer. On reaching the top pointer address the read pointer will wrap around to continue reading from the base address. If the read pointer is equal to the write pointer then no data will be read.

For channel 0, if the buffer is not being read by any of the DMA channels 1 to 3, then the ST20 software must perform these checks, i.e. that:

- if the read address is not less than the top address, then the read address must be wrapped round to the base address;
- if the read address is equal to the write address then the buffer is empty and data should not be read.

PTI_DMA*n*Top DMA buffer top address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x1010	DMA0Top
0x1014	DMA1Top
0x1018	DMA2Top
0x101C	DMA3Top

Address: *PTIBaseAddress* + 0x1010 to 0x101C
Type: Write only

Description

Each of these registers holds the top address of the DMA buffer for the corresponding DMA channel. If the buffer size is not zero then this address must be 1 less than a 16-byte boundary, so bits 3:0 must be written as 1. Bits 31:30 of this address must be the same as bits 31:30 of the corresponding **DMA*n*Base** address register.

For channel 0 only, if the buffer size is zero then this address must be equal to the corresponding **DMA*n*Base** address. The DMA channel 0 register would be set up for each PID in the PID data structure in the PTI data SRAM.

PTI_DMA*n*Write DMA buffer write addres

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x1020	DMA0Write
0x1024	DMA1Write
0x1028	DMA2Write
0x102C	DMA3Write

Address: *PTIBaseAddress* + 0x1020 to 0x102C

Type: Write only except **DMA0Write** which is read/write
Reset value: Undefined

Description

Each of these registers holds the write address within the DMA buffer for the corresponding DMA channel. The address in the register is always a pointer to the next location for a byte to be written. The pointer must remain between the addresses defined by the base and top registers. The DMA channel 0 register would be set up for each PID in the PID data structure in the PTI data SRAM and an updated write pointer would be read from the same data structure.

The write pointer is initialized to be equal to the same address as the read pointer. As data is written into the buffer the write pointer will be updated and after reaching the top pointer address will wrap around to write the next byte at the base pointer address. In the case of channel 0 the DMA hardware will update the write pointer, the TC will copy this back to the data SRAM at the end of a packet, and the CPU should read the pointer from the SRAM. In the case of channels 1-3 the write pointer will be updated by the TC if channel 0 is writing to the buffer for that channel or by the CPU in the case that the CPU is writing the data into the buffer.

PTI_DMACDAddr DMA CD FIFO page address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14	13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x1060	DMACDAddr	

Address: *PTIBaseAddress* + 0x1060
Type: Write only

Description

DMA channels 1 to 3 must all write to the same 16 Kbyte page in memory. The **DMACDAddr** register holds the address of the base of the 16 Kbyte page. The page must be 16 Kbyte aligned, so the least significant 14 bits of **DMACDAddr** must be 0.

For DMA channels 1 to 3, certain aspects of the output are fixed in the DMA channels. The destination addresses for the written data are at fixed offsets from the output base address given by **DMACDAddr**. The offsets are shown in the table below. This table also shows which decoder is connected to each DMA channel.

DMA Channel	**Write address offset from DMACDAddr**	**Target decoder**	**DMA request signal**
1	#2000	Video decoder	notCDReq1
2	#3000	Audio decoder	notCDReq2
3	#0000	Sub-picture decoder	notCDReq3

PTI_DMAEnable DMA Enable

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9	8	7	6	5
0x1050		DMA3 Enable	DMA2 Enable	DMA1 Enable	DMA0 Enable

Address: *PTIBaseAddress* + 0x1050
Type: Write only

Description

This register controls the enabling of the four DMA channels. Disabling channels 1-3 will not result in any data being lost inside the DMA controller but data may be lost by buffer overflow. Disabling channel 0 may result in lost data depending on the input data rate to the PTI and the length of time that the channel is disabled.

Bit field	Function
DMA0Enable	Enable (‘1’) or disable(‘0’) channel 0.
DMA1Enable	Enable (‘1’) or disable(‘0’) channel 1.
DMA2Enable	Enable (‘1’) or disable(‘0’) channel 2.
DMA3Enable	Enable (‘1’) or disable(‘0’) channel 3.

20.2 PTI input interface registers

IPTI_IFAltFifoCount IIF alternative FIFO count

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

	31–8	7–0
0x2004		Count

Address: *PTIBaseAddress* + 0x2004
Type: Read only
Reset value: Undefined

Description

The number of bytes in the alternative FIFO. This FIFO does not overflow since there is flow control on its input.

PTI_IIFAltLatency IIF alternative output latency

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

	31–8	7–0
0x2010		Latency

Address: *PTIBaseAddress* + 0x2010
Type: Write only

Description

The number of byte clock cycles from a transport packet header being latched at input to data being available at the alternative output.

PTI_IIFFifoCount IIF count

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

	31–8	7	6–0
0x2000		Full	Count

Address: *PTIBaseAddress* + 0x2000
Type: Read only
Reset value: Undefined

Description

Bit field	Function
Count	The number of bytes in the input FIFO.
Full	A full flag which is set when the FIFO becomes full. It is reset when the ST20 reads this register.

PTI_IIFFifoEnable IIF FIFO enable

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x2008		Count

Address: *PTIBaseAddress* + 0x2008
Type: Read/write
Reset value: Undefined

Description

Allow data into the input FIFO. When this field is zero, the input FIFO is reset. After the ST20 completes the initialization sequence, it should set this field to 1.

PTI_IIFSyncDrop IIF sync drop

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x2018		DropPackets

Address: *PTIBaseAddress* + 0x2018
Type: Write only

Description

The number of successive erroneous sync bytes found before the lock is treated as lost.

PTI_IIFSyncLock IIF sync lock

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x2014		LockPackets

Address: *PTIBaseAddress* + 0x2014
Type: Write only

Description

The number of successive correct sync bytes found before the sync detection is locked.

20.3 PTI configuration registers

PTI_AudPTS Audio Presentation Time Stamp

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

0x40	AudPTS[31:0]	
0x44		AudPTS [32]

Address: *PTIBaseAddress* + 0x40 and 0x44
Type: Read only
Reset value: Undefined

Description

The System Time Clock value is latched into this register at the beginning of audio frame output.

The audio and video PTS registers can be used by driver software to synchronize the audio and video. The time stamps are 33 bit values, so the most significant bit is held at a separate address.

PTI_IntAck PTI interrupt acknowledgment

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x20		IntAck0
0x24		IntAck1
0x28		IntAck2
0x2C		IntAck3

Address: *PTIBaseAddress* + 0x20 to 0x2C
Type: Write only

Description

Acknowledge the corresponding interrupt bit when a bit is written as a 1.

PTI_IntEnable PTI interrupt enable

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x10		IntEnable0
0x14		IntEnable1
0x18		IntEnable2
0x1C		IntEnable3

Address: *PTIBaseAddress* + 0x10 to 0x1C
Type: Read/write
Reset value: Undefined

Description

The corresponding interrupt is enabled when a bit is 1.

PTI_IntStatus0-3 PTI interrupt status

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x00		IntStatus0
0x04		IntStatus1
0x08		IntStatus2
0x0C		IntStatus3

Address: *PTIBaseAddress* + 0x00 to 0x0C
Type: Read only
Reset value: Undefined

Description

An interrupt is set when any bit of one of these registers is 1. The **PTIIntStatus0-3** registers are used by the TC to raise an interrupt to the ST20 by writing a 1 to a bit in one of the registers. Each of the four registers has 16 bits. All 64 bits are ORed together to produce a single interrupt for the PTI.

Once the TC has set bits to 1, each bit will stay set until the interrupt is acknowledged by the ST20 software by writing 1 to the corresponding bit of the corresponding **PTIIntAck** register. An interrupt can be masked by writing 0 to the corresponding enable bit of the corresponding **PTIIntEnable** register.

PTI_VidPTS Video Presentation Time Stamp

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x48	VidPTS[31:0]	
0x4C		VidPTS [32]

Address: *PTIBaseAddress* + 0x48 and 0x4C
Type: Read only
Reset value: Undefined

Description

The System Time Clock value is latched into this register at VSYNC.

The audio and video PTS registers can be used by driver software to synchronize the audio and video. The time stamps are 33 bit values, so the most significant bit is held at a separate address.

PTI_STCTimer Set STC timer

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
0x50	STCTimer[31:0]	
0x54		STCTimer[32]

Address: *PTIBaseAddress* + 0x50 to 0x54
Type: Write only

Description

These registers, when written, load a new value into the STC timer counter. The load is performed on the next clock edge of the 27 MHz clock after the write.

The most significant bit of the value to be loaded must be written to the register containing **STCTimer[32]** before writing to the **STCTimer[31:0]** register, as the write to the **STCTimer[31:0]** causes the update of the 33-bit counter value.

20.4 PTI section filter registers

PTI_SFFilterData*n* Section filter data

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x4000	SFFilterData0[31:0]
0x4008	SFFilterData0[63:32]
0x4010	SFFilterData1[31:0]
0x4018	SFFilterData1[63:32]
...	...
0x41F0	SFFilterData31[31:0]
0x41F8	SFFilterData31[63:32]

Address: *PTIBaseAddress* + 0x4000, 0x408, 0x4010, ... and 0x41F8
Type: Read/write
Reset value: Undefined

Description

The **SFFilterData** registers are 64-bit registers holding the filter data. There is one **SFFilterData** register for each of the 32 section filters. Each register is split into two 32-bit registers; **SFFilterData[31:0]** holds the least significant word, and **SFFilterData[63:32]** holds most significant word. This enables the least significant or most significant word to be written independently without affecting the other word.

Only one word of a filter can be updated at a time, so updates of the section filters should be carried out when one of the following is true:

- a false match will not matter;
- the filter is not in use for any PIDs;
- the PIDs that use the filter have been disabled.

Filtering is performed by testing whether:

packet BITAND *FilterMask* = *FilterData* BITAND *FilterMask*

If this is true then the packet passes through the filter.

PTI_SFFilterMask*n* Section filter mask

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x4004	SFFilterMask0[31:0]
0x400C	SFFilterMask0[63:32]
0x4014	SFFilterMask1[31:0]
0x401C	SFFilterMask1[63:32]
...	...
0x41F4	SFFilterMask31[31:0]
0x41FC	SFFilterMask31[63:32]

Address: *PTIBaseAddress* + 0x4004, 0x400C, 0x4014, ... and 0x41FC
Type: Read/write
Reset value: Undefined

Description

The **SFFilterMask** registers are 64-bit registers holding the filter masks. There is one **SFFilterMask** register for each of the 32 section filters. Each register is split into two 32-bit registers; **SFFilterMask[31:0]** holds the least significant word, and **SFFilterMask[63:32]** holds most significant word. This enables the least significant or most significant word to be written independently without affecting the other word.

Bits set to 1 in the **SFFilterMask** registers enable the corresponding bits in the **SFFilterData** registers to be compared with the packet. Bits set to 0 in the mask disable comparison between the corresponding bits, and may be regarded as '*do not care*' bits.

Only one word of a filter can be updated at a time, so updates of the section filters should be carried out when one of the following is true:

- a false match will not matter;
- the filter is not in use for any PIDs;
- the PIDs that use the filter have been disabled.

Filtering is performed by testing whether:

packet BITAND *FilterMask* = *FilterData* BITAND *FilterMask*

If this is true then the packet passes through the filter.

20.5 PTI transport controller mode register

PTI_TC mode Transport controller mode

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3	2	1	0
0x30		TC Single Step	TC Reset Iptr	TC Enable

Address: *PTIBaseAddress* + 0x30
Type: Read/write
Reset value: Undefined

Description

The **TCMode** register allows the ST20 software to control the transport controller. Under normal operation the software should only set the **TCEnable** bit to start the TC executing code.

Bit field	Function
TCEnable	‘1’ enables the TC, ‘0’ disables the TC disabling instruction fetch: reading a zero after a single step means the TC has finished executing the previous instruction.
TCResetIptr	‘1’ resets the instruction pointer of the TC (level sensitive).
TCSingleStep	‘1’ means reset **TCEnable** after each instruction.

If the ST20 software needs to reset the TC instruction pointer, the steps are:

1 clear the **TCEnable** bit,

2 set the **TCReset** bit,

3 clear the **TCReset** bit and

4 set the **TCEnable** bit.

To single-step the TC through its instructions:

1 set the **TCEnable** and **TCSingleStep** bits and

2 wait for the **TCEnable** bit to be cleared.

Repeat this procedure for further single steps.

21 SmartCard interface (SC)

Sc*n*ClkCon SmartCard *n* clock control

	31–2	1	0
0x04		Enable	Source

Address: *SmartCard*n*BaseAddress* + 0x04
Type: Write only
Reset value: 0

Description

The **Sc*n*ClkCon** register controls the source of the clock and determines whether the SmartCard clock output is enabled. The programmable divider and the output are reset when the enable bit is set to 0.

Bit field	Function
Source	Selects source of SmartCard clock. 0: Selects global clock. 1: Selects external pin.
Enable	SmartCard clock generator enable bit. 0: Stop clock, set output low and reset divider. 1: Enable clock generator.

Sc*n*ClkVal SmartCard *n* clock

	31–5	4–0
0x00		ScClkVal

Address: *SmartCard*n*BaseAddress* + 0x00
Type: Write only
Reset value: 0

Comment: ...but 0 is an illegal value!

Description

The **Sc*n*ClkVal** register determines the SmartCard clock frequency. The 5-bit value given in the register is multiplied by 2 to give the division factor of the input clock frequency. For example, if **Sc*n*SlkVal** is 8 then the input clock frequency is divided by 16. The value zero must not be written into this register.

The divider is updated with the new value for the divider ratio on the next rising or falling edge of the output clock.

22 PWM and counter module (PWC)

PWM_*n*CaptureEdge PWM *n* Capture event definition

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1 0
0x30		PWM_0CaptureEdge
0x34		PWM_1CaptureEdge
0x38		PWM_2CaptureEdge
0x3C		PWM_3CaptureEdge

Address: *PWMBaseAddress* + 0x30 (PWM_0CaptureEdge) to 0x3C (PWM_3CaptureEdge)
Type: Read/write
Reset value: Undefined

Description

The code in register PWM_nCaptureEdge defines what constitutes an event on input pin **CaptureIn***n*. Possible events are rising edge, falling edge, both or neither (in other words, disabled).

Bitfield	Description
CaptureEdge	01: Capture on rising edge 10: Capture on falling edge 11: Capture on rising or falling edge 00: Capture disabled

PWM_*n*CaptureVal PWM *n* capture value

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x10	PWM0CaptureVal
0x14	PWM1CaptureVal
0x18	PWM2CaptureVal
0x1C	PWM3CaptureVal

Address: *PWMBaseAddress* + 0x10 (PWM0CaptureVal0) to 0x1C (PWM3CaptureVal)
Type: Read only
Reset value: Undefined

Description

Each of the four capture value registers holds the 32-bit counter value at the time of the last event occurring at the corresponding CaptureIn pin. When an input event occurs on input CaptureInn, the value of the counter in register PWM_CaptureCount at that time is captured in register PWM_nCaptureVal. The value can be any 32-bit value.

When an input event occurs, an interrupt is generated provided the CapturenIntEn bit of the PWM_IntEnable register is set to 1. Bit CaptureIntn of register PWM_IntStatus becomes 1, and can be reset by writing 1 to bit CaptureIntAckn of register PWM_IntAck.

The counter is not stopped nor reset by any of these events.

PWM_*n*CompareOutVal PWM *n* compare output value

31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	0
	Compare OutVal2

Address: *PWMBaseAddress* + 0x40 to 0x4C
Type: Read/write
Reset value: Undefined

Description

Register PWM_CompareOutValN holds the value which will be written to the PWM_CompareOutN pin when the compare value in PWM_CompareValN matches the counter value PWM_CaptureCount.

PWM_*n*CompareVal PWM *n* compare value

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x20	PWM0CompareVal
0x24	PWM1CompareVal
0x28	PWM2CompareVal
0x2C	PWM3CompareVal

Address: *PWMBaseAddress* + 0x20 (PWM0CompareVal) to 0x2C (PWM3CompareVal)
Type: Read/write
Reset value: Undefined

Description

Each of the four compare registers PWM_nCompareVal in the module can be set to any 32-bit value. When the counter in register PWMCaptureCount reaches the value of register PWM_nCompareVal, two things happen:

1 An interrupt is generated provided the PWM_nCompareIntEn bit of the PWM_IntEnable register is set to 1. Bit PWM_CompareIntN of register PWM_IntStatus becomes 1, and can be reset by writing 1 to bit PWM_nCompareIntAck of register PWM_IntAck.

2 Pin PWM_nCompareOut takes on the value set in register PWM_nCompareOutVal.

The counter is not stopped nor reset by any of these events.

PWM_*n*Val PWM *n* pulse width

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9	8 7 6 5 4 3 2 1 0
0x00		PWM0Val
0x04		PWM1Val
0x08		PWM2Val
0x0C		PWM3Val

Address: *PW*MBaseAddress + 0x00 (PWM_0Val) to 0x0C (PWM_3Val)
Type: Read/write

Reset value: Undefined

Description

These registers hold the counter values, which are used to determine the width of the pulse generated on the PWMOut pin.

PWMOut pulse width = (PWMVal + 1) x prescaled clock period

If PWMVal is 255 then PWMOut does not go low.

PWM_CaptureCount PWM capture/compare counter

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>28</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="32">CaptureCount</td></tr>
</table>

Address: *PWMBaseAddress* + 0x64
Type: Read/write
Reset value: Undefined

Description

This register holds the shared capture/compare counter used by all the capture and compare functions.

The capture/compare counter is clocked from the prescaled system clock. The prescaling factor, and therefore the period represented by one count, is determined by the value of field CaptureClkValue in register PWM_Control. The factor can be from 1 to 32.

The counter is enabled by setting the PWM_CaptureEnable bit of the PWM_Control register to 1. When it is disabled (PWM_CaptureEnable is 0), none of the capture or compare functions work. PWM_CaptureCount can be read or written at any time.

When the capture/compare counter reaches its maximum count of #FFFFFFFF, it wraps round to count up from zero again.

PWM_Control PWM control register

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>28</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="21"></td><td>Capture Enable</td><td>PWM Enable</td><td colspan="5">Capture ClkValue</td><td colspan="4">PWM ClkValue</td></tr>
</table>

Address: *PWMBaseAddress* + 0x50
Type: Read/write
Reset value: Undefined

Description

Bitfield	Description
PWMClkValue	PWM clock prescale factor 0-15 (divide clock by value + 1)
CaptureClkValue	Capture/compare clock prescale factor 0-31 (divide clock by value + 1)
PWMEnable	Enables PWM counter when = 1
CaptureEnable	Enables capture/compare counter when = 1

PWM_Count PWM output counter

31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																								Count							

Address: *PWMBaseAddress* + 0x60
Type: Read/write (but see text)
Reset value: Undefined

Description

PWM output counter. The counter (in register PWM_Count) is enabled by setting the PWM_Enable bit of the PWM_Control register to 1. When it is disabled (PWM_Enable is 0), pin PWM_Out is forced low. PWM_Count is writable at any time but can have a synchronization latency.

PWM_IntAck PWM interrupt acknowledge

31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																							CmplA3	CmplA2	CmplA1	CmplA0	CptIA3	CptIA2	CptIA1	CptIA0	IntAck

Address: *PWMBaseAddress* + 0x5C
Type: Write only

Description

Bitfield	Description
IntAck	Interrupt acknowledge: write 1 to reset PWMInt to 0
CmplA0	Compare 0 interrupt acknowledge: write 1 to reset status bit
CmplA1	Compare 1 interrupt acknowledge: write 1 to reset status bit
CmplA2	Compare 2 interrupt acknowledge: write 1 to reset status bit
CmplA3	Compare 3 interrupt acknowledge: write 1 to reset status bit
CptIA0	Capture 0 interrupt acknowledge: write 1 to reset status bit.
CptIA1	Capture 1 interrupt acknowledge: write 1 to reset status bit
CptIA2	Capture 2 interrupt acknowledge: write 1 to reset status bit
CptIA3	Capture 3 interrupt acknowledge: write 1 to reset status bit

PWM_IntEnable PWM interrupt enable

31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																							CmplE3	CmplE2	CmplE1	CmplE0	CptIE3	CptIE2	CptIE1	CptIE0	IntEn

Address: *PWMBaseAddress* + 0x54
Type: Read/write
Reset value: Undefined

Description

Bitfield	Description
IntEn	PWM counter overflow interrupt enable
CptIE0	Capture 0 interrupt enable: 1 = enabled
CptIE1	Capture 1 interrupt enable: 1 = enabled
CptIE2	Capture 2 interrupt enable: 1 = enabled
CptIE3	Capture 3 interrupt enable: 1 = enabled
CmpIE0	Compare 0 interrupt enable: 1 = enabled
CmpIE1	Compare 1 interrupt enable: 1 = enabled
CmpIE2	Compare 2 interrupt enable: 1 = enabled
CmpIE3	Compare 3 interrupt enable: 1 = enabled

PWM_IntStatus PWM interrupt status

31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																							Cmp3	Cmp2	Cmp1	Cmp0	Cpt3	Cpt2	Cpt1	Cpt0	Int

Address: *PWMBaseAddress* + 0x58
Type: Read only
Reset value: Undefined

Description

Bitfield	Description
Int	1 means PWM counter overflow
Cpt0	Capture 0 interrupt: 1 = interrupt
Cpt1	Capture 1 interrupt: 1 = interrupt
Cpt2	Capture 2 interrupt: 1 = interrupt
Cpt3	Capture 3 interrupt: 1 = interrupt
Cmp0	Compare 0 interrupt: 1 = interrupt
Cmp1	Compare 1 interrupt: 1 = interrupt
Cmp2	Compare 2 interrupt: 1 = interrupt
Cmp3	Compare 3 interrupt: 1 = interrupt

23 Sub-picture decoder (SPD)

SPD_CTL1 Control Register 1

	7	6	5	4	3	2	1	0
0x00	BOT_TOP		P	B	H	V	D	S

Address: *SubPictureBaseAddress* + 0x00
Type: Read/write
Reset value: 0
Synchronization: VSYNC (field)

Description

This register contains control bits for the subpicture decoder.

Bitfield	Description
S	*Sub-Picture Decoder Start command* When this bit is set it indicates the start of a new subpicture unit. The subpicture decoder will then reset the local time reference counter. The state is sampled on each VSYNC.
D	*Decoder Active* This bit is set by the decoder when, at shadow register update, the Sub-picture start bit is sampled high. When the decoder active bit is reset decoding is disabled and can only be re-enabled by the decoder start bit.
V	*Display Active* When reset, the sub-picture display is turned off. However, decoding continues even when the display is disabled. When decoding is disabled using the ‘Decoder active’ bit then the display is automatically disabled.
H	*Highlight Enable* When set, highlighting is turned on inside the sub-picture display area.
B	*Bypass* When set, the run-length decoder is put into transparent mode. This allows standard 2-bit-per-pixel bitmaps to be fed into the sub-picture decoder.
P	*Sub-Picture Pause* When set, the 90KHz clock in sub-picture is paused.
BOT_TOP	00: Subpicture commands are executed on any field 01: A new subpicture command is taken into account on a top field 10: A new subpicture command is taken into account on a bottom field 11: Reserved

SPD_CTL2 Control Register 2

	7	1	0
0x02			Rc

Address: *SubPictureBaseAddress* + 0x02
Type: Read/write
Reset value: 0
Synchronization: None

Description

This register contains control bits for the subpicture decoder.

Rc: *Reset Lut #2*. This bit, when set, resets the auto-increment address counter.

SPD_HCN Highlight Region Contrast

	7 ... 4	3 ... 0
0x16	e2[3:0]	e1[3:0]
0x17	p[3:0]	b[3:0]

Address: *SubPictureBaseAddress* + 0x16 and 0x17
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the highlight contrast map values.

SPD_HCOL Highlight Region Color

	7 ... 4	3 ... 0
0x14	e2[3:0]	e1[3:0]
0x15	p[3:0]	b[3:0]

Address: *SubPictureBaseAddress* + 0x14 and 0x15
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the highlight color map values.

SPD_HLEX Highlight Region End X

	7 ... 2	1 ... 0
0x10		SPD_HLEX[9:8]
0x11	SPD_HLEX[7:0]	

Address: *SubPictureBaseAddress* + 0x10 and 0x11
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Highlight region end position X-coordinate.

SPD_HLEY Highlight Region End Y

	7	2	1	0
0x12			SPD_HLEY[9:8]	
0x13	SPD_HLEY[7:0]			

Address: *SubPictureBaseAddress* + 0x12 and 0x13
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Highlight region end position Y-coordinate.

SPD_HLSX Highlight Region Start X

	7	2	0
0x0C			SPD_HLSX[9:8]
0x0D	SPD_HLSX[7:0]		

Address: *SubPictureBaseAddress* + 0x0C and 0x0D
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Highlight region start position X-coordinate.

SPD_HLSY Highlight Region Start Y

	7	6	5	4	3	2	1	0
0x0E							SPD_HLSY[9:8]	
0x0F	SPD_HLSY[7:0]							

Address: *SubPictureBaseAddress* + 0x0E and 0x0F
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Highlight region start position Y-coordinate.

SPD_LUT Main Lookup Table

	7	0
0x03	SPD_LUT[7:0]	

Address: *SubPictureBaseAddress* + 0x03
Type: Write
Reset value: 0
Synchronization: None

Description

This register allows input of the main lookup table. Writing to this register auto-increments the address in the lookup table. For each color, starting with color 0, the Y component (8-bit) is written first followed by U and V. The process continues for each color up to 16 colors.

SPD_SPR Soft Reset

	7	0
0x01		SPD_SPR

Address: *SubPictureBaseAddress* + 0x01
Type: Read/write
Reset value: 0
Synchronization: None

Description

This register resets the subpicture decoder.

SPR: *Sub Picture Reset*. When this bit is set the sub-picture decoder and its FIFOs are reset. Note that the sub-picture decoder is also reset by a hard reset or a global soft reset.

SPD_SXD0 Sub-picture display area

	7	6	5	4	3	2	1	0
0x24							SPD_SXD0[9:8]	
0x25	SPD_SXD0[7:0]							

Address: *SubPictureBaseAddress* + 0x24 and 0x25
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the parameters which define the subpicture display area within the subpicture decode area. The value represents an offset from the corresponding parameter used to define the subpicture decode area.

For example: The true horizontal start position of the subpicture display will be equal to XDO + SXDO.

SPD_SXD1 Subpicture Display Area

	7	6	5	4	3	2	1	0
0x28							SPD_SXD1[9:8]	
0x29	SPD_SXD1[7:0]							

Address: *SubPictureBaseAddress* + 0x28 and 0x29
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the parameters which define the subpicture display area within the subpicture decode area. The value represents an offset from the corresponding parameter used to define the subpicture decode area.

SPD_SYD0 Subpicture Display Area

	7	6	5	4	3	2	1	0
0x26							SPD_SYD0[9:8]	
0x27	SPD_SYD0[7:0]							

Address: *SubPictureBaseAddress* + 0x26 and 0x27
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the parameters which define the subpicture display area within the subpicture decode area. The value represents an offset from the corresponding parameter used to define the subpicture decode area.

SPD_SYD1 Subpicture Display Area

	7	6	5	4	3	2	1	0
0x2A							SPD_SYD1[9:8]	
0x2B	SPD_SYD1[7:0]							

Address: *SubPictureBaseAddress* + 0x2A and 0x2B
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

These registers contain the parameters which define the subpicture display area within the subpicture decode area. The value represents an offset from the corresponding parameter used to define the subpicture decode area.

SPD_XD0 Sub-picture X Offset

	7	6	5	4	3	2	1	0
0x04							SPD_XD0[9:8]	
0x05	SPD_XD0[7:0]							

Address: *SubPictureBaseAddress* + 0x04 and 0x05
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register value sets the horizontal position of the left hand side of the active subpicture decode region. The position is measured in number of pixels from the left hand edge of the screen.

The minimum values for **SPD_XD0** and **SPD_YD0** are 3 and 2 respectively. If the programmed values are less than this, no subpicture is displayed.

SPD_YD0 Sub-picture Y Offset

	7	6	5	4	3	2	1	0
0x06							SPD_YD0[9:8]	
0x07	SPD_YD0[7:0]							

Address: *SubPictureBaseAddress* + 0x06 and 0x07
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register value sets the vertical position of the top of the active subpicture decode region. The position is measured in number of pixels from the top edge of the screen.

The minimum values for **SPD_XD0** and **SPD_YD0** are 3 and 2 respectively. If the programmed values are less than this, no subpicture is displayed.

24 Synchronous serial controller (SSC)

SSC*n*BRG SSC *n* baud rate generation

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	BRG															

Address: *SSC*n*BaseAddress* + 0x00
Type: Read/write
Reset value: 1

Description

This address is dual purpose. When reading, the current 16-bit counter value is returned. When a value is to this address, the 16-bit reload register is loaded with that value.

SSC*n*Con SSC *n* control

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0C						LPB	EN	MS	SR	PO	PH	HB	BM			

Address: *SSC*n*BaseAddress* + 0x0C
Type: Read/write
Reset value: 0

Description

Bit Field	Function
BM	SSC Data Width Selection 0000: Reserved. Do not use this combination. 0001: 2 bits 0010: 3 bits ... 1111: 16 bits
HB	SSC Heading Control Bit 0: LSB first, 1: MSB first
PH	SSC Clock Phase Control Bit 0:pulse in second half cycle, 1:pulse in first half cycle
PO	SSC Clock Polarity Control Bit 0:clock idles at logic 0, 1: clock idles at logic 1
SR	SSC Software Reset 0:device is not reset, 1:all functions are reset while this bit is set
MS	SSC Master Select Bit 0:slave mode, 1:master mode
EN	SSC Enable Bit 0:transmission and reception disabled, 1: ransmission and reception enabled
LPB	SSC Loopback Bit 0:disabled, 1:shift register output is connected to shift register input

Comment: Reset value for BM is illegal.

SSC*n*I2C SSC *n* I²C control

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x18												AD10	ACKG	STOPG	STRTG	I2CM

Address: *SSCnBaseAddress* + 0x18
Type: Read/write
Reset value: 0

Description

Bit Field	Function
I2CM	SSC I²C control bit 0: disabled, 1: enable I²C features
STRTG	SSC I²C generate START condition 0: disabled, 1: generate a START condition
STOPG	SSC I²C generate STOP condition 0: disabled, 1: generate a STOP condition
ACKG	SSC I²C generate acknowledge bits 0: disabled, 1: generate acknowledge bits when receiving
AD10	SSC I²C 10-bit addressing control 0: disabled, 1: use 10 bit addressing

SSC*n*IEn SSC *n* interrupt enable

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x10								ARBLEN	STOPEN	AASEN		PEEN	REEN	TEEN	TIEN	RIEN

Address: *SSCnBaseAddress* + 0x10
Type: Read/write
Reset value: 0

Description

This register holds the interrupt enable bits, which can be used to mask the interrupts.

Bit Field	Function
RIEN	Receiver Buffer Full Interrupt Enable 1: receiver buffer interrupt enabled
TIEN	Transmitter Buffer Empty Interrupt Enable 1: transmitter buffer empty interrupt enabled
TEEN	Transmit Error Interrupt Enable 1: transmit error interrupt enabled
REEN	Receive Error Interrupt Enable 1: receive error interrupt enabled
PEEN	Phase Error Interrupt Enable 1: phase error interrupt enabled
AASEN	I^2C Addressed As Slave Interrupt Enable 1: addressed as slave interrupt enabled
STOPEN	I^2C Stop Condition Interrupt Enable 1: stop condition interrupt enabled
ARBLEN	I^2C Arbitration Lost Interrupt Enable 1: arbitration lost interrupt enabled

SSC*n*RBuf SSC *n* receive buffer

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x08	**RD[15:0]**															

Address: *SSC*n*BaseAddress* + 0x08
Type: Read only
Reset value: 0

Description

Receive buffer data **D15** to **D0**.

SSC*n*SlAd SSC *n* slave address

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x1C							**SL[9:7]**			**SL[6:0]**						

Address: *SSC*n*BaseAddress* + 0x1C
Type: Write only
Reset value: 0

Description

The slave address is written into this register. If the address is a 10-bit address it is written into bits 9:0. If the address is a 7-bit address then it is written into bits 6:0.

SSC*n*Stat SSC *n* status

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x14							BUSY	ARBL	STOP	AAS	CLST	PE	RE	TE	TIR	RIR

Address: *SSC*n*BaseAddress* + 0x14
Type: Read only
Reset value: 2, i.e. all active bits clear except **TIR**.

Description

Bitfield	Function
RIR	Receiver buffer full flag 1: receiver buffer full
TIR	Transmitter buffer empty flag 1: transmitter buffer empty
TE	Transmit error flag 1: transmit error set
RE	Receive error flag 1:receive error set
PE	Phase error flag 1:phase error set
CLST	I^2C clock stretch flag 1:clock stretching in operation
AAS	I^2C addressed as slave flag 1:addressed as slave device
STOP	I^2C stop condition flag 1:stop condition detected
ARBL	I^2C arbitration lost flag 1:arbitration lost
BUSY	I^2C bus busy flag 1:I^2C bus busy

SSC*n*TBuf SSC *n* transmit buffer

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x04	**TD[15:0]**															

Address: *SSC*n*BaseAddress* + 0x04
Type: Write only
Reset value: 0

Description

Transmit buffer data **D15** to **D0**.

25 Still picture plane (TDL)

TDL_DCF TDL Configuration

	7	6	5	4	3	2	1	0
0xF4				FORMAT	UND	UDS	DSR	ETDL

Address: *VideoBaseAddress* + 0xF4
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Bitfield	Description
ETDL	*TDL Enable video display* When this bit is reset, the still picture plane has a constant value of Y=16, C_B=C_R=128. In this case no fading is possible between still picture plane and video picture.
DSR	*Disable SRC.* When this bit is set, both luminance and chrominance SRCs (sample rate converters) are disabled.
UDS	*Up / Down Scaling* When this bit is set, enable the vertical resizing of the still picture plane.
UND	*Up Not Down* This bit is valid only if UDS is set. When UND = 1 then a vertical size of the still picture plane is multiply by 2 by repeating each line. When UND is reset then the vertical size of the still picture plane is divide by 2 by line dropping.
FORMAT	The still picture is stored in 4:2:2 raster scan format, the format order is set by this bit: 0: Format order is: Cb Y Cr Y Cb Y Cr Y 1: Format order is: Y Y Y Y Cb Cb Cr Cr

TDL_LSR SRC Luma/Chroma Resolution

	7 ... 0	
0xEB	TDL_LSR[7:0]	
0xED		TDL_LSR[8]

Address: *VideoBaseAddress* + 0xEB, 0xED
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the upsampling factor of the luminance SRC (sample rate converter). It applies on chroma also. The upsampling factor is equal to 256/LSR. Below are given some examples of upsampling factors, where in each case the displayed picture has a nominal width of 720 pels[1]. Also shown are the numbers of valid pels generated, “N”, calculated as shown in the STi5512 datasheet

Decoded Picture Width	LSR	N
640	228	715
640	227	718
544	193	717
544	192	721
480	170	717
480	169	722
352	125	713
352	124	719
704	250	717
704	249	720

Table 46 Upsampling factors

Note *Downsampling is not supported*

TDL_SCN Still picture plane scan vector

	7	6	5	4	3	2	1	0
0xB0	TDL_HSCN							TDL_SCN[8]
0xB1	TDL_SCN[7:0]							

Address: *VideoBaseAddress* + 0xB0 to 0xB1
Type: Read/Write
Reset value: 0
Synchronization: VSYNC

Description

Bitfield	Description
TDL_SCN	This value sets the location of the first displayed line of the still picture plane, with respect to the decoded picture. For example, if TDL_SCN=11, then the 11 top lines of each field of the still picture plane are not displayed.
TDL_HSCN	This value sets the location of the first displayed pixel of the still picture plane, with respect to the decoded picture. For example, if TDL_HSCN=11, then the 11 left-most pixels of each field of the still picture plane are not displayed.

TDL_SWT Switch Line Number

	7		0
0xC8			TDL_SWT[8]
0xC9	TDL_SWT[7:0]		

Address: *VideoBaseAddress* + 0xC8 - 0xC9

1. Displayed picture widths other than 720 can of course be supported.

Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the value of the line on which we will switch from **TDL_TEP** to **TDL_TEP2** or **TDL_TOP** to **TDL_TOP2**. This is the unrolling function.

TDL_TDW Third Display Width

	7	5 ... 0
0x86		TDL_TDW[6:0]

Address: *VideoBaseAddress* + 0x86
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a value equal to the width in macroblocks of the picture plane.

TDL_TEP TDL Even Pointer

	7 ...	0
0x83		TDL_TEP[16]
0x84	TDL_TEP[15:8]	
0x85	TDL_TEP[7:0]	

Address: *VideoBaseAddress* + 0x83 to 0x85
Type: Read/write
Reset value: undefined

Description

This register holds the source address of the even field of the third display layer. **TDL_TEP[16:0]** is defined in units 128bytes (64 pixels). Therefore, the two LSBs must be programmed to zero.

TDL_TEP2 TDL Even Pointer 2

	7 ...	0
0xC3		TEP2[16]
0xC4	TDL_TEP2[15:8]	
0xC5	TDL_TEP2[7:0]	

Address: *VideoBaseAddress* + 0xC3 to 0xC5
Type: Read/write
Reset value: undefined

Description

This register holds the second source address of the even field of the picture plane. See unrolling feature for the right usage of **TDL_TEP** and **TDL_TEP2**. **TDL_TEP2[16:0]** is defined in units 128bytes (64 pixels). Therefore, the two LSBs must be programmed to zero.

TDL_TOP TDL Odd Pointer

	7	0
0x80		TDL_TOP[16]
0x81	TDL_TOP[15:8]	
0x82	TDL_TOP[7:0]	

Address: *VideoBaseAddress* + 0x80 to 0x82
Type: Read/write
Reset value: undefined

Description

This register holds the source address of the odd field of the picture plane. **TDL_TOP[16:0]** is defined in units 128bytes (64 pixels). Therefore, the two LSBs must be programmed to zero.

TDL_TOP2 TDL Odd Pointer

	7	0
0xC0		TOP2[16]
0xC1	TDL_TOP2[15:8]	
0xC2	TDL_TOP2[7:0]	

Address: *VideoBaseAddress* + 0xC0 to 0xC2
Type: Read/write
Reset value: undefined

Description

This register holds the second source address of the odd field of the picture plane. See unrolling feature for the right usage of **TDL_TOP** and **TDL_TOP2**. **TDL_TOP2[16:0]** is defined in units 128bytes (64 pixels). Therefore, the two LSBs must be programmed to zero.

TDL_XDO TDL X Offset

	7	1 0
0xF0		TDL_XDO[9:8]
0xF1	TDL_XDO[7:0]	

Address: *VideoBaseAddress* + 0xF0 to 0xF1
Type: Read/write
Reset value: 0

Synchronization: VSYNC

Description

TDL_XDO[9:0] is set up with a number defining the beginning of the left-hand border of the display. This offset is measured from the active (first) edge of HSYNC and is specified in units of PIXCLK cycles. The horizontal offset, *XDO*', is given by: *XDO*' = TDL_XDO + 11

TDL_XDS TDL X End

	7 … 2	1 … 0
0xF2		TDL_XDS[9:8]
0xF3	TDL_XDS[7:0]	

Address: *VideoBaseAddress* + 0xF2 to 0xF3
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a number defining the right-hand boundary of the picture display window, expressed in units of PIXCLK cycles. The value of XDS[9:0] must be equal to **TDL_XDO[9:0]** plus the width of the display window. The actual offset, *XDS*', is given by: *XDS*' = TDL_XDS + 11

TDL_YDO TDL Y Offset

	7 … 1	0
0xEE		TDL_YDO[8]
0xEF	TDL_YDO[7:0]	

Address: *VideoBaseAddress* + 0xEE to 0xEF
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a number defining the first line of the picture display, i.e. the top edge of the display window. Lines are counted from the active (first) edge of VSYNC. In an interlaced display, the same value of **TDL_YDO** would be used for both fields.

TDL_YDS TDL Y End

	7 … 1	0
0xC6		TDL_YDS[8]
0xC7	TDL_YDS[7:0]	

Address: *VideoBaseAddress* + 0xC6 to 0xC7
Type: Read/write

Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a number defining the bottom line of the display window counted in one field. In an interlaced display, the same value of **TDL_YDS** would be used for both fields.

26 Teletext interface (Ttxt)

Ttxt_Abort Teletext Abort

Address: *TtxtBaseAddress* + 0x24
Type: Write only

Description

The **TtxtAbort** register is write only. Any write to this address causes the teletext interface to abort the current operation. The state of the teletext input or output operation is reset, and the teletext data transfer is interrupted. The DMA engine is reset only after the current word read or write is complete.

TtxtAckOddEven Teletext Acknowledge odd or even

Address: *TtxtBaseAddress* + 0x20
Type: Write only

Description

This register acknowledges the odd/even toggle interrupt. Any write to the **TtxtAckOddEven** register clears the **Odd** and **Even** bits of the **TtxtIntStatus** register.

TtxtDmaAddress Teletext DMA address

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
0x00	DmaAddress

Address: *TtxtBaseAddress* + 0x00
Type: Read/write
Reset value: Undefined

Description

The **TtxtDmaAddress** register is a 32-bit register, which specifies the base address in memory for the DMA transfer to or from memory.

Bit field	**Function**
DmaAddress	The base address for DMA transfer of data to or from memory.

TtxtDmaCount Teletext DMA count

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11	10 9 8 7 6 5 4 3 2 1 0
0x04		DmaCount

Address: *TtxtBaseAddress* + 0x04
Type: Read/write
Reset value: Undefined

Description

The **TtxtDmaCount** register specifies the number of bytes to be transferred to or from memory during the DMA operation. A write to this register also starts the teletext input or output operation.

- For teletext output operation, this value must be:46 bytes x *number_of_teletext_lines_to_send*
- For teletext input operation, the value must be: 42 bytes x *number_of_teletext_lines_to_receive*

TtxtInCbDelay Teletext Input color burst delay

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x10																								CbDelay								

Address: *TtxtBaseAddress* + 0x10
Type: Read/write
Reset value: 270

Description

This register is used during teletext input mode to specify the delay from a rising edge of HSYNC to when the teletext interface starts to look for the framing code. This delay is measured in 27 MHz cycles, and is used to mask out the color burst present at the beginning of every line. The default value is 270 (0x10E), which provides a delay of 10 µs.

TtxtInStartLine Teletext Input start line

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0C																								StartLineIn								

Address: *TtxtBaseAddress* + 0x0C
Type: Read/write
Reset value: Undefined

Description

This register is used to specify the first line number to input teletext data. Its value is the delay from the toggle in **TtxtEvennotOdd** to the first valid teletext line.

TtxtIntEnable Teletext Interrupt enable

	7	6	5	4	3	2	1	0
0x1C						EvenEnable	OddEnable	InOutCompleteEn

Address: *TtxtBaseAddress* + 0x1C
Type: Read/write
Reset value: Undefined

Description

The **TxtIntEnable** register allows masking of the **TxtIntStatus** register.

Bitfield	Function
InOutCompleteEn	Enable teletext input or output operation completed interrupt.
OddEnable	Enable odd field interrupt.
EvenEnable	Enable even field interrupt.

TxtIntStatus Teletext Interrupt status

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3	2	1	0
0x18		Even	Odd	InOut Complete

Address: *TxtBaseAddress* + 0x18
Type: Read only
Reset value: Undefined

Description

The **TxtIntStatus** register gives the current state of the teletext operations. If the appropriate bits in the interrupt enable register are set then interrupts can be driven by the state of this register.

Bitfield	Function
InOutComplete	Teletext input or output operation completed.
Odd	Current (video encoder) field is ODD.
Even	Current (video encoder) field is EVEN.

TxtMode Teletext Mode

	31 30 29 28 28 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2	1	0
0x14		Odd Even	Mode

Address: *TxtBaseAddress* + 0x14
Type: Read/write
Reset value: Undefined

Description

The **TxtMode** register sets the mode of the teletext interface to teletext data output or teletext data input. It also specifies whether teletext data input memory is for odd or even fields.

Bit field	Function
Mode	Teletext interface mode: 0: teletext output enabled, 1: teletext input enabled.
OddEven	Specify odd or even fields of teletext data. 0: teletext data to or from memory is for EVEN fields 1: teletext data to or from memory is for ODD fields.

TtxtOutDelay Teletext Output delay

	31	30	29	28	28	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x08																								Delay								

Address: *TtxtBaseAddress* + 0x08
Type: Read/write
Reset value: Undefined

Description

The **TtxtOutDelay** register is used to program the delay, in 27 MHz clock periods, from the rising edge of **TtxtRequest** to the first valid teletext data bit, i.e. **TtxtData** starting to transmit.

27 Block move (USD)

USD_BMC Block Move Control

	7							0
0x9E	Num_Line[7:0]							
0x9F		D_A	NIR	CPAT	RST	VSE	VSO	EXE

Address: *VideoBaseAddress* + 0x9E and 0x9F
Type: Read/write
Reset value: 0x00

Description

This register controls the 2D block move engine.

Bitield	Description
EXE	*Execute.* This bit has to be set to launch the Block Move. This bit is internally cleared when the Block Move begins.
VSO	*Vertical Synchronization Odd.* When this bit is set the Block Move waits for the Odd Vsync to begin. This can be combined with line number. If you set VSE and VSO, the Block Move begins at next Vsync.
VSE	*Vertical Synchronization Even.* When this bit is set the Block Move waits for the Even Vsync to begin. This can be combined with line number.
RST	*Reset Block Move.* When this bit is set the block move is reset. this bit must be reset before continuing (after a minimum of 200 ns).
CPAT	*Copy Pattern.* When this bit is set the data written is specified by register. See **USD_PAT**.
NIR	*No Increment Read.* When this bit is set, the read address never changes. By this way you can fill the memory with a pattern stored in memory.
D_A	*Decrement Address.* 0: Increment the address by one after each read or write 1: Decrement the address by one after each read or write This has to be combined with the right skip value to generate a Block Move “from right bottom to left top”.
Num_Line[7:0]	When *Num_Line* is reached, the block move starts.

USD_BMH Block Move Height

	7				3	2		0
0xA2						USD_BMH[9:8]		
0xA3	USD_BMH[7:0]							

Address: *VideoBaseAddress* + 0xA2 and 0xA3
Type: Read/write
Reset value: Undefined

Description

When a 2D block move is to be executed, this register holds the number of lines of the block to be moved.

USD_BMW Block Move Width

	7 ... 0
0xA0	USD_BMW[15:8]
0xA1	USD_BMW[7:0]

Address: *VideoBaseAddress* + 0xA0 to 0xA1
Type: Read/write
Reset value: 0

Description

This register holds the number of 64-bit words to be moved in a block move operation.

The write to the higher address A1 (the least significant byte) with **USD_BMW** non-zero enables (but does not launch) the 2D block move mode. This register must be written before the associated **USD_BRP**

The byte pointer is reset by a hardware reset.

USD_BRP Memory Read Pointer

	7 ... 3	2 ... 0
0x88	Reserved	USD_BRP[19:16]
0x89	USD_BRP[15:8]	
0x8A	USD_BRP[7:0]	

Address: *VideoBaseAddress* + 0x88 to 0x8A
Type: Read/write
Reset value: Undefined

Description

This register holds the source address of the block to be moved. **USD_BRP** defines a 19-bit memory word address (a memory word is 64 bits).

USD_BSK Block Skip

	7 ... 3	2 ... 0
0xA4	Reserved	USD_BSK[19:16]
0xA5	USD_BSK[15:8]	
0xA6	USD_BSK[7:0]	

Address: *VideoBaseAddress* + 0xA4, 0xA5, 0xA6
Type: Read/write
Reset value: Undefined

Description

When a 2D block move is to be executed, this register holds the number of 64-bits words to skip before finding the beginning of the next line to move.

As this register has 19 bits, it is able to contain negative values.

USD_BWP Memory Write Pointer

	7 … 3	2 … 0
0x8C	Reserved	USD_BWP[19:16]
0x8D	USD_BWP[15:8]	
0x8E	USD_BWP[7:0]	

Address: *VideoBaseAddress* + 0x8C to 0x8E
Type: Read/write
Reset value: Undefined

Description

When a block move is to be executed, this register holds the destination address where the block is to be moved. **USD_BWP** defines a 19-bit memory word address, where a memory word is 64 bits.

USD_PAT Block Move Pattern

0xA7	PATTERN[63:56]
0xA8	PATTERN[55:48]
0xA9	PATTERN[47:40]
0xAA	PATTERN[39:32]
0xAB	PATTERN[31:24]
0xAC	PATTERN[23:16]
0xAD	PATTERN[15:8]
0xAE	PATTERN[7:0]

Address: *VideoBaseAddress* + 0xA7 to 0xAE
Type: Read/write
Reset value: Undefined

Description

This register is to be used to fill a part of memory with a pattern. It is a 64-bit pattern.

28 Video decoder (VID)

VID_ABG Start of audio bit buffer

	7		0
0x1C		ABG[13:8]	
0x1D	ABG[7:0]		

Address: *VideoBaseAddress* + 0x1C and 0x1D
Type: Read/write
Reset value: 0

Description

The register holds the starting address of the audio bitstream buffer, defined in units of 2 Kbits. If the audio bit buffer starts at address 0, then this register does not need to be set up, since its reset state is 0. When all the registers corresponding to the settings of the bit buffer are done (VID_ABG, VID_ABS), an audio soft reset must be done for values to be taken in account. In other words, it must only be changed before the first compressed data of a new sequence is input, and never during the decoding of a sequence.

VID_ABL Audio bit buffer level

	7		0
0x1E		ABL[13:8]	
0x1F	ABL[7:0]		

Address: *VideoBaseAddress* + 0x1E and 0x1F
Type: Read only
Reset value: 0

Description

This register holds the current level of occupation of the audio bit buffer, defined in units of 2 Kbits. It can be read at any time for the monitoring of the audio bit buffer level. When **VID_ABL** is greater than or equal to the value held in the **VID_ABT** register, the status bit **VID_STA.ABF** (audio bit buffer full) becomes set. When **VID_ABL** is zero, the status bit **VID_STA.ABE** (audio bit buffer empty) becomes set.

VID_ABS Audio bit buffer stop

	7		0
0x20		ABS[13:8]	
0x21	ABS[7:0]		

Address: *VideoBaseAddress* + 0x20 and 0x21
Type: Read/write
Reset value: 0

Description

This register holds the address of the top of the audio bit buffer, defined in units of 2 Kbits. The space allocated to the audio bit buffer starts at the address defined by the **VID_ABG** register, or, by default, 0. The end address of the audio bit buffer is:

(128 x ABS) + 127

VID_ABS must only be changed before the first compressed data of a new sequence is input, and never during the decoding of a sequence.

VID_ABT Audio bit buffer threshold

	7	0
0x22		ABT [13:8]
0x23	ABT [7:0]	

Address: *VideoBaseAddress* + 0x22 and 0x23
Type: Read/write
Reset value: 0
Synchronization: Edge Triggered

Description

This register holds the level of occupancy of the audio bit buffer, in units of 2 Kbits, which when reached causes the status bit **VID_STA.ABF** to become set.

If the bit **CFG_CCF.PBO** is set, then transfer of data to the audio bit buffer is prevented if the bit buffer level is at or above the level defined in the **VID_ABT** register. If **VID_ABT** is set to a value equal to the top of the bit buffer, then this automatic mechanism will ensure that overflow never occurs.

VID_BFC Backward chroma pointer

	7	0
0x5E		BFC [13:8]
0x5F	BFC [7:0]	

Address: *VideoBaseAddress* + 0x5E and 0x5F
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the chrominance buffer of the backward prediction frame picture, defined in units of 256 bytes.

VID_BFP Backward frame pointer

	7 ... 0
0x12	BFP [13:8]
0x13	BFP [7:0]

Address: *VideoBaseAddress* + 0x12 and 0x13
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the luminance buffer of the backward prediction frame picture, defined in units of 256 bytes.

VID_CDcount Bit buffer input counter

	7 ... 0
1st cycle	CDcount [23:16]
2nd cycle	CDcount [15:8]
3rd cycle	CDcount [7:0]

Address: *VideoBaseAddress* + 0x67
Type: Serial read only
Reset value: 0

Description

These three registers are accessed serially. They hold the number of bytes input to the bit buffer.

The byte pointer is reset by a hardware reset, a global soft reset or a video soft reset.

VID_CSO SRC chrominance offset

	7 ... 0
0x6C	CSO[7:0]

Address: *VideoBaseAddress* + 0x6C
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a value calculated from the fractional part of the pan vector. If no pan vector is defined, this register can be left in its reset (default) state. The method of calculation of the CSO value is given in the **VID_PAN** register description.

VID_CTL Decoding Control

	7	6	5	4	3	2	1	0
0x02	ERU	ERS				PRS	SRS	EDC

Address: *VideoBaseAddress* + 0x02
Type: Read/write
Synchronization: Edge triggered

Description

Bitfield	Description
ERU	Enable pipeline reset on picture decode error. When this bit is set, a pipeline reset is automatically generated in case of Picture Decode Error (less than DFS macroblocks decoded).
ERS	Enable pipeline reset on severe error. When this bit is set, a pipeline reset is automatically generated in case of Severe Error (more than DFS macroblocks decoded).
PRS	Pipeline reset. In order to generate a pipeline reset, this bit must be kept set for a duration of at least 4 SDRAM clock cycles (40 ns with a 100 MHz SDRAM clock).
SRS	Soft reset. In order to generate a soft reset, this bit must be kept set for a duration of at least 54 SDRAM clock cycles (540 ns with a 100 MHz primary clock).
EDC	Enable decoding. This bit must be set to allow decoding.

VID_DCF Display Configuration

	7	6	5	4	3	2	1	0
0x74			BLL	BFL	FNF	FLY	ORF	FRZ
0x75			EVD		DSR	OSD_POL	STP_CPU	STP_DEC

Address: *VideoBaseAddress* + 0x74 and 0x75
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Bitfield	Description
STP_DEC	Set this bit 1 in Zoom-outx4 mode. This prevents the MPEG video decoder from performing reconstruction accesses to the SDRAM during the video picture display, and therefore prevents picture degradation. This bit should only be used in Zoom-outx4 mode.
STP_CPU	When set, this bit prevents the CPU from accessing the SDRAM during every line the video picture is being displayed. This option must be used with care, and only in Zoom-outx4 mode.
OSD_POL	When set, this bit inverts the internal Enot0 signal used by the OSD.
DSR	Disable SRC. When this bit is set, both luminance and chrominance SRCs (sample rate converters) are disabled. In this case no horizontal filtering can occur, as would be required when the horizontal resolution of the decoded picture is equal to the horizontal resolution of the display.
EVD	Enable video display. When this bit is reset, the video output has a constant value of Y=16, C_B=C_R=128. OSD is still displayed.
FRZ	This bit is used to freeze the display. When bit 0 is set, the current decoded field is displayed continuously on both fields until this bit is reset by the user. This bit is also used to control three-to-two pull-down operation. Note that the actual effect of this bit is to simply freeze the current polarity of the internal B/notT signal.

Bitfield	Description
ORF	One Row per Frame. This bit is only active when bit **VID_DCF.FNF** is set. It defines the number of rows to be stored in the block to row RAM during a frame picture display. If it is set, one field macroblock line is stored (8 video lines). Otherwise, two field macroblock lines are stored (16 video lines). This bit is for test purposes. This bit must be reset for normal operation.
FLY	On The Fly. When this bit is set, the current picture is displayed directly from the pipeline. Otherwise, it is loaded from external memory. Before performing a pipe-reset when decoding a B-picture on-the-fly, quit the “fly” mode by resetting this bit.
FNF	Frame not Field. This bit is only used during on-the-fly decoding. The bit must be set if a frame picture is going to be displayed, or reset if field picture are to be decoded on-the-fly. For classical decoding (field or frame) the bit is always 1.
BFL	Blank First Line. If this bit is set the first active line of a picture is blanked. (Used in letter box format display).
BLL	Blank Last Line. If this bit is set, the last active line of a picture is blanked. (Used in letter box format display).

VID_DFC Displayed chroma frame pointer

	7	6	5	4	3	2	1	0
0x58			VID_DFC[13:8]					
0x59	VID_DFC[7:0]							

Address: *VideoBaseAddress* + 0x58 and 0x59
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the start address, defined in units of 256 bytes, of the chroma frame which is currently being displayed. When a new value is written this is used at the start of the next field. When **VID_DFC** is set to same value as **VID_RFC** (i.e. the decoder is writing the reconstructed picture into the buffer which is being displayed), bit **VID_TIS.OVW** must be set.

VID_DFP Displayed luma frame pointer

	7	6	5	4	3	2	1	0
0x0C			VID_DFP[13:8]					
0x0D	VID_DFP[7:0]							

Address: *VideoBaseAddress* + 0x0C and 0x0D
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the start address, defined in units of 256 bytes, of the luma frame which is currently being displayed. When a new value is written this is used at the start of the next field.

When **VID_DFP** is set to the same value as **VID_RFP** (i.e. the decoder is writing the reconstructed picture into the buffer which is being displayed), bit **VID_TIS.OVW** must be set.

VID_DFS Decoded frame size

	7	6	5 ... 0
1st cycle		CIF	DFS[13:8]
2nd cycle	DFS[7:0]		

Address: *VideoBaseAddress* + 0x24
Type: Serial read/write
Reset value: 0
Synchronization: DSYNC

Description

The circularity is reset by a software reset.

Field	Description
DFS	This register field is set up with a value equal to the number of macroblocks in the decoded picture. This is derived from the *horizontal_size* and *vertical_size* values transmitted in the sequence header.
CIF	The CIF mode. When the CIF mode bit is set the display frame macroblocks are read into the block to row converter in consecutive frame lines to enable the vertical filter to perform inter-field filtering. The frames are read in once for each field to be displayed. CIF mode must only be used for half resolution (i.e. CIF) pictures. When the CIF bit is reset the vertical filter performs intra-field filtering and the frame is only read once.

VID_DFW Decoded frame width

	7 ... 0
0x25	DFW[7:0]

Address: *VideoBaseAddress* + 0x25
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register is set up with the width in macroblocks of the decoded picture. This is derived from the *horizontal_size* value transmitted in the sequence header.

VID_DIS Video configuration

	7	6	5	4	3	2	1	0
0xD6	Reserved	CPU_priority		display_priority		ZoomOutx2	FBS	

Address: VideoBaseAddress + 0xD4
Type: R/W
Reset value: 0

Description

Bitfield	Description
CPU_priority/ display_priority	These bits set priority for the memory accesses of the CPU, still picture plane and MPEG video planes. The register can be set according to the application. In the default (5510 compatible) configuration, the CPU takes high priority and the still picture and video take lower priority. In all other configurations, still picture-plane memory accesses take high priority. "x" indicates that the bit value is not important.
ZoomOutx2	When set, this bit can be used with a block-to-row filter (set by registers VID_VFL and VID_VFC) to increase the zoom-out by X2, without increasing the bandwidth. This bit sets-up a line drop, where alternate lines are read into the memory.
FBS	Selects frame-based or field-based vertical filtering. These bits are used in conjunction with registers VID_VFC and VID_VFL to set the vertical filter mode.

CPU_priority/display_priority:

CPU_ priority	display_ priority	Still picture plane priority	MPEG video plane priority	CPU priority
xx	00 (default)	med	med	high
1x	01	high	low	high
01	01	high	low	medium
00	01	high	low	low
1x	10	high	med	high
01	10	high	med	medium
00	10	high	med	low
1x	11	high	high	high
01	11	high	high	medium
00	11	high	high	low

FBS:

bitfield	luma	chroma	type
00:	field	field	field-based filtering
01:	field	frame	luma field-based, chroma frame-based filtering
10:	frame	field	luma frame-based, chroma field-based filtering
11:	frame	frame	frame-based filtering

VID_FFC Forward chroma frame pointer

	7 … 0
0x5C	FFC[13:8]
0x5D	FFC[7:0]

Address: *VideoBaseAddress* + 0x5C and 0x5D
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the forward prediction chroma frame picture buffer, defined in units of 256 bytes.

VID_FFP Forward Luma Frame Pointer

	7		0
0x10		FFP[13:8]	
0x11	FFP[7:0]		

Address: *VideoBaseAddress* + 0x10 and 0x11
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the forward prediction luma frame picture buffer, defined in units of 256 bytes.

VID_HDF Header Data FIFO

	7 ... 0
1st cycle	HDF[15:8]
2nd cycle	HDF[7:0]

Address: *VideoBaseAddress* + 0x66
Type: Serial read only
Reset value: Undefined

Description

When the start code detector has found a start code, the header data FIFO must be read in order to identify the start code and if required to obtain the header data. The start code identification procedure is described in the STi5512 datasheet.

Before reading the header FIFO, status bit **VID_STA.HFE** should be checked to ensure that it is not empty. If bit **VID_STA.HFF** is set then the header FIFO contains at least 66 bytes of data.

As the start code detection is managed with 16-bit words, two successive reads are needed to access the whole 16-bit word; you must always perform an even number of reads before start code search restart (by software or with DSYNC signal).

The byte pointer is reset by a hardware reset, a global soft reset or a video reset.

VID_HDS Header Search

	7	3	2	1	0
Read cycle		SCM			
Write cycle			SOS	QMI	HDS

Address: *VideoBaseAddress* + 0x69
Type: Read only bit SCM, write only HDS, QMI and SOS.
Reset value: 0
Synchronization: None

Description

This register controls the header search when it is written to, and returns the location of the start code when read.

Field	Description
SCM	Start Code on MSB. This bit indicates in which byte the start code is located. If this bit is set then **VID_HDF[15:8]** contains the start code; otherwise **VID_HDF[7:0]** contains the start code.

Field	Description
HDS	Writing a 1 to this bit starts a header search. Completion of the header search is indicated by the setting of bit **VID_STA.SCH**.
QMI	This bit is used to control access to the inverse quantizer tables. 1: select the intra table, 0: select the non-intra table. For example, to write a new intra table, write **VID_HDS.QMI** = 1 then write 64 weights to **VID_QMW.**
SOS	Stop on First Slice. This bit when set allows the start code detector to stop on the first slice start code of a picture (0x00000101). To allow mismatches, this bit must be used in conjunction with **VID_HDS.SCM**.

VID_ITM Interrupt Mask

	7	6	5	4	3	2	1	0
0x3C	NDP	ERR			ABF	SFF	AFF	ABE
0x60	PDE	SER	BMI	HFF	PNC	ERC	DEI	PII
0x61	PSD	VST	VSB	BBE	BBF	HFE	CFF	SCH

Address: *VideoBaseAddress* + 0x3C, 0x60 and 0x61
Type: Read/write
Reset value: 0 (all interrupts disabled)
Synchronization: None

Description

Any bit set in this register will enable the corresponding interrupt. An interrupt is generated whenever a bit in the **VID_STA** register changes from 0 to 1 and the corresponding mask bit is set.

VID_ITS Interrupt Status

	7	6	5	4	3	2	1	0
0x3D	NDP	ERR			ABF	SFF	AFF	ABE
0x62	PDE	SER	BMI	HFF	PNC	ERC	DEI	PII
0x63	PSD	VST	VSB	BBE	BBF	HFE	CFF	SCH

Address: *VideoBaseAddress* + 0x3D, 0x62 and 0x63
Type: Read only
Reset value: 0

After the clocks have been enabled, the state changes to be the same as that of **VID_STA**.

Description

When a bit in the **VID_STA** register changes from 0 to 1, the corresponding bit in the **VID_ITS** register is set, irrespective of the state of **VID_ITM**. If the corresponding bit of **VID_ITM** is set, the interrupt is asserted. Reading the most significant byte of **VID_ITS** clears it, leaving IRQ in its de-asserted (high) state.

See the STi5512 datasheet for more information on interrupt handling.

VID_LDP Load Pointer

	7	0
0x3F		LDP

Address: *VideoBaseAddress* + 0x3F
Type: Read/write
Reset value: 0
Synchronization: None

Description

When this bit is set, the current start code detector pointer is stored in an internal register. When a PSC hit occurs, the current start code detector pointer has to be stored for further use by the VLD (if a B frame is to be processed on the fly). This bit has to be **set then reset** before the PSD interrupt corresponding to the picture which has to be decoded on the fly (i.e. during the SCH interrupt).

VID_LSO SRC Luminance Offset

	7 ... 0
0x6A	LSO[7:0]

Address: *VideoBaseAddress* + 0x6A
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a value calculated from the fractional part of the pan vector. If no pan vector is defined, this register can be left in its reset (default) state.

The method of calculation of the LSO value is given in the **VID_PAN** register description.

VID_LSR SRC Luma/Chroma Resolution

	7	0
0x6B	LSR[7:0]	
0x6D		LSR[8]

Address: *VideoBaseAddress* + 0x6B and 0x6D
Type: Read/write
Reset value: 0

Synchronization: VSYNC

Description

This register holds the upsampling factor of the luminance SRC (sample rate converter). It applies on chroma also.

The upsampling factor is equal to 256/LSR. The table below gives some examples of upsampling factors, where in each case the displayed picture has a nominal width of 720 pels. Also shown are the numbers of valid pels generated, “N”, calculated as shown in the STi5512 datasheet. Displayed picture widths other than 720 are supported.

Decoded Picture Width	**LSR**	**N**
640	228	715
640	227	718
544	193	717
544	192	721
480	170	717
480	169	722
352	125	713
352	124	719
704	250	717
704	249	720

VID_MWS Mix Weight Still Picture

	7	0
0x9C	MWS[7:0]	

Address: *VideoBaseAddress* + 0x9C
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the value of the mix between the background color and the still picture plane. When **VID_MWS** = 0xFF then the background color is being displayed and at 0 the still picture plane is being displayed.

VID_MWSV Mix Weight Still / Video

	7	0
0x9D	MWSV[7:0]	

Address: *VideoBaseAddress* + 0x9D
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the value of the mix between the video and the still picture plane. When **VID_MWSV** = 0xFF then the still picture plane is being displayed and at 0 the video is being displayed.

VID_MWV Mix Weight Video

	7 … 0
0x9B	MWV[7:0]

Address: *VideoBaseAddress* + 0x9B
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register holds the value of the mix between the background color and the video. When **VID_MWV** = 0xFF then the background color is being displayed and at 0 the video is being displayed.

VID_OUT Output of 4:2:2 display

		3	2	1 … 0
0x90		NOS	SPO	LAY

Address: *VideoBaseAddress* + 0x90
Type: Read/write
Reset value: 1100_2
Synchronization: VSYNC

Description

This register controls the content of the 4:2:2 output from the display mixing unit and the order of the OSD and sub-picture planes.

Bitfield	**Description**
LAY	The number of layers to be displayed in front of the video in the 4:2:2 output.
SPO	Place the sub-picture plane in front of the OSD plane.
NOS	Do not include the OSD in the 4:2:2 output.

VID_PAN Pan/Scan Horizontal Vector Integer Part

	7 … 3	2 … 0
0x2C		PAN[10:8]
0x2D	PAN[7:0]	

Address: *VideoBaseAddress* + 0x2C and 0x2D
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with the integer part of the horizontal pan/scan vector. The horizontal pan/scan vector defines, in the decoded picture, the location of the first displayed luminance sample relative to the first luminance sample in the line.

The **VID_LSO** and **VID_CSO** registers are set up with the fractional part of the horizontal pan/scan vector, as follows:

- PSV = ⌊horizontal pan/scan vector⌋, where ⌊x⌋ indicates the integer part of x.
- VID_LSO = 256 x (horizontal pan/scan vector - PSV)
- VID_CSO = VID_LSO / 2 (+ 1 if PSV is odd)

VID_PFH Picture F-parameters Horizontal

	7 4	3 0
0x04	BFH[3:0]	FFH[3:0]

Address: *VideoBaseAddress* + 0x04
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register contains parameters of the picture to be decoded. These parameters are extracted from the bit stream.

Field	MPEG-1	MPEG-2
FFH	Set to *forward_f_code* of the picture header.	Set to *forward_horizontal_f_code* of the picture coding extension.
	Set to *full_pel_forward_vector* of the picture header.	
BFH	Set to *backward_f_code* of the picture header.	Set to *backward_horizontal_f_code* of the picture coding extension.
	Set to *full_pel_backward_vector* of the picture header.	

VID_PFV Picture F-parameters Vertical

	7 4	3 0
0x05	BFV[3:0]	FFV[3:0]

Address: *VideoBaseAddress* + 0x05
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register contains parameters of the picture to be decoded. These parameters are extracted from the bitstream.

In MPEG-1 mode (i.e. when **VID_PPR2.MP2** is reset), **VID_PFV** is not used.

Field	Description
FFV	The *forward_vertical_f_code* of the picture coding extension.
BFV	The *backward_vertical_f_code* of the picture coding extension.

VID_PPR1 Picture Parameters 1

	7	6	5	4	3	2	1	0
0x06		OTF	PCT[1:0]		DCP[1:0]		PST[1:0]	

Address: *VideoBaseAddress* + 0x06
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register contains parameters of the picture to be decoded. These parameters are extracted from the bit stream.

In MPEG-1 mode (i.e. when **VID_PPR2.MP2** is reset), only PCT has to be set; the other bits must be reset.

Field	Description
OTF	When set this bit directs the decoded data in the block-to-row memory to be displayed directly. The picture data is not reconstructed in memory.
PCT	Set to the two least significant bits of *picture_coding_type* in the picture header.
DCP	Set equal to *intra_dc_precision* of the picture coding extension. The value “11”, defining a precision of 3 bits, is not allowed.
PST	Set to the *picture_structure* bits of the MPEG-2 picture coding extension. 00: Frame picture. This value is illegal in the MPEG-2 variable. 01: Top field. 10: Bottom field. 11: Frame picture.

VID_PPR2 Picture Parameters 2

	7	6	5	4	3	2	1	0
0x07		MP2	TFF	FRM	CMV	QST	IVF	AZZ

Address: *VideoBaseAddress* + 0x07
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register contains parameters of the picture to be decoded. These parameters are extracted from the bitstream.

In MPEG-1 mode, all bits must be reset to 0.

Field	Description
AZZ	This bit is set equal to the *alternate_scan* bit of the picture coding extension.
IVF	This bit is set equal to the *intra_vlc_format* bit of the picture coding extension.
QST	This bit is set equal to th*e q_scale_type* bit of the picture coding extension.
CMV	This bit is set equal to the *concealment_motion_vectors* bit of the MPEG-2 picture coding extension. It indicates that motion vectors are coded for intra macroblocks.
FRM	This bit is set equal to the *frame_pred_frame_dct* bit of the picture coding extension.
TFF	This bit is set equal to the *top_field_first* bit of the MPEG-2 picture coding extension.
MP2	*MPEG-2 mode*. When this bit is set, the STi5512 expects an MPEG-2 video bitstream. If it is reset, then an MPEG-1 bitstream is expected.

VID_PTH Panic threshold

	7	6	5	4	3	2	1	0
0x2E			FPAN	PEN	NFW	PTH[10:8]		
0x2F	PTH[7:0]							

Address: *VideoBaseAddress* + 0x2E and 0x2F
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

The panic threshold defines a block-to-row DRAM fullness level. In the block-to-row converter, there are two counters indicating the number of luma and chroma words stored. If the luma counter pointer is less than PTH or the chroma counter pointer is less than PTH divided by 2, a PANIC flag is set.

The panic threshold is programmed in units of DRAM cells.

Field	Description
PTH	Panic mode threshold.
NFW	Near Forward. This bit allows the user to select which prediction direction will be retained for bidirectional macroblocks in panic mode. Forward if set, backward otherwise.
PEN	Panic mode enable. When set this bit allows the decoding to set the panic flag and enter the panic mode.
FPAN	Force Panic Mode. This bit, when set, forces panic mode while decoding. For test purposes only.

VID_QMW Quantization Matrix Data

	7 … 0
0x76	QMW[7:0]

Address: *VideoBaseAddress* + 0x76
Type: Write only
Reset value: Undefined
Synchronization: None

Description

This address is used to load the quantization coefficients in the order in which they appear in the bit stream, i.e. zig-zag order. The bit **VID_HDS.QMI** defines which matrix (Intra or Inter) is written.

For example, to write a new intra table, write **VID_HDS.QMI** = 1, and then write 64 weights to **VID_QMW**.

VID_REV Revision

	7 … 0
0x78	REV[7:0]

Address: *VideoBaseAddress* + 0x78
Type: Read only

Description

This register holds the version number of the device. The contents of this register match the marking on the package.

VID_LCK Memory configuration register lock

	7	6	5	4	3	2	1	0
0x7B								LCK

Address: VideoBaseAddress + 0x7B
Type: R/W
Reset value: 0

Description

Bitfield	Description
LCK	When set, this bit locks the current values of registers CFG_DRC, CFG_MCF and CFG_CCF.

VID_RFC Reconstructed Chroma Frame Pointer

	7	5	0
0x5A		RFC[13:8]	
0x5B	RFC[7:0]		

Address: *VideoBaseAddress* + 0x5A and 0x5B
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the reconstructed (decoded) chroma frame picture buffer, defined in units of 256 bytes.

VID_RFP Reconstructed Frame Pointer

	7	5	0
0x0E		RFP[13:8]	
0x0F	RFP[7:0]		

Address: *VideoBaseAddress* + 0x0E and 0x0F
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register holds the start address of the reconstructed (decoded) luma frame picture buffer, defined in units of 256 bytes.

VID_SCDcount Bit Buffer Output Counter

	7 ... 0
1st cycle	SCDcount [23:16]
2nd cycle	SCDcount [15:8]
3rd cycle	SCDcount [7:0]

Address: *VideoBaseAddress* + 0x68
Type: Serial read only
Reset value: 0

Description

These three registers are accessed serially. This register holds the number of 16-bit words output from the bit buffer into the Start Code Detector.

The byte pointer is reset by a hardware reset, a global soft reset or a video reset.

VID_SCN Pan/scan vertical vector

	7	6	5	4	3	2	1	0
0x87			SCN[8:3]					

Address: *VideoBaseAddress* + 0x87
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register defines the position of the top left-hand corner of the displayed picture with respect to the decoded picture. The position is defined by this register in macroblocks, i.e. 2 fields of 8-lines. Register bits VID_YDO.SCN[4:0] define the line within the macroblock, where bits VID_YDO.SCN[2:0] are used in standard mode and VID_YDO.SCN[4:3] are used in zoom modes.

VID_SPB Sub-picture Buffer Begin

	7 ... 3	2 ... 0
0x50		SPB[10:8]
0x51	SPB[7:0]	

Address: *VideoBaseAddress* + 0x50 and 0x51
Type: Read/write
Reset value: 0
Synchronization: none

Description

This register holds the start address of the sub-picture circular buffer and is programmed in units of 64 bytes. The buffer should be aligned on a 1 Kbyte boundary.

VID_SPE Sub-picture Buffer End

	7	2	0
0x52		SPE[10:8]	
0x53	SPE[7:0]		

Address: *VideoBaseAddress* + 0x52 and 0x53
Type: Read/write
Reset value: 0
Synchronization: none

Description

This register holds the stop address of the sub-picture circular buffer and is programmed in units of 64 bytes. The buffer should be aligned on a 1 Kbyte boundary.

VID_SPRead SubPicture Read Pointer

	7	2	0
1st cycle		SPR[18:16]	
2nd cycle	SPR[15:8]		
3rd cycle	SPR[7:0]		

Address: *VideoBaseAddress* + 0x4E
Type: Serial read/write
Reset value: 0
Synchronization: VSYNC

Description

These three registers are accessed serially. The byte pointer is reset by a hardware reset. This is the absolute address in the memory. This register is used when the software needs to set the read address of the sub-picture decoder and is programmed in units of 64-bit words. This value is taken into account after a VSYNC.

VID_SPWrite SubPicture Write Pointer

	7	2	0
1st cycle		SPW[18:16]	
2nd cycle	SPW[15:8]		
3rd cycle	SPW[7:0]		

Address: *VideoBaseAddress* + 0x4F
Type: Serial read/write
Reset value: 0
Synchronization: None

Description

These three registers are accessed serially. The byte pointer is reset by a hardware reset. This is the absolute address in the memory. This register is used when the software needs to set the write address of the sub-picture decoder and is programmed in units of 64-bit words.

It is recommended to stop loading sub-picture data to the sub-picture decoder FIFO before changing the value of this register.

VID_SRA Audio Soft Reset

	7 ... 1	0
0x35		SRA

Address: *VideoBaseAddress* + 0x35
Type: Read/write
Reset value: 0
Synchronization: None

Description

In order to generate an audio soft reset (bit buffer), this bit must be kept set for a duration of at least 54 SDRAM clock cycles (540 ns with a 100 MHz primary clock).

VID_SRV Video Soft Reset

	7 ... 1	0
0x39		SRV

Address: *VideoBaseAddress* + 0x39
Type: Read/write
Reset value: 0
Synchronization: none

Description

In order to generate a video soft reset, this bit must be kept set for a duration of at least 54 SDRAM clock cycles (540 ns with a 100 MHz primary clock).

VID_STA Status

	7	6	5	4	3	2	1	0
0x3B	NDP	ERR			ABF	SFF	AFF	ABE
0x64	PDE	SER	BMI	HFF	PNC	ERC	DEI	PII
0x65	PSD	VST	VSB	BBE	BBF	HFE	BFF	SCH

Address: *VideoBaseAddress* + 0x3B, 0x64 and 0x65
Type: Read only
Reset value: See below.

Description

This register contains a set of bits which represent the status of the decoder at any instant. Any change from 0 to 1 of any of these bits sets the corresponding bit of the **VID_ITS** register, and can thus potentially cause an interrupt.

The status vector is sampled internally at the start of the read cycle accessing the most significant byte of **VID_STA** (address 0x3B). VST, VSB and PSD are pulses and are unlikely ever to be read as a 1.

The status bits are described in the table below. The reset column shows the state after clocks have been enabled.

Bitfield	Description	Reset
SCH	Start Code Hit. This bit is set whenever the first 16-bit word available in the header FIFO contains one of the start codes recognized (see the STi5512 datasheet). While data is being read from the header FIFO, this bit can be tested to determine whether the next word contains a start code.	0
BFF	Video Compressed Data (bit stream) FIFO Full. This bit is set when the video CD FIFO is full. This bit is equivalent to the signal $\overline{CDREQ}$.	0
HFE	Header FIFO Empty. This bit is set when the header FIFO is empty.	1
BBF	Video bit buffer full. This bit is set when the bit buffer level (= **VID_VBL**) is greater than or equal to the value loaded into the **VID_VBT** register.	1
BBE	Video bit buffer empty. This bit is set when the bit buffer contains no data.	1
VSB	VSYNC Bottom. This bit is set for a short time at the beginning of the bottom field, corresponding to the rising edge of the B/$\overline{T}$ signal.	0
VST	VSYNC Top. This bit is set for a short time at the beginning of the top field, corresponding to the falling edge of the B/$\overline{T}$ signal.	0
PSD	Pipeline Starting to Decode. This bit is set for a short period at the instant the pipeline starts decoding a picture.	0
PII	Pipeline Idle. This bit is set when the STi5512 is ready to decode a picture, i.e. when the pipeline is inactive and it has found the next picture start code. It becomes low when the decoding of a picture starts and high when picture decoding is complete and the next picture start code has been found by to pipeline.	1
DEI	Decoder Idle. This bit is set when the STi5512 is not in the course of decoding a picture, i.e. when the pipeline is inactive. It becomes low when the decoding of a picture starts and high when picture decoding is complete.	1
ERC	Error Concealment. This bit is set when an error is detected in the bit stream and the mechanism of error concealment is active.	0
PNC	Panic. This bit is set when decoding is in late compare to display.	0
HFF	Header FIFO Full. This bit is set when the header FIFO contains at least 66 bytes.	0
BMI	Block Move Idle. This bit is set when a block move operation has terminated. It is automatically reset at the start of a block move.	1
SER	Severe Error or overflow error. This bit is set when more than the programmed number of macroblocks (defined by DFS) have been decoded, either due to a data or a programming error. Decoding is halted automatically when this error condition is detected. This bit is reset by all three types of reset.	0
PDE	Picture Decoding Error or underflow error. This bit is set when less than the programmed number of macroblocks (defined by DFS) have been decoded, either due to a data or a programming error. Decoding is halted automatically when this error condition is detected. This bit is reset by all three types of reset.	0
ABE	Audio bit-buffer empty. This bit is set when the audio bit buffer contains no data.	0
AFF	Audio compressed data (bit stream) FIFO full. This bit is set when the audio CD FIFO is full.	0
SFF	Sub-picture compressed data (bit stream) FIFO full. This bit is set when the sub-picture CD FIFO is full.	0
ABF	Audio bit-buffer full.	0
ERR	Inconsistency error in PES parser.	0
NDP	New discarded packet.	0

VID_TIS Task Instruction

	7	6	5	4	3	2	1	0
0x03			SKP[1:0]		OVW	FIS	RPT	EXE

Address: *VideoBaseAddress* + 0x03
Type: Write only
Reset value: 0
Synchronization: VSYNC

Description

This register contains 6 bits of the decoding task instruction.

Field	Description
EXE	When this bit is not set, no decoding or skipping task is executed for one or two VSYNC periods, depending on the state of **VID_TIS.RPT**. If set, the next task (decoding or skipping) is executed. **EXE** is internally cleared when the task starts its execution, i.e. setting this bit activates only one execution.
RPT	When this bit is set, the task duration is two VSYNC periods. When the frame display rate is equal to the picture decoding rate, **RPT** will generally always be high. In case of a task launched by **VID_TIS.FIS**, **RPT** is not taken in account.
FIS	Force instruction: if this bit is set, the task described by this register will be launch immediately. This bit is reset after its action. Its effect is the same as the effect of a VSYNC.
OVW	This bit must be set when the displayed picture and the reconstructed picture share the same buffer (i.e. **VID_DFP** = **VID_RFP**). It enables the overwrite mode which ensures that the reconstructed picture does not overwrite data which has not yet been displayed.
SKP	These bits are part of the instruction register, and are thus synchronized with VSYNC. They define the skipping of one or two pictures. If skipping is required, then these bits must be set up as part of the instruction for the picture which will be decoded immediately after the skipped pictures. 00: No skip (default) 01: Skip one picture, decode next 10: Skip two pictures, decode next 11: Skip one picture then stop

VID_TRF Temporal Reference

	7	6	5	4	3	2	1	0
0x56					DC2	DTR	TRF[9:8]	
0x57	TRF[7:0]							

Address: *VideoBaseAddress* + 0x56 and 0x57
Type: Read/write
Reset value: 0
Synchronization: DSYNC

Description

This register contain extensions of the decoding task instruction. It is used only when decoding a B frame on the fly.

Field	Description
TRF	*Temporal reference.* This field holds the temporal reference of the current decoded B frame. On every B frame re-decode the VLD uses the temporal reference field **TRF** to resynchronize on the previously decoded picture. **TRF** is used only when bit **DTR** is reset.
DTR	*Disable temporal reference comparison.* This bit is used only when decoding a B frame on the fly. On every B frame re-decode the VLD uses the temporal reference field **VID_TRF.TRF[9:0]** to resynchronize on the previously decoded picture. When this bit is set this comparison mechanism is disabled.
DC2	*Re-decode same B Frame twice.* When this bit is set it signals the VLD that the next picture will have to be decoded twice. This bit is used when decoding a B Frame on the fly. It has to be set by the user **before** the first PSD interrupt corresponding to a B frame and reset on the according PSD interrupt.

VID_VBG Start of Video Bit Buffer

	7	5	0
0x14		VBG[13:8]	
0x15	VBG[7:0]		

Address: *VideoBaseAddress* + 0x14 and 0x15
Type: Read/write
Reset value: 0
Synchronization: None

Description

The register holds the starting address of the video bit buffer, defined in units of 2 Kbits. If the video bit buffer starts at address 0, then this register does not need to be set up, since its reset state is 0. When all the registers corresponding to the settings of the bit buffer are done (VID_VBG, VID_VBS), a video soft reset must be done for values to be taken in account. In other words it must only be changed before the first compressed data of a new sequence is input, and never during the decoding of a sequence.

VID_VBL Video Bit Buffer Level

	7	5	0
0x16		VBL[13:8]	
0x17	VBL[7:0]		

Address: *VideoBaseAddress* + 0x16 and 0x17
Type: Read only
Reset value: 0

Description

This register holds the current level of occupation of the video bit buffer, defined in units of 2 Kbits. It can be read at any time for the monitoring of the video bit buffer level. When **VID_VBL** is greater than or equal to the value held in the **VID_VBT** register, the status bit **VID_STA.BBF** (video bit buffer full) becomes set. When **VID_VBL** is zero, the status bit **VID_STA.BBE** (video bit buffer empty) becomes set.

VID_VBS Video Bit Buffer Stop

	7	6	5	4	3	2	1	0
0x18			VBS[13:8]					
0x19	VBS[7:0]							

Address: *VideoBaseAddress* + 0x18 and 0x19
Type: Read/write
Reset value: 0
Synchronization: None

Description

This register holds the address of the top of the video bit buffer, defined in units of 2 Kbits. The space allocated to the video bit buffer starts at the address defined by the **VID_VBG** register, or, by default, 0. The end address of the video bit buffer is (in units of 64 bits words):

(32 x **VID_VBS**) + 31

VID_VBS must only be changed before the first compressed data of a new sequence is input, and never during the decoding of a sequence.

VID_VBT Video Bit Buffer Threshold

	7	6	5	4	3	2	1	0
0x1A			VBT[13:8]					
0x1B	VBT[7:0]							

Address: *VideoBaseAddress* + 0x1A and 0x1B
Type: Read/write
Reset value: 0
Synchronization: Edge triggered

Description

This register holds the level of occupancy of the video bit buffer, in units of 2 Kbits, which when reached causes the status bit **VID_STA.BBF** to become set, i.e. if **VID_VBL** $\geq$ **VID_VBT**, then **VID_STA.BBF** is set.

If the bit **CFG_CCF.PBO** is set, then transfer of data from the video CD FIFO to the bit buffer is prevented if the bit buffer level is at or above the level defined in the **VID_VBT** register. If **VID_VBT** is set to a value equal to the top of the bit buffer, then this automatic mechanism will ensure that overflow never occurs.

VID_VFC Chroma vertical filter

	7	6	5	4	3	2	1	0
0x55				VFC[4:0]				

Address: *VideoBaseAddress* + 0x55
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

The value in this register selects the vertical filter mode used for the chroma data. Filter modes are selected by setting registers VID_VFC and VID_VFL, field-based or frame-based filtering must also be selected by setting register VID_DIS. The different filter modes and register configurations are described in the section "*Block-to-row converter (vertical filtering)*" in the STi5512 Data Sheet.

VID_VFL Luma vertical filter

	7	6	5	4	3	2	1	0
0x54				VFL[4:0]				

Address: *VideoBaseAddress* + 0x54
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

The value in this register selects the vertical filter mode used for the luma data. Filter modes are selected by setting registers VID_VFC and VID_VFL. Field-based or frame-based filtering must also be selected by setting register VID_DIS. The different filter modes and register configurations are described in the section "*Block-to-row converter (vertical filtering)*" in the STi5512 Data Sheet.

VID_XDO Display X Offset

	7		1	0
0x70			XDO[9:8]	
0x71	XDO[7:0]			

Address: *VideoBaseAddress* + 0x70 to 0x71
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

VID_XDO is set up with a number defining the beginning of the left-hand border of the display. This offset is measured from the active (first) edge of HSYNC and is specified in units of PIXCLK cycles.

The horizontal offset, *XDO*', is given by: *XDO*' = **VID_XDO** + 4

The offset, *XDO*' cannot be less than 153 video decoder clock cycles.

VID_XDS Display X End

	7		1	0
0x72			XDS[9:8]	
0x73	XDS[7:0]			

Address: *VideoBaseAddress* + 0x72 to 0x73

Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a number defining the right-hand boundary of the picture display window, expressed in units of PIXCLK cycles. The value of **VID_XDS** must be equal to **VID_XDO** plus the width of the display window.

The actual offset, *XDS*', is given by:

***XDS*' = VID_XDS + 4**

VID_XFWDisplayed Frame Width

	7	6	5	4	3	2	1	0
0x28	XFW[7:0]							

Address: *VideoBaseAddress* + 0x28
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a value equal to the width in macroblocks of the displayed picture. This is derived from the *horizontal_size* value transmitted in the sequence header.

VID_YDO Display Y offset

	7	6	5	4	3	2	1	0
0x6E			SCN[4:0]					YDO[8]
0x6F	YDO[7:0]							

Address: *VideoBaseAddress* + 0x6E to 0x6F
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

Bitfield	Description
YDO[8:0]	This value defines the first line of the picture display, i.e. the top edge of the display window. Lines are counted from the active (first) edge of VSYNC. In an interlaced display, the same value of VID_YDO must be used for both fields.
SCN[4:0]	These bits define the line within the macroblock set by register VID_SCN. Bits 0 to 2 only should be used in normal (non zoom) modes Bits 3 to 4 can be used in zoom modes where the video is upscaled by 2 or 4, to give continuous video scan.

VID_YDS Display Y End

	7	6	5	4	3	2	1	0
0x46								YDS[8]
0x47	YDS[7:0]							

Address: *VideoBaseAddress* + 0x46 to 0x47
Type: Read/write
Reset value: 0
Synchronization: VSYNC

Description

This register is set up with a number defining the bottom line of the display window. The value of YDS must be equal to:

VID_YDO + (display size / 2) -1

In an interlaced display, the same value of **VID_YDS** must be used for both fields.

29 Revision history

29.1 Changes to 7112462A to create 7112462B

Section	Change
Audio	Register AUD_SFR on page 47: reset changed from "Undefined" to "0"
OSD	Registers OSD_BOT on page 125 and OSD_TOP on page 126: Changed from 14 bits to 15.
Video	Register VID_YDO **on page 199**: SCN bits 3 ad 4 added for use in zoom modes. Comment made in VID_SCN **on page 191** and register map Table 29 on page 23 updated.
TDL	Register TDL_TDW on page 164 expanded to 6 bits to increase the width in macroblocks of the picture plane. Register map Table 26 on page 22 updated.
EMI	Table 11: *EMI configuration registers (EMI) on page 9*: bank and register numbers corrected. External memory interface (EMI) **on page 91**: Register EMI_SDRAMNOPGEN removed, corresponding change made to register map. EMI_ConfigPadlogic on page 98: Notes added to bitfield descriptions.
ASC	Asynchronous serial controller (ASC) on page 26: New register ASC_n_Retries added, bits ASC_n_Control bits BaudMode and CTSenable added.
Reg map	Table 7: *Cache Control Registers (CAC) on page 7*, Table 20: *Parallel input/output registers (PIO) on page 13* and Table 19: *PES parser registers (PES) on page 13*: Correction of discrepancies and repetitions in the register map.

Table 47 7112462B-7112462A

29.2 Changes to STi5510 to create the STi5512 doc 7112462A

The following changes have been made between this and the STi5510 Register Manual, version 7112462A. The changes listed below are hypertext linked into the document where they are also identified by change bars.

Register block base variables on page 2: Changes to the base variable values are listed in this table.

Cache control (CAC) on page 54: This section has been re-written to describe the new and changed cache control registers. Corresponding changes have been made to register map Table 7 on page 7.

CFG_DRC on page 60: New bit added to select 16Mbit or 64Mbit SDRAM. Corresponding change made to register map Table 8 on page 7.

Digital encoder (DEN) on page 66: Registers DEN_REG64 to DEN_REG71 are new and the section has been rewritten for clarification. Corresponding changes have been made to register map Table 10 on page 8.

External memory interface (EMI) on page 91: This section has been re-written to describe the new and changed External Memory Interface registers. Corresponding changes have been made to register map Table 11 on page 9.

OSD_CFG on page 126: Bit OSD_CFG.LLG added to double the linked list granularity. This doubles the SDRAM area available to the OSD linked-list pointer. Corresponding changes have been made to register map Table 18 on page 12.

SPD_CTL1 on page 152: Register bit SPD_CTL.BOT_TOP added to enable selection of top or bottom field for subpicture command execution. Corresponding changes have been made to register map Table 24 on page 20.

TDL_DCF on page 162: Register bit TDL_DCF.FORMAT added to enable selection of still picture format for storage in memory. Corresponding change made to register map Table 26 on page 22.

TDL_SCN on page 163: TDL_SCN.HSCN bits added to enable selection of the location of the first displayed pixel of the still picture plane, with respect to the decoded picture. Corresponding changes have been made to register map Table 26 on page 22.

VID_SCN on page 191 and **VID_YDO on page 199**: These two registers are now used in conjunction to define the position of the top-left-hand corner of the displayed picture, with respect to the decoded picture. The position is defined in macroblocks by VID_SCN, and by line within the macroblock by the new bits VID_YDO.SCN. Corresponding changes have been made to register map Table 29 on page 23.

VID_DIS on page 180: This is a new register for video configurations - CPU and display priority setting, zoom-out by X2, and used in conjunction with VID_VFC and VID_VFL for field or frame-based vertical filter selection. Corresponding changes have been made to register map Table 29 on page 23.

VID_LCK on page 190: This is a new register which locks the current values of registers CFG_DRC, CFG_MCF and CFG_CCF. Corresponding changes have been made to register map Table 29 on page 23.

30 Index of registers

STMicroelectronics GROUP OF COMPANIES
Australia - Brazil - Canada - China - France - Germany - Italy - Japan - Korea - Malaysia - Malta - Mexico - Morocco
The Netherlands - Singapore - Spain - Sweden - Switzerland - Taiwan - Thailand - United Kingdom - U.S.A.

http://www.st.com